Chapter 8
Cardiovascular System

Abstract The components and operation of the human circulatory system are overviewed, followed by a detailed discussion about the properties, pressure, and flow of blood in it. This leads to analysis of the physical aspects of strokes and aneurysms. Models of the circulatory and systems are developed and applied to the normal and malfunctioning heart.

There are three components of the cardiovascular system. (a) Blood is the vehicle for transport. It transports fuel from the digested food to the cells, transports oxygen from the air in the lungs so it can combine with fuel to release energy, and it disposes of waste products—such as carbon dioxide from the fuel engine and other metabolic wastes. (b) The circulatory system is the distribution system, and consists of a series of branched blood vessels. (c) The heart is the four-chambered pump composed mostly of cardiac muscle that enables this circulatory flow. General descriptions of the cardiovascular system can be found in [5, 7, 10, 11, 14, 16, 23, 24, 27, 29, 36, 37, 40, 46, 57, 58, 66].

8.1 Overview of the Circulatory System and Cardiac Cycle

8.1.1 Circulation

Blood flow from the heart branches into two separate systems (Fig. 8.1). In the *pulmonary circulation* system, the right side of the heart pumps oxygen-poor ("blue") blood to the lungs to be oxygenated; oxygen-rich ("red") blood then returns to the left side of the heart. In the *systemic circulation* system, the left side of the heart pumps this oxygen-rich ("red") blood to the rest of the body where it is used; oxygen-poor ("blue") blood then returns to the right side of the heart. This occurs in a system of arteries that conducts the blood from the heart to the lungs and other organs and components, and a system of veins that returns the blood to the heart.

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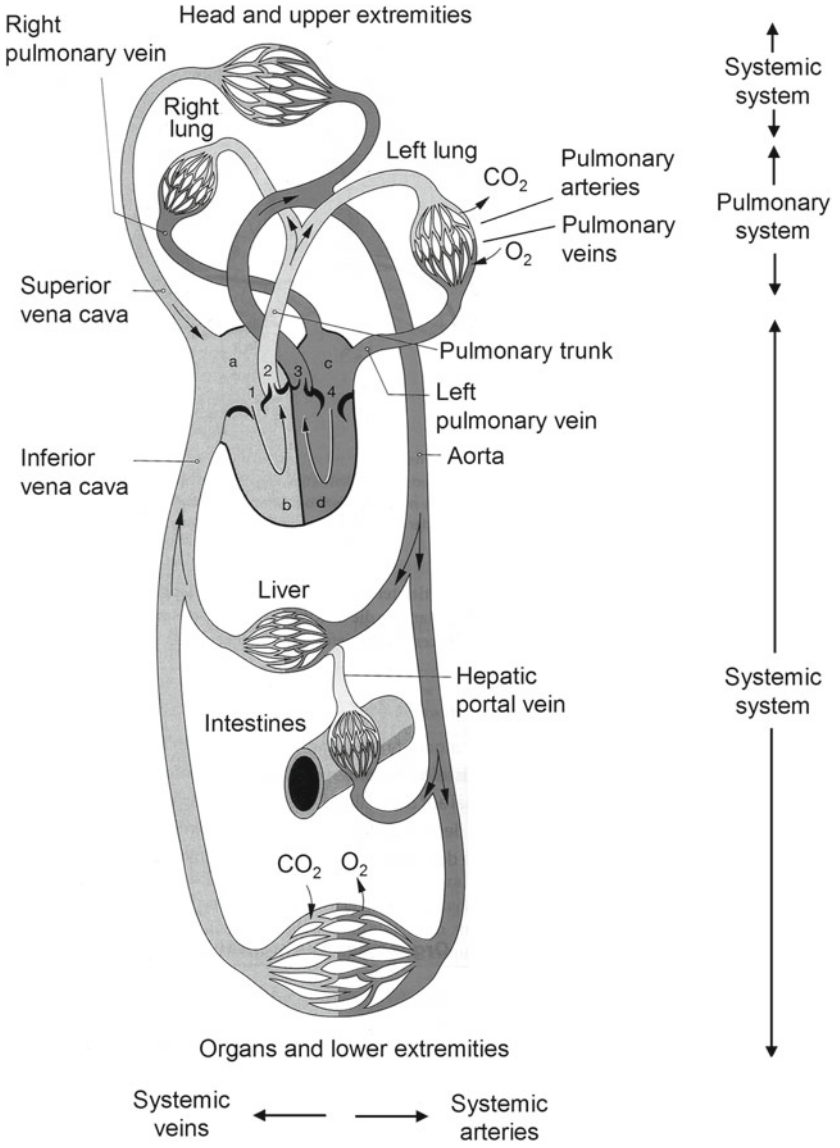


Fig. 8.1 Blood circulation system, and labeled within the heart: the **a** right atrium, **b** right ventricle, **c** left atrium, **d** left ventricle, *1* right atrioventricular (tricuspid) valve, *2* pulmonary semilunar valve, *3* aortic semilunar valve, *4* left atrioventricular (bicuspid, mitral) valve (From [65])

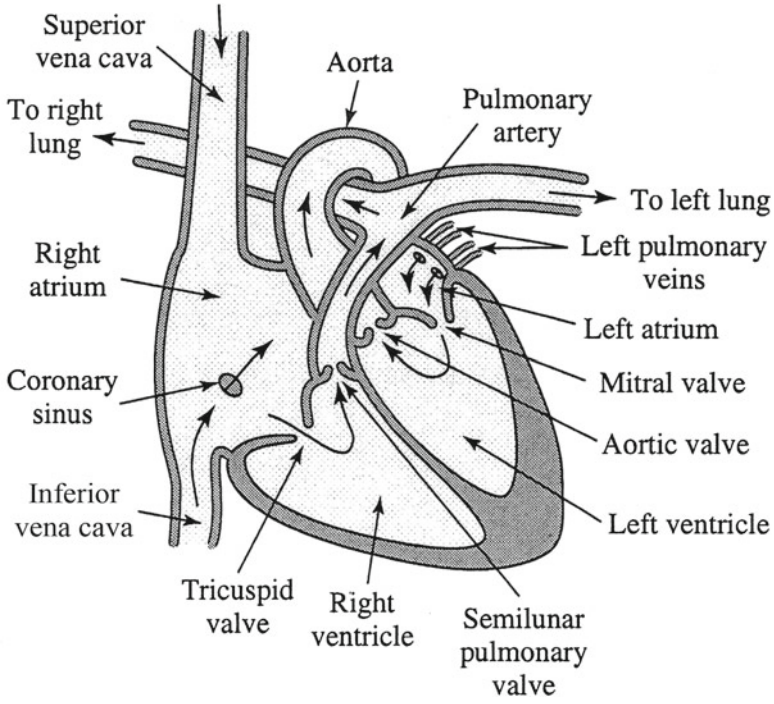


Fig. 8.2 Diagram of the heart, with its principle chambers, valves, and vessels (From [4])

In the pulmonary system (Fig. 8.1), blood enters the *right atrium* (RA) of the heart (Fig. 8.2) through the *inferior and superior vena cava(e)* (vee'-na cae'vuh). The blood passes through the *right atrioventricular (or tricuspid) valve* to enter the *right ventricle* (RV). Blood is first pumped through the *pulmonary semilunar valve* to the pulmonary arteries, which branch out into a series of more minor arteries and arterioles, and then into capillaries in the lungs. These pulmonary capillaries combine into venules (veen'-yools), then into more major veins, and finally into the pulmonary veins.

In the systemic system (Fig. 8.1), blood enters the *left atrium* (LA) of the heart through the pulmonary veins. The blood passes through the *left atrioventricular (or bicuspid or mitral) valve* to enter the *left ventricle* (LV). Blood is pumped through the *aortic semilunar valve* to the *aorta*, which first branches out into a series of major and then minor arteries (with smaller diameters, the arterioles), and finally into a series of capillaries in the systems where gas exchange and diffusion occur. These systemic capillaries combine into venules, then more major veins, and finally into the superior (from above the heart) and inferior (from below the heart) vena cavae.

(A useful mnemonic for the flow of blood in the heart comes from knowing that the author once lived on Rahlves Drive in Castro Valley, California—a town approximately 20 miles south of Berkeley. The whole heart cycle starts with blood

Table 8.1 Normal resting values of blood pressure, with system volumes

	P (mmHg)	V (L)
Systemic arteries	100	1.0
Systemic veins	2	3.5
Pulmonary arteries	15	0.1
Pulmonary veins	5	0.4

flowing into the right atrium (RA) and then getting oxygenated in the lungs, returning to the heart (H), and then continuing with the blood leaving the left ventricle (LV) and exiting (E) for the systems (S). Put together this spells RAHLVES. The most important concept here is that deoxygenated blood enters the heart through the right atrium (RA) and eventually oxygenated blood leaves through the left ventricle (LV) of the heart to be used by the body for metabolism. For some, it may be easier to remember that an American Daron Rahlves was the winner of the super-G downhill skiing competition in the 2001 World Championships.)

The systemic and pulmonary systems have similarities and differences. They have the same volumetric flow rate Q averaged over a heart beat. (If they were not equal, blood would have to pile up or leak out somewhere.) It is ∼5–6 L/min, and it is also equal the average rate of blood pumped by the heart, the *cardiac output*. In the systemic system the blood disposes of oxygen and receives carbon dioxide, while in the pulmonary system the blood disposes of carbon dioxide and receives oxygen. Table 8.1 shows that the systemic system has higher pressures (in the arteries) and larger volumes than the pulmonary system (even with the same Q). This difference in pressure makes sense because the blood vessels need to be longer to get to more distant regions in the body in the systemic system. The left heart (LA + LV) is bigger (and is a larger pump) than the right heart (RA + RV) because of this need to generate higher pressure for systemic circulation. The heart walls consist mostly of the thick middle muscle layer, the *myocardium*, which is lined internally by a thin layer of tissue, the *endocardium*, and externally by a membrane, the *epicardium*. The two sides of the heart are separated by a wall called a *septum*. The difference in volume is due to the longer distance of travel and the much higher number of systems that receive blood in the systemic system. Table 8.1 also shows that arteries have higher pressure than the corresponding veins, whereas the veins have larger volumes. The total volume of blood is ≃5 L.

For a person at rest, 12% of the blood is in the heart chambers, 2% in the aorta, 8% in the arteries, 1% in the arterioles, 5% in the capillaries, 50% in the systemic veins, and 18% in the pulmonary circulation.

Major arteries and veins are shown in Figs. 8.3 and 8.4. Tables 8.2 and 8.3 provide a very approximate quantification of the vessels in the circulatory system. Table 8.4 provides more specific information about some of the major or large arteries [57]. There is a taper along the length of most large arteries (Table 8.4), where the artery

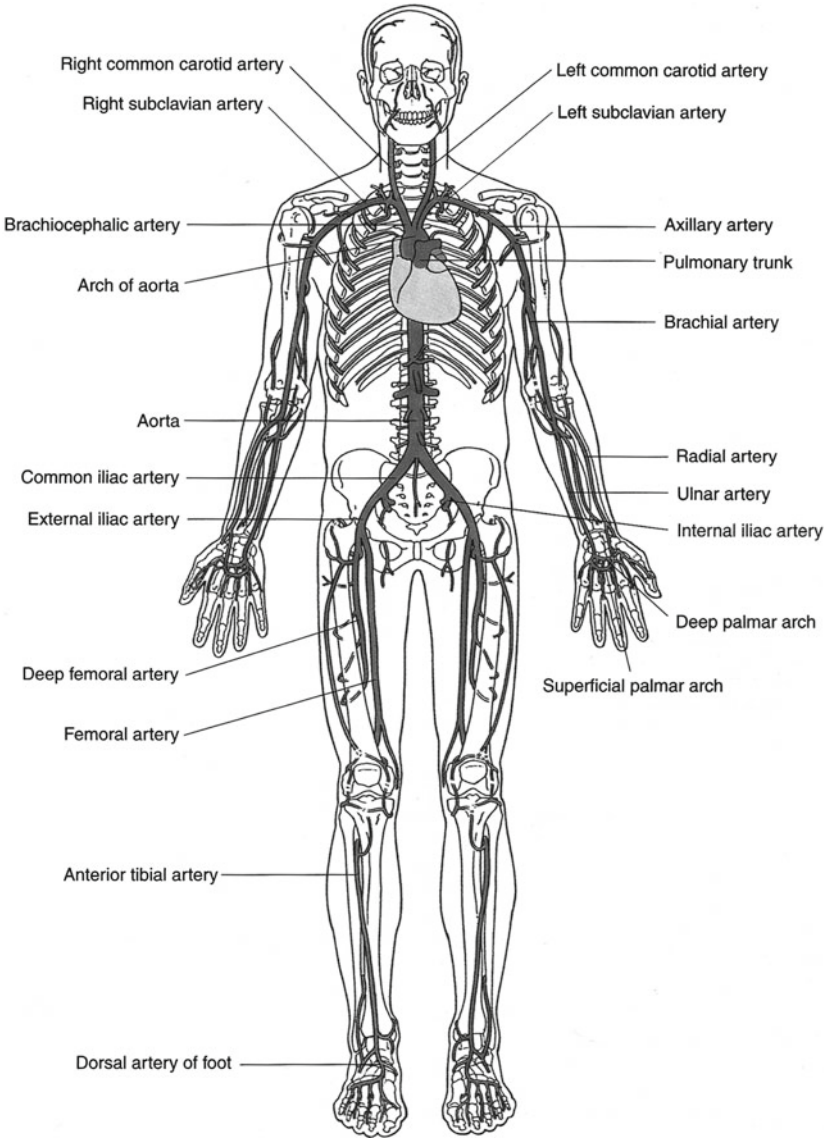


Fig. 8.3 Major arteries in the body. Arteries carry blood away from the heart in the systemic and pulmonary system. Many come in pairs, such as the right and left radial arteries (From [53]. Used with permission)

radius decrease can be modeled as a slow exponential decay [57]. Small arteries can be modeled assuming no taper. The length L of a smaller artery can be estimated from its diameter d by $L/d \sim 25$.

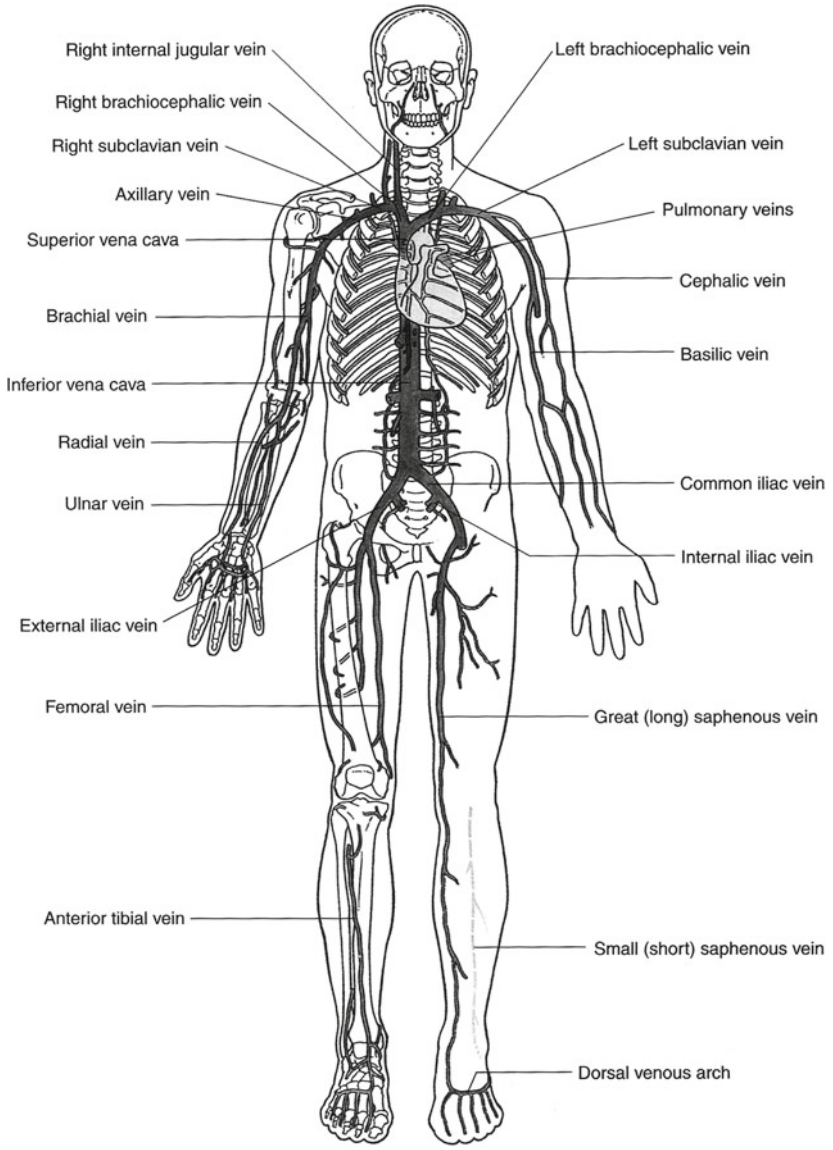


Fig. 8.4 Major veins in the body. Only the superficial veins are shown in the left limbs and only the deep veins are shown in the right limbs. Veins carry blood back to the heart in the systemic and pulmonary system. Many come in pairs, such as the right and left radial veins (From [53]. Used with permission)

In the arterial tree, the arteries bifurcate into two smaller ones, with the cross-sectional area of each daughter vessel being smaller than that of the parent vessel, but with their sum being larger [57]. Bifurcation is usually not symmetric, with

Table 8.2 Approximate quantification of individual vessels in the human circulatory system

Vessel	Diameter (mm)	Length (mm)	Wall thickness (μm)	Pressure (mmHg)
Aorta	25.0	400	1,500	100
Large arteries	6.5	200	1,000	100
Main artery branches	2.4	100	800	95
Terminal artery branches	1.2	10	125	90
Arterioles	0.1	2	20	60
Capillaries	0.008	1	1	30
Venules	0.15	2	2	20
Terminal venules	1.5	10	40	15
Main venous branches	5.0	100	500	15
Large veins	14.0	200	800	10
Vena cava[a]	30.0	400	1,200	5
Heart chambers	–	–	–	120

Using data from [21]
This is for a 30-yr-old male, with mass 70 kg and 5.4 L blood volume
[a]There are really two vena cavae

Table 8.3 Approximate quantification of total vessel systems in the human circulatory system

Vessel	Number	Total length (mm)	Total surface area (mm^2)	Total blood volume (mm^3)
Aorta	1	400	31,400	200,000
Large arteries	40	8,000	163,000	260,000
Main artery branches	500	50,000	377,000	220,000
Terminal artery branches	11,000	110,000	415,000	120,000
Arterioles	4,500,000	9,000,000	2,800,000	70,000
Capillaries	19,000,000,000	19,000,000,000	298,000,000	375,000
Venules	10,000,000	20,000,000	9,400,000	355,000
Terminal venules	11,000	110,000	518,000	190,000
Main venous branches	500	50,000	785,000	1,590,000
Large veins	40	8,000	352,000	1,290,000
Vena cava[a]	1[a]	400	37,700	280,000
Heart chambers	–			450,000
Total		∼19,000 km	312,900,000	5,400,000

Using data from [21]
This is for a 30-yr-old male, with mass 70 kg and 5.4 L blood volume
[a]There are really two vena cavae

the larger daughter vessel being a continuation of the parent vessel. In the ∼26 generations of bifurcations, the total cross-sectional area of the arteries increases roughly a hundred fold from the aorta to the small arterioles (0.003 cm diameter), with the first five-fold increase occurring from the aorta to large arteries. For small

Table 8.4 Length and beginning and ending radii of several of the large arteries shown in Fig. 8.3 (of different sections of the aorta and, for the last eight arteries, either the corresponding right or left artery)

Artery	Length (cm)	Beginning radius (cm)	Ending radius (cm)
Ascending aorta	4.0	1.525	1.420
Aortic arch	7.0	1.420	1.246
Thoracic aorta	16.0	1.246	0.924
Abdominal aorta	28.75	0.924	0.550
Subclavian	3.5	0.425	0.407
Brachial	39.75	0.407	0.250
Radial	22.0	0.175	0.175
Ulnar	22.25	0.175	0.175
Internal iliac	4.50	0.200	0.200
External illiac	5.75	0.400	0.370
Femoral	58.75	0.370	0.200
Deep femoral	11.25	0.200	0.200

Using data from [57]

arteries, the radii of the parent artery r_p and daughter vessels r_{d1} and r_{d2} have been related using three relations (which are not independent of each other): $r_p^\xi = r_{d1}^\xi + r_{d2}^\xi$ with $\xi = 2.76$, $\gamma = (r_{d2}/r_{d1})^2 = 0.41$, and $\eta = (r_{d1}^2 + r_{d2}^2)/r_p^2 = 1.16$ [57]. (Also see Problems 8.34–8.37.)

The circulatory system in the fetus is very similar to this but not identical [40]. The fetal right and left ventricles are the nearly the same size, and so the left heart has too much flow capacity because the fetus receives oxygenated blood from the umbilical cord and placenta, and does not need (and does not get) oxygen from exchange through the lungs. In fact, the fetal lungs are collapsed and flow resistance is so high that only 12% of all blood from the right ventricle flows through the lungs (as opposed to 100% after birth). The body accomplishes this by shunting pulmonary blood in two ways. The rest of the blood leaving the right ventricle is shunted from the pulmonary artery to the descending aorta (which branches into the systemic arteries) though the *ductus arteriosus* blood vessel. Also, there is a small opening in the septum between the left and right atria, the *foramen ovale*, and a small flap on the left side that allows blood to flow only from the right atrium to the left. (Flow through the heart valves is also unidirectional, but this is achieved in a different way, as is shown below in Fig. 8.8.) At birth the lungs expand when the newborn first breathes, so the lungs expand and their flow resistance decreases (and also the umbilical cord constricts). The foramen ovale closes right after birth, the ductus arteriosus normally closes two days after birth [41], and the left ventricle gradually grows, and faster than does the right ventricle, to accommodate increased demand.

The fetal systemic system includes the flow back through the umbilical cord and the placenta. A vein passes through the umbilical cord carrying blood that has passed

through the placenta to become oxygenated and nutrient rich, and splits in the fetus into one vessel that goes to the liver and a second one (the *ductus venosus*) that flows into the inferior vena cava. Two arteries branch from the internal iliac arteries, carrying deoxygenated, nutrient-depleted blood back through the umbilical cord and to the placenta. (As in the pulmonary system, blood in these fetal arteries (from the heart) is deoxygenated and that in these fetal veins (to the heart) is oxygenated.)

In the mother, flow from spiral arteries (∼0.6 L/min) from the uterine arteries enters the intervillus spaces of the placenta to enable exchange of oxygen, carbon dioxide, nutrients and waste by diffusion (but not direct flow) between maternal blood and the umbilical veins and arteries. Maternal blood in these spaces is exchanged 2–3 times/min, returning to the mother through the uterine veins. Because of this flow, the maternal blood flow volume and cardiac output increase during pregnancy.

8.1.2 Cardiac Cycle

There is a highly controlled timing cycle in well-functioning hearts, the cardiac cycle, which lasts a time τ (Fig. 8.5). In the first stage of *diastole* (die-as'-toe-lee), the veins fill up both the right and left atria, while the right and left ventricles are relaxed. In the second stage, the cardiac muscle (myocardium) of the right and left atria contract and pump blood through the atrioventricular valves, into the right and left ventricles, respectively, at the same time $t = 0$. (This is actually a gross simplification of ventricular filling, because ∼75% of this blood flows into the ventricles from the atria before atrial contraction.) In the first step of *systole* (sis'-toe-lee, which has the same cadence as Sicily), both ventricles contract (isovolumetrically) at the same time Δ. In the second stage, they eject blood through the respective semilunar valves: the right ventricle into the pulmonary arteries and the left ventricle into the aorta. The *systolic* (sis-stah'-lic) *blood pressure* occurs in this second stage of systole, while the *diastolic* (die-uh-stah'-lic) *pressure* is that during diastole. (One way to measure the flow of blood ejected by the left ventricle is *ballistocardiography*, which is described in Problem 8.51. Another method is Doppler ultrasonography echocardiography, which is used more often clinically; it is described in Problem 10.23.)

The right and left hearts must work at exactly the same time to keep the flow rate Q the same in both systems. There is a timing mechanism in place to do this and to set the contraction times 0 and Δ for one beat, followed by τ and $\tau + \Delta$ for the next, 2τ and $2\tau + \Delta$ for the next, and so on. There is a heart pacemaker at the sinoatrial or sinus node (see the conducting system in Fig. 8.6), which sends an electrical signal to the atrial cardiac muscle of both atria for simultaneous atrial contraction. This electrical signal then travels to the atrioventricular or AV node, is delayed there for a time Δ, and then the node sends a signal to the ventricular cardiac muscle of both ventricles for simultaneous ventricular contraction.

The electrocardiogram (EKG or ECG) is a measurement of these electrical signals, and their timing, as measured by probes on the body [17, 25, 45]. Figure 8.6 shows the EKG during one ∼1 s long heart beat (also see Fig. 12.28). The P wave is

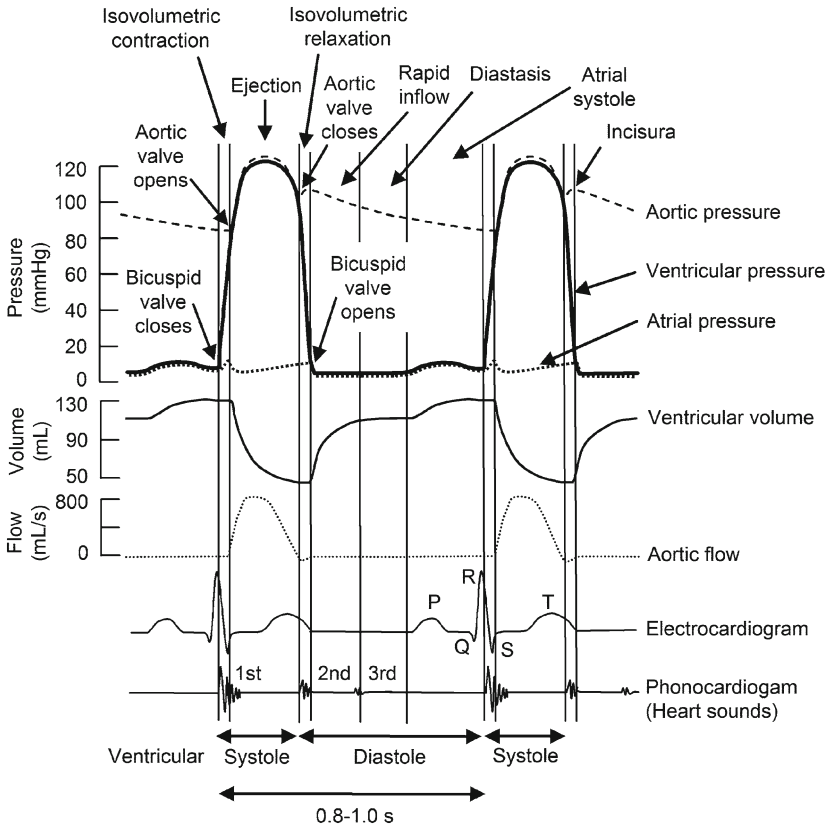


Fig. 8.5 The left ventricular and atrial pressures are plotted along with the left ventricular volume, aortic pressure and flow rate, the electrocardiogram and the phonocardiogram (which is the signal from heart sounds) in this Wiggers diagram. The opening and closing times of the aortic semilunar and bicuspid (mitral) valves are also shown (Based on [29, 63, 66])

due to atrial depolarization (which is atrial contraction). The QRS complex is due to ventricular depolarization (contraction). The T wave is due to ventricular repolarization (relaxation). The atrial repolarization (relaxation) signal is masked by the larger QRS complex. Depolarization and repolarization, the electrical properties of the heart, and the use of EKGs are discussed more in Chap. 12.

There are several cardiac arrhythmias (i.e., timing irregularities) of varying degrees of concern that affect this cardiac timing mechanism. In an *atrioventricular block*, there is injury to the atrioventricular (AV) fibers from the AV node to the ventricle or to the AV node itself (Fig. 8.6). In an incomplete AV block, the conduction time through the AV junction increases from the normal 0.16 to 0.25–0.50 s and there are dropped ventricular beats; these sometimes lead to 2:1, 3:2, or 3:1 rhythms of atrial to ventricular beats. In a complete AV block, a person may faint until ventricular beats develop (with 40/min, compared to 100 beats/min in the atria). There can

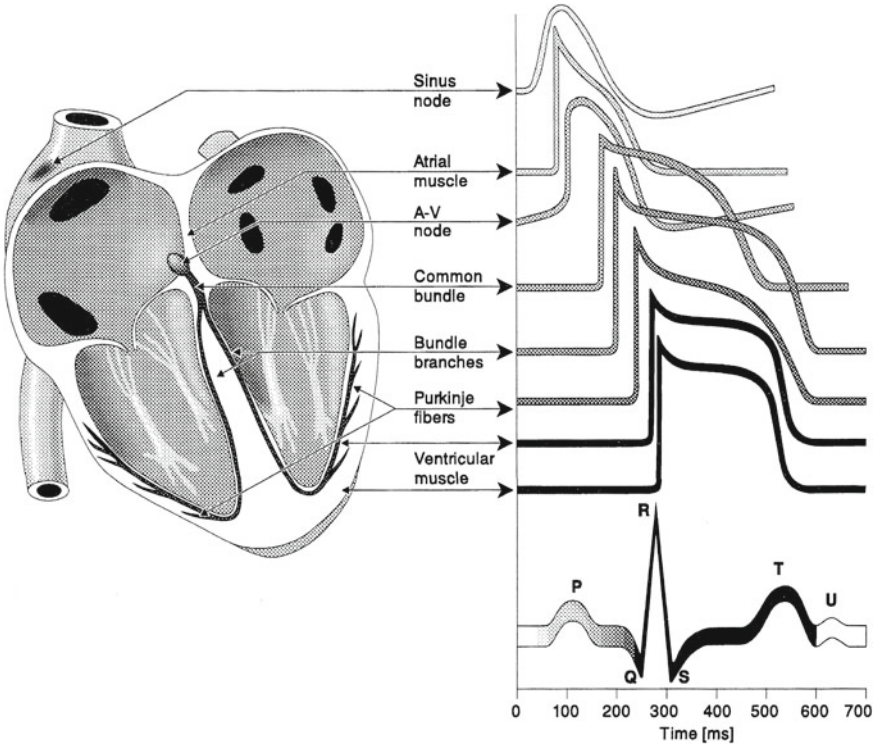


Fig. 8.6 Conducting system of heart with the sinoatrial/sinus and atriventricular/AV nodes, along with the electrical waveforms of the activity of each—including the EKG on the *bottom*. Also see Fig. 12.28 (From [49])

also be *premature contractions* of the atria or ventricles. In *paroxysmal tachycardia*, there are sudden increases in the heart rate, say from 95 to 150 beats/min in the atria or ventricles, which can cause serious ventricular (not atrial) damage. *Ventricular fibrillation* is the most serious arrhythmia and is fatal if not treated immediately. It can be caused by 60-cycle AC. There is uncoordinated muscle contraction of the ventricles, and so parts of them contract while other parts relax; this leads to little or no pumping of blood. Unconsciousness occurs in 4–5 s and the death of tissues begins in a few minutes. *Atrial fibrillation* involves similar uncoordinated muscle contraction, but it is less serious because most blood flows passively from the atria to ventricles. Blood flow decreases by only ~20–30%. Resuscitation after *cardiac arrest* can occur in many cases by cardiac pulmonary resuscitation (CPR) [29].

Cardiac muscle is similar to the skeletal muscle described in Chap. 5 (Fig. 5.1b). In particular, the basic building block is the sarcomere with its sliding actin and myosin filaments. There are some differences, however. At the resting muscle length, the maximum tension for skeletal muscle is ~20 N/cm^2 or more, while it is only ~7 N/cm^2 for cardiac muscle. Also, the resting, passive tension is fairly large (and

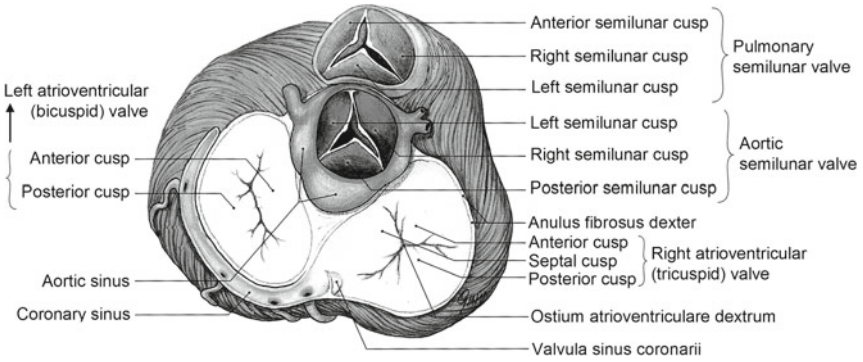


Fig. 8.7 Drawing of the four major heart valves, showing the cusps (flaps) (From [68])

relatively larger) at the length of peak tension in cardiac muscle (as is depicted in Fig. 5.21a). Both of these differences can be attributed in part to the greater fraction of noncontractile tissue in heart muscle, which contains collagen and other fibrotic tissue. The first difference also arises from the nonparallel nature of cardiac muscle fibers. Another difference between skeletal and cardiac muscle is that it is usually not possible to tetanize cardiac muscle. The twitches merge only partially at very high stimulation frequency.

8.1.3 Valves

There are four major valves in the heart (Fig. 8.7). The right atrioventricular valve controls flow between the right atrium and right ventricle. It has three flaps (or cusps) and is therefore also called the tricuspid valve. The pulmonary semilunar valve controls blood flow from the right ventricle to the left and right pulmonary arteries. The left atrioventricular valve controls flow from the left atrium to the left ventricle. It has two flaps and is therefore also called the bicuspid valve. Another name for this valve is the mitral valve, because it looks like a miter. The aortic semilunar valve controls flow from the left ventricle to the aorta.

These four valves share some common traits. They are one-way valves (Fig. 8.8) that allow blood flow in the described direction under some conditions, but never in the opposite direction (unless they are defective). We can imagine a flapped unidirectional valve that will not allow any back flow. With this type of valve we can see how the flaps will open, allowing this unidirectional flow, when the pressure in front of the valve exceeds that on the other side. In this way, the valve is closed until the pressure in the chamber increases due to contraction to a value greater than that after the valve. However, such a valve cannot withstand very much back pressure. Backward opening of the atrioventricular valves is also prevented by the papillary muscles on the ventricular side that contract when the valve is closed, making the

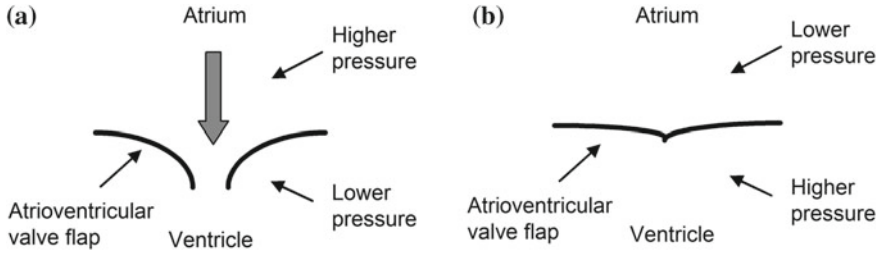


Fig. 8.8 Schematic of the unidirectional flow of an atrioventricular heart valve. **a** The pressure in the atrium exceeds that in the corresponding ventricle and the valve opens, as shown, with a jet of blood rushing in. Toward the end of diastole, the jet is broken. The deceleration of the blood creates a pressure, which tends to close the valve. **b** The valve is normally closed (Based on [66])

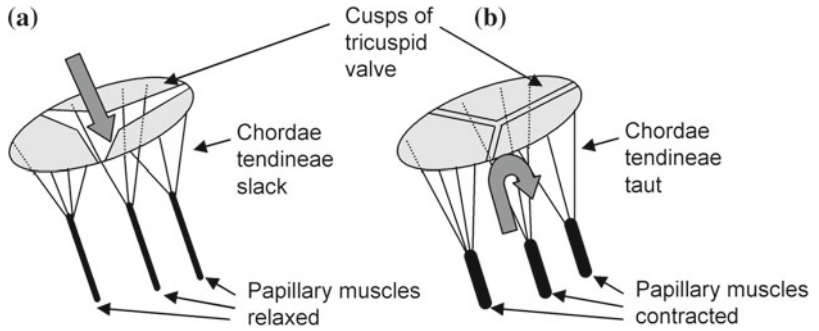


Fig. 8.9 The papillary muscles on the ventricular side of the atrioventricular valves contract when the valves are closed, making the chordae tendineae taut and preventing backward flow, as in (**b**). The muscles and the chordae tendineae are relaxed when the valves are open and forward blood flow is allowed, as in (**a**) (Based on [66])

chordae tendineae that are attached to the flaps taut (Fig. 8.9). This prevents the flaps from bending backward, so there is no backward flow of blood.

Let us consider the cycle for the aortic semilunar valve. During ventricular relaxation the pressure in the left ventricle is ~0 mmHg. In the aorta the pressure is ~120 mmHg during systole and then decreases to ~80 mmHg during diastole, just before ventricular contraction. The valve is still closed. During ventricular contraction the pressure in the LV increases to 80 mmHg, continuing up to ~120 mmHg. Because the pressure in the aorta is ~80 mmHg, the aortic semilunar valve opens once the pressure in the LV exceeds 80 mmHg and then remains open. During this flow, the pressure in the LV and aorta become equal, ~120 mmHg, and then the valve closes as the flow cycle comes to an end.

The measurement of systolic and diastolic pressure by listening to Korotkoff sounds is described in Chap. 7. This method is the standard way of determining blood pressure even though it routinely underestimates systolic pressure by 5–20 mmHg and overestimates diastolic pressure by 12–20 mmHg [15]. (It has the precision needed for a diagnostic, even though it is not very accurate.)

Cardiac valve openings and closings and the flow of blood in the heart create sounds that can be heard with a stethoscope, and are described in Chap. 10. (Also see Fig. 8.5.) Echocardiography is the use of ultrasound to diagnose heart disorders and blood flows. It and related methods are also described briefly in Chap. 10 (and in Problems 10.22 and 10.23).

8.2 Physics of the Circulation System

We now examine the circulation system in more detail. First, we will examine how the blood pressure varies with distance along the arteries and veins, including within the capillaries. We then investigate the consequences of nonuniformities in arteries, such as clogged arteries and aneurysms—in the context of the strength of the artery walls. We next calculate the work done by the heart, to see how this contributes to the metabolic needs of the body. In the last section of this chapter we will develop a model of the entire circulatory system and the heart.

8.2.1 Properties of Blood

Blood is a non-Newtonian fluid, in part because of its complex, inhomogeneous composition [40]. The blood solution consists of plasma, red blood cells (erythrocytes; 5 million/mm^3; 45% of total blood volume), white blood cells (leukocytes; 0.3%), and platelets (0.15%). (The red blood cell volume fraction is called the hematocrit.) The red blood cells are biconcave disks that are toroidal in shape with the center partially filled in, and have a diameter of 7.5 μm and maximum thickness of 2 μm. Their diameter is about the same as the inner diameter of capillaries, but they can deform and flow in even smaller tubes. White blood cells are spherical, with a diameter of 7 μm, while the platelets are much smaller. The blood plasma is 90% water and behaves like a Newtonian fluid with a viscosity of 0.0012 Pa-s. The blood rheology is greatly altered by the red blood cells, and not much by the white blood cells or platelets because they comprise very small fractions of the blood volume. (Rheology is the study of the deformation and flow of materials, particularly unusual materials.) The blood viscosity increases with the hematocrit, as seen in Fig. 8.10.

The effective viscosity of blood decreases as the shear rate increases (Fig. 8.11). For very slow shear rates, this viscosity is more than 100× that of water, while at the high shear rates characteristic of flow in larger vessels it is about 4× that of water, with a value of 0.004–0.005 Pa-s.

The viscosity of some fluids changes even while the strain rate is constant. Blood is a thixotropic fluid, for which the shear stress decreases while the strain rate is constant. Still, for our purposes it will be adequate to treat blood as a Newtonian fluid, even though the velocity flow profile is not the ideal parabolic form for a Newtonian fluid (Fig. 7.13).

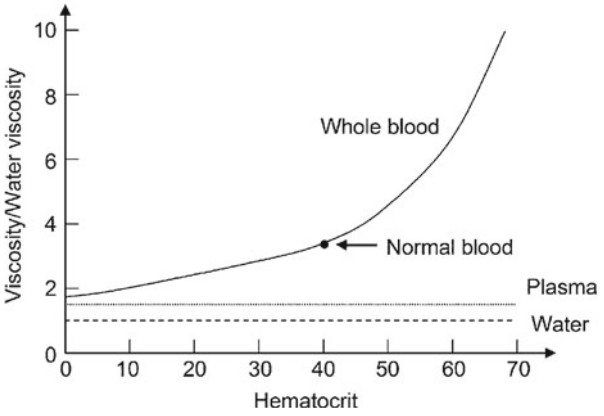


Fig. 8.10 Blood viscosity versus hematocrit (Based on [29])

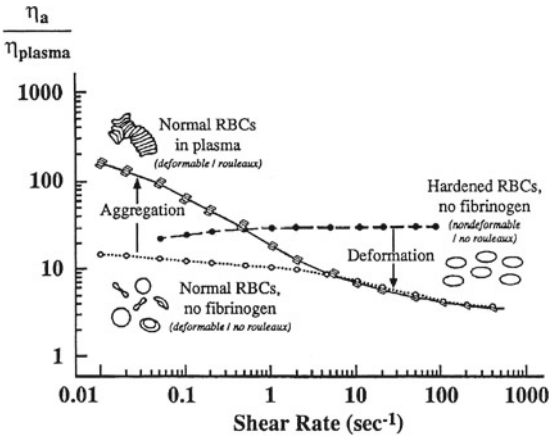


Fig. 8.11 Blood viscosity versus shear rate for a hematocrit of 45%, at 310 K. (From [21]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com, based on [13])

8.2.2 Blood Pressure and Flow in Vessels

Structure of Blood Vessels

Arteries contain inner layers that are 1–2 endothelial (lining) cells thick—along with elastic issue (composed of collagen and elastic proteins). This innermost region surrounding the opening—the lumen—is known as the tunica intima. Next in the wall comes a layer of circular, smooth muscle fibers interspersed with elastic tissue (the tunica media) and finally connective tissue (the tunica adventitia) (Fig. 8.12, also see Fig. 8.43 below). The walls of veins have a thickness w that is typically $\sim d/10$, where d is the lumen diameter; they are thinner than the walls in arteries of corresponding diameters, for which $w \sim d/5$. The aorta and other large arteries contain much elastic tissue and stretch during systole and recoil during diastole. The walls of the arterioles contain less elastic tissue and more smooth muscles, and

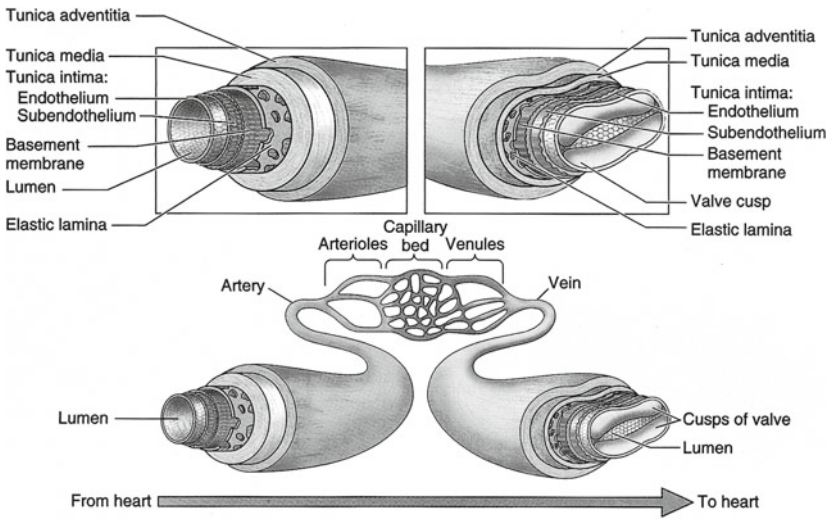


Fig. 8.12 Schematic of the walls of arteries and veins (From [53]. Used with permission)

stretch relatively little. The walls of the capillaries are composed of a single layer of endothelial cells. The diameters of the capillaries are so small that red blood cells can barely pass through them.

Approximately 70% of the walls of arteries and veins is composed of water, which is not elastic, except in how it withstands compression. The other 30% consists of the dry mass: elastin, collagen, and smooth muscle fibers, each having different materials properties. As discussed in Chap. 4, elastin is rubber-like and has a Young's modulus of $\sim 3 \times 10^5$ Pa; it can be stretched to twice its relaxed length. Collagen is much stiffer, having a Young's modulus of $\sim 1 \times 10^8$ Pa. Elastin has an ultimate tensile stress (UTS) less than 5% of that of collagen. Smooth muscle has a Young's modulus more like that of elastin, with $Y \sim 1 \times 10^5$ Pa when relaxed and $\sim 2 \times 10^6$ Pa when active. About half the dry mass in vessels is elastin and collagen, with more elastin than collagen in the aorta ($\sim 1.5\times$) and relatively less elastin in other arteries ($\sim 0.5\times$) and veins ($\sim 0.3\times$). Veins contain less elastin than arteries. The fraction of smooth muscle in the dry mass averages to $\sim$50%, and is $\sim$25% in the aorta, and increases to $\sim$60% more peripherally in the arteries and arterioles. The mechanical properties of these vessels (Chap. 4) also depend on the tissue to which they are attached. This is particularly significant for capillaries, because the vessel walls are essentially a single layer of endothelial cells.

Blood Pressure

Blood pressure is needed to push blood flow. Figure 8.13 is a schematic of the mean arterial and venous blood pressure at different positions in the circulation cycle, for a person lying horizontally. The oscillations in blood pressure along the major arteries in systemic circulation reflect the oscillating pressure of this blood as it leaves

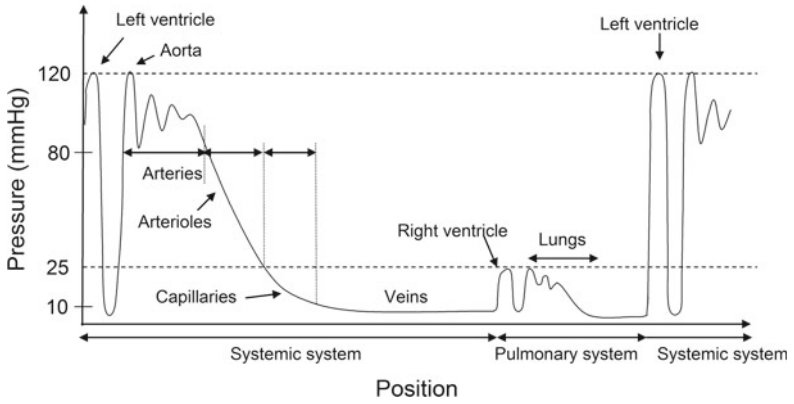


Fig. 8.13 Blood pressure along the circulatory system for a person lying horizontally (Based on [9])

the aorta, at pressures between ~80 mmHg (P_{diastole}, at diastole) and ~120 mmHg (P_{systole}, at systole) (Fig. 8.13). Because systole lasts for about 1/3 of the cycle and diastole for about 2/3, the *mean blood pressure* is a weighted sum,

$$P_{\text{mean}} = \frac{P_{\text{systole}} + 2P_{\text{diastole}}}{3}, \tag{8.1}$$

or (1/3)120 mmHg + (2/3)80 mmHg ~ 94 mmHg in this example. This difference in pressure of 40 mmHg between systole and diastole is the *arterial pulse pressure* P_{pulse}. We will see that blood flows at a speed of ~20 cm/s in these systemic arteries, so with a heart rate of about 1 Hz = 1 cycle/s it is reasonable that there are quasiperiodic pressure variations every 20 cm or so. Much of the pressure drop in the arterial system is in the arterioles (small arteries) and the capillaries. We will see that this can be attributed to viscous flow. There is very low pressure in the veins. It is too little pressure to pump the blood back to the heart—even with the large diameters of the veins and consequently low resistance to flow (7.24). There is a peristaltic pumping mechanism by muscles surrounding the large veins that assists the return of venous blood to the heart, with one-way valves to prevent backflow (Fig. 8.14). (Similarly, blood flow in the capillaries is usually not continuous, but is turned on and off every few seconds or minutes, due to sphincter muscles that can contract the feeding arterioles (*vasomotion*).) The pulmonary system mirrors this systemic circulation, except the pressures are all lower. Figures 8.13, 8.15, and 8.16 show the blood pressure and flow speed at different points in the arterial tree.

The pumping cycle sets up a pressure pulse wave in addition to the hydrostatic pressure variation. This pulse wave speed is independent of the speed of blood flow—and is faster than the blood flow speed—and is 4 m/s in the aorta, 8 m/s in the large arteries, and 16 m/s in the small arteries of young adults.

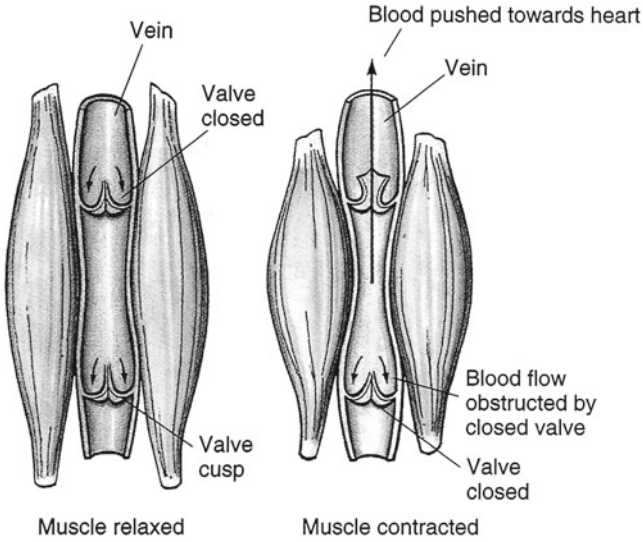


Fig. 8.14 Musculovenous pump of veins, with outward expansion of the bellies of contracting muscles pumping the blood back to the heart against gravity and distal valves closing to prevent backflow (From [53]. Used with permission)

When you stand upright vertically (Fig. 8.17) there is an additional pressure ρgh, where h is the height relative to the heart. This is approximately the height in the upper arm where blood pressure measurements are made. For $\rho = 1.06\,\mathrm{g/cm^3} = 1{,}060\,\mathrm{kg/m^3}$, $g = 9.8\,\mathrm{m/s^2}$ and $h = 1\,\mathrm{m}$, this pressure is $10{,}400\,\mathrm{N/m^2} = 10{,}400\,\mathrm{Pa} = 79\,\mathrm{mmHg}$ (with $1\,\mathrm{MPa} = 7{,}600\,\mathrm{mmHg}$). At any given height, the driving pressure difference from the arteries to the veins is unchanged. Also, this pressure change is not important when considering pressure changes between the inside and outside of a vessel because ρgh is added both inside and outside the vessel.

Still, this effect of gravity can be significant. The blood pressure at the aorta has to be high enough to pump the blood to the top of your brain. This distance is about $h = 40\,\mathrm{cm}$, so the pressure drop is about 30 mmHg (compared to the diastolic pressure of ~80 mmHg). Problems 8.1 and 8.2 explore what happens to cranial blood circulation in humans in rapidly climbing jets and on more massive planets with higher g, and also in giraffes. One manifestation of this effect of gravity is potential fainting when you stand. When you stand up, the volume of blood in the leg veins increases and the pressure in the veins pumping blood back to the heart decreases. This can decrease the cardiac output and the flow of blood to the brain. This rarely happens because there is a reflex constriction of the veins in the legs (due to a contraction of the skeletal muscle surrounding the veins, Fig. 8.14) that limits the blood pool and an arteriolar constriction that increases flow resistance and lessens the decrease in arterial blood pressure. The effect of gravity on humans—who normally stand upright—also explains why standing on your head for long periods is

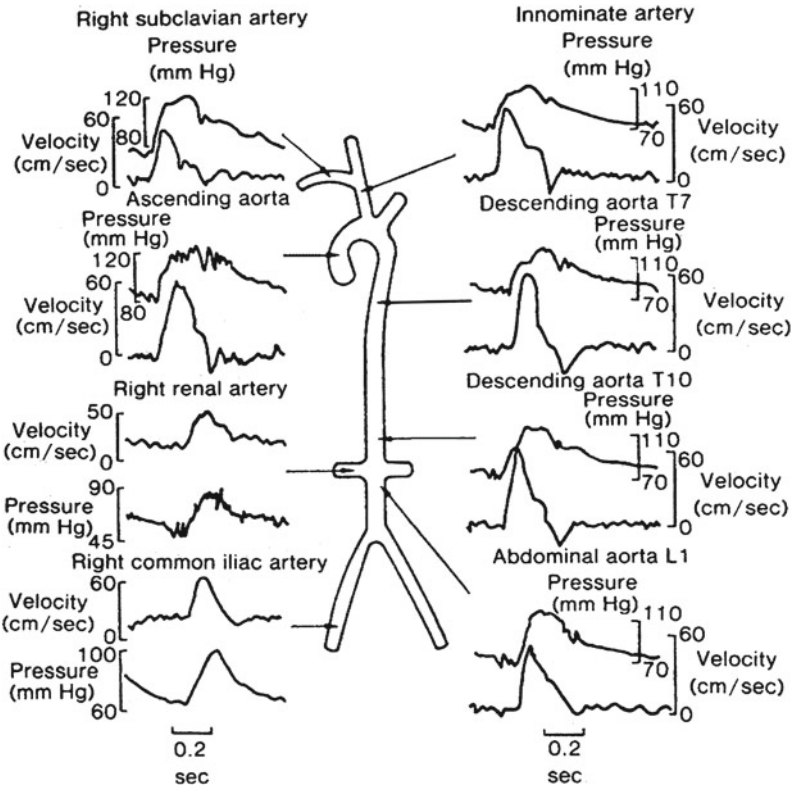


Fig. 8.15 Simultaneous pressure and flow velocity at different points in the human arterial tree for a person lying horizontally. All data were taken from one patient except for the right renal artery and the right common iliac artery (From [31]. Adapted from [52])

not advisable (Problem 8.3). The veins in your head are not designed to pump blood back to the heart (as are those in the lower body). Also, your feet would stop getting blood. It also explains why varicose veins are worse when you stand upright, because blood then needs to be pumped up.

Body control of blood pressure is briefly described in Chap. 13.

Measuring Flow in Blood Vessels

The mechanical properties of blood vessels can be measured under two types of conditions. (1) The length of a given vessel can be kept constant, while its diameter is measured as a function of the *distending pressure*. This leads to a tensile stress on the wall, directed around the circumference, which is called the circumferential or hoop stress. (2) The diameter of a vessel can be kept constant, while its length is measured as it is stretched longitudinally. Examples of both are shown in Fig. 8.18.

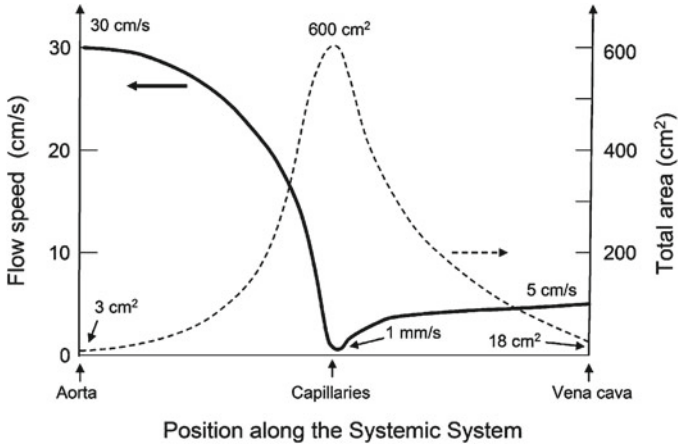


Fig. 8.16 Flow speed (*solid curve*) and total area (*dashed curve*) in the systemic circulation system (Based on [9])

The flow of blood in arteries is affected by changes in the heart beat rate and stroke volume (which is the volume pumped per beat), and also by changes in the arteries themselves that control their diameters by chemical and neural mechanisms.

Modeling Flow in Blood Vessels

Figure 8.19 is a schematic describing the flow in a vessel of length L, with a volumetric flow rate Q_1 in and Q_2 out. In steady state $Q_1 = Q_2 = Q$, which is averaged over a heart beat. At the beginning of the vessel the pressure is P_1 and at the end it is P_2. The pressure outside the vessel is P_{ext}, which can be taken to be 0 when using a gauge pressure. The volume of the vessel is V. The relation between these various parameters depends on the properties of the vessels.

There are two physical attributes of blood vessels in steady state. (a) They have a *resistance* to flow, and so they need a pressure difference along the length of the vessel to drive the blood flow. (b) They have a *compliance* in response to a distending pressure. This is much like a balloon expanding when the pressure inside increases much above that outside.

One special case is a rigid vessel with constant volume V, which is called a *resistance vessel* (Fig. 8.20). Equation (7.25) applies to this vessel, so

$$P_1 - P_2 = R_{\text{flow}} Q \tag{8.2}$$

or

$$Q = \frac{1}{R_{\text{flow}}}(P_1 - P_2), \tag{8.3}$$

where the vascular resistance is $R_{\text{flow}} = 8\eta L/\pi r^4$ for a tube with radius r. The former equation has the same form as Ohm's Law $V_{\text{elect},1} - V_{\text{elect},2} = R_{\text{elect}} I$, which

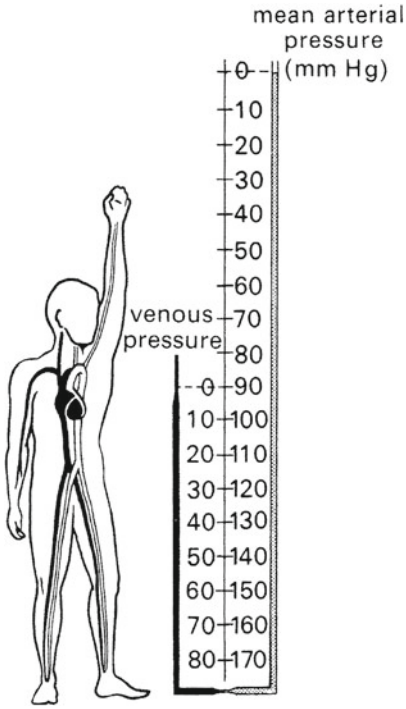


Fig. 8.17 Mean arterial and venous (gauge) pressures for a vertical person (From [10], after [61]. Used with permission of Oxford University Press)

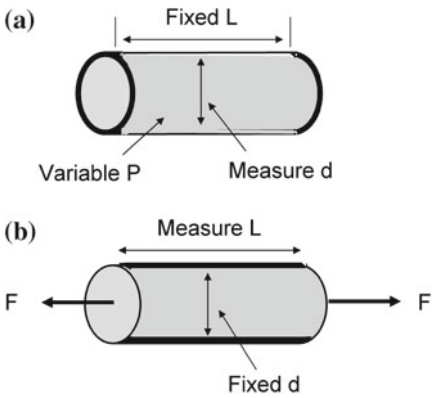


Fig. 8.18 Measuring the mechanical properties of blood vessels by fixing the vessel **a** length (which is called *inflation*) or **b** diameter (which is called *extension*). Some investigators do combined loading, with combinations of inflation, extension, and torsion

relates the drop in voltage, V_{elect}, when a current of charges I traverses a structure with electrical resistance R_{elect}.

A second special case is an elastic vessel that has no noticeable resistance, which is called a *compliance vessel*. There is no pressure drop, so $P_1 = P_2 = P$. One model (Fig. 8.21) of the properties of such a vessel is

$$V(P) = C_{\text{flow}}(P - P_{\text{ext}}) = C_{\text{flow}}P, \tag{8.4}$$

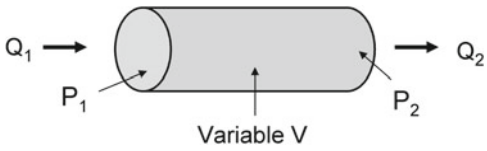


Fig. 8.19 Blood flow: general vessel

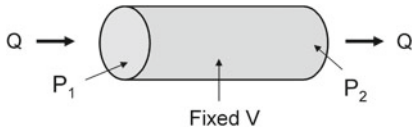


Fig. 8.20 Blood flow: ideal resistance vessel

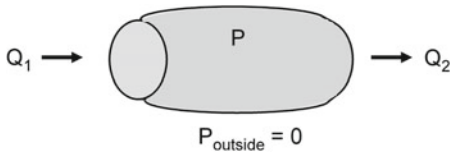


Fig. 8.21 Blood flow: ideal compliance vessel

where C_{flow} is the compliance and P_{ext} is taken to be 0. Because the vessel usually has a volume with no pressure, called the dead volume V_{d}, a better relation is

$$V(P) = V_{\text{d}} + C_{\text{flow}} P. \tag{8.5}$$

Note that a property of the resistance vessel, namely the flow rate, is affected by the pressure drop along (and inside the vessel), while a property of the compliance vessel, its volume, is affected by the pressure difference between the inside and outside of the vessel. We can also describe compliance by changes in the area A

$$A(P) = A_{\text{d}} + c_{\text{flow}} P, \tag{8.6}$$

where A_{d} is the area with no pressure difference and c_{flow} is the compliance per unit length, or by changes in the radius r

$$r(P) = r_{\text{d}} + \frac{C'_{\text{flow}}}{2} P, \tag{8.7}$$

where r_{d} is the radius with no pressure difference. Because $V = AL = \pi r^2 L$ and $V_{\text{d}} = A_{\text{d}} L = \pi r_{\text{d}}^2 L$ for a vessel of radius r and length L, these formulations can be interrelated, with $c_{\text{flow}} = C_{\text{flow}}/L$ and $C'_{\text{flow}} = C_{\text{flow}}/\pi r L = c_{\text{flow}}/\pi r$ (the latter set being true for small changes in r or V).

Real vessels have some attributes of both types of vessels. Still, the aorta, large arteries and large veins are much like compliance vessels. We will see that the pressure drops along them are relatively small. Arterioles, capillaries, and venules act like resistance vessels. We will see that they, and in particular the arterioles, are the main

sites of the pressure drop, and this is the reason why the heart needs to pump blood to such high pressures.

Pressure Drops in Arteries and Resistive Vessels

We will use Poiseuille's Law $\Delta P = \left(8\eta L/\pi r^4\right) Q$ (7.25) to estimate the pressure drop $\Delta P = P_1 - P_2$ across the aorta, large arteries, arterioles, and capillaries, and compare these results to the plot in Fig. 8.13 (and also see Fig. 8.15). First we calculate the resistance $R_{\text{flow},0} = 8\eta L_0/\pi r_0^4$ for a standard radius $r_0 = 1\,\text{cm}$ and standard length $L_0 = 1\,\text{cm}$ and scale the results for each specific case. With the viscosity $\eta = 4.0 \times 10^{-3}\,\text{Pa-s} = 4.0 \times 10^{-3}\,(\text{N-s/m}^2) = 4.0 \times 10^{-2}$ poise for whole blood at 37 °C

$$R_{\text{flow},0} = \frac{8 \times (4.0 \times 10^{-3}\,\text{N-s/m}^2)(1\,\text{cm})}{\pi(1\,\text{cm})^4} = \frac{1.02 \times 10^{-2}\,\text{N/m}^2}{\text{cm}^3/\text{s}} \tag{8.8}$$

$$= \frac{1.02 \times 10^{-8}\,\text{N/mm}^2}{\text{cm}^3/\text{s}} = \frac{7.7 \times 10^{-5}\,\text{mmHg}}{\text{cm}^3/\text{s}}, \tag{8.9}$$

where we have used $0.1\,\text{N/mm}^2 = 1\,\text{atm.} = 760\,\text{mmHg}$. The units in (8.9) are mmHg-s/cm^3. This is the resistance when the pressure difference is 1 mmHg and the flow rate is 1 mL/s, and is also known as a PRU, a peripheral resistance unit. This unit is commonly used in physiology.

The resistance of a vessel of an arbitrary length and radius is

$$R_{\text{flow}} = R_{\text{flow},0}\frac{L/L_0}{(r/r_0)^4} = R_{\text{flow},0}\frac{L(\text{in cm})}{r(\text{in cm})^4} \tag{8.10}$$

and so

$$\Delta P = R_{\text{flow}} Q = 7.7 \times 10^{-5}\,\text{mmHg}\,\frac{L(\text{in cm})}{r(\text{in cm})^4}\,Q\ (\text{in cm}^3/\text{s}). \tag{8.11}$$

The total flow rate, Q_t, from the aorta, enters the large arteries, and the whole flow from the large arteries enters the arterioles, and finally this whole flow enters the capillaries. In each level of flow we will model the arteries as n parallel vessels of roughly equivalent length and radius carrying roughly the same flow (Fig. 8.22), where n increases for each successive level of flow. So in a given level of flow with n vessels, the flow rate in each vessel is $\sim Q_\text{t}/n$.

With $Q_\text{t} = 80\,\text{cm}^3/\text{s}$ and $Q = Q_\text{t}/n$, we get

$$\Delta P = 7.7 \times 10^{-5}\,\text{mmHg}\,\frac{L(\text{in cm})}{r(\text{in cm})^4}\,\frac{80}{n} \tag{8.12}$$

$$= \frac{0.0062\,\text{mmHg}}{n}\,\frac{L(\text{in cm})}{r(\text{in cm})^4}. \tag{8.13}$$

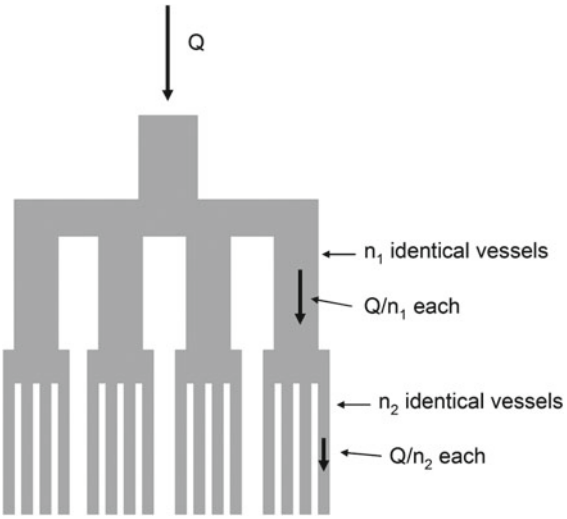


Fig. 8.22 Schematic of blood flow in idealized branching vessels

This is the pressure drop across any vessel in a given level of flow and, because they are in parallel, it is the pressure drop across the entire given level of arterial flow. We now determine this for the various levels of arteries.

Aorta. There is one aorta ($n = 1$) with $r \sim 1.25\,\text{cm}$ and $L \sim 10\,\text{cm}$, and so ΔP across the aorta is 0.025 mmHg, which is insignificant.

Largest arteries. There are about 200 large arteries with $r \sim 0.2\,\text{cm}$ and $L \sim 75\,\text{cm}$, and so ΔP is 1.4 mmHg, which is pretty insignificant.

Smallest arteries and arterioles. There are about 5×10^5 arterioles with $r \sim 30\,\mu\text{m}$ and $L \sim 0.6\,\text{cm} = 6\,\text{mm}$, and so ΔP is 91 mmHg, which is very significant.

Capillaries. There are about 10^{10} capillaries with $r \sim 3.5\,\mu\text{m}$ and $L \sim 0.2\,\text{cm} = 2\,\text{mm}$, and so ΔP is 8.2 mmHg, which is fairly significant.

We could have just plugged the parameters for each vessel directly into (7.25), without calculating $R_{\text{flow},0}$, but our scaling approach does provide some new insight.

These estimates agree with what we would expect from Fig. 8.13. Also, arterioles and capillaries are seen to be well modeled as resistance vessels. The aorta and large arteries have very small pressure drops across them, and behave more like compliance vessels. Veins have larger radii than the corresponding arteries, and consequently much lower resistances and pressure drops across them.

Along any vessel there is obviously a linear pressure drop with distance x along the vessel. This is seen from Poiseuille's Law, (7.25), $(\Delta P/L) = \left(8\eta/\pi r^4\right) Q$ or, expressing this in the differential limit and recognizing that this change in pressure is negative,

$$\frac{\mathrm{d}P}{\mathrm{d}x} = -\frac{8\eta}{\pi r^4} Q = -\frac{8\pi\eta}{A^2} Q, \tag{8.14}$$

where this has also been given in terms of the area A. This represents a "distributed" or "transmission-line" view of blood flow, in which flow is analyzed per unit length along the vessel, whereas in (8.2) and (8.11) flow was analyzed with the vessel as a "lumped" parameter (see Appendix D).

We can study the pressure drop in clogged arteries. There could be a larger pressure drop for the same Q or a smaller Q for the same pressure drop if r decreases, as occurs with clogged arteries, or with fewer vessels. This can stimulate an increase in blood pressure to maintain the flow rate or lead to a reduction in flow at a given inlet pressure, which is what actually happens in coronary artery disease.

How can we "optimize" the design of resistive vessels and how such vessels bifurcate and otherwise branch into smaller vessels (and what does it really mean to optimize the design)? See Problems 8.34–8.37.

Radial Profile of Blood Flow

Blood flow in vessels is not uniform. We have implicitly been assuming that the blood flow is parabolic ((7.40) and Fig. 7.11) because we are modeling blood as a Newtonian fluid. However, the parabolic profile of blood flow speed in a resistive vessel from (7.40) is not quite accurate. Because whole blood is not Newtonian, the steady-state profile is not parabolic, as is seen in Fig. 7.13. Moreover, we have assumed that flow is always steady state (ignoring the pulsatile nature of blood flow); however, it enters the steady state regime only a certain distance after a bifurcation (7.54)–(7.56). This approach to steady-state flow is depicted for a specific initial condition in Fig. 7.11.

Properties of a Compliance Vessel

In Chap. 7 we showed that (8.4) and (8.5) are reasonable models of an elastic compliance vessel. We can determine the compliance C_{flow} of such vessels and analyze how blood pressure can increase the volume of vessels.

For large arteries, the Young's modulus Y, radius r and wall thickness w are interrelated by $Yw/r = k_1 \exp(-k_2 r) + k_3$, with $k_1 = 2.0\,\text{MPa}$, $k_2 = 22.53/\text{cm}$, and $k_3 = 86.5$ kPa [57]. Using (7.46), the compliance of a vessel per unit volume is $2r/wY$, so this fit shows that the compliance per unit volume is the same for large arteries with larger radius ($>0.2\,\text{cm}$), but decreases as radius decreases for smaller radius arteries in this range. The deviations between this fit and measured values are due to variations of the arterial compliances within the body.

How large is vessel expansion each pulse? The pressure in the aorta and large arteries is $120\,\text{mmHg} = 0.0158\,\text{MPa}$ during systole. The value of Y for such vessels is about 1 MPa (Table 4.2) and so $P/Y = 0.0158 \sim 1.6\%$. The thickness of arterial walls is typically 1/5 of the radius, so $r_0/w = 5$. Using (7.43), this means that the fractional increase in radius of these vessels due to this internal pressure is $\sim$8% and the fractional increase in volume (and area) is $\sim$16%—both sizeable fractions. (We did not need to explicitly determine the compliance to calculate this.) Also, this predicts that the radius changes by $\sim$3% during each heart beat during the changes between systolic (120 mmHg) and diastolic (80 mmHg) pressure. Veins are also compliance vessels.

Pulmonary arteries in the lungs are more compliant than those in the systemic system. Veins have compliances about ∼24× those of arteries because they have larger radii and more compliant walls. Because of this and their larger radii, they can store much blood, especially when the venous pressure is increased only a little. When the pressure in arteries and arterioles becomes very low, and below that of the fluid outside them, they can collapse and prevent flow of the blood plasma for 5–10 mmHg and for red blood cells below ∼20 mmHg [40].

Flow with Resistance and Compliance

If a vessel is resistive and compliant [40, 50], the change in pressure with distance is

$$\frac{\mathrm{d}P}{\mathrm{d}x} = \frac{\mathrm{d}P}{\mathrm{d}A}\frac{\mathrm{d}A}{\mathrm{d}x} = \frac{1}{c_{\mathrm{flow}}}\frac{\mathrm{d}A}{\mathrm{d}x}, \tag{8.15}$$

using (8.6). Setting this equal to $\mathrm{d}P/\mathrm{d}x$ from (8.14) and bringing the A terms to the left and the x terms to the right, gives

$$A^2\mathrm{d}A = -8\pi c_{\mathrm{flow}}\eta Q\,\mathrm{d}x, \tag{8.16}$$

where the flow rate Q is a constant. After integrating over a vessel length from $x = 0$ to $x = L$, we get

$$(A(x=0))^3 - (A(x=L))^3 = 24\pi c_{\mathrm{flow}}\eta QL \tag{8.17}$$

and after using (8.6)

$$(A_{\mathrm{d}} + c_{\mathrm{flow}}P(x=0))^3 - (A_{\mathrm{d}} + c_{\mathrm{flow}}P(x=L))^3 = 24\pi c_{\mathrm{flow}}\eta QL. \tag{8.18}$$

(See Appendix C.)

Both terms on the left side can be expanded to four terms. The first terms are both A_{d}^3, which cancel, and for relatively small compliance ($c_{\mathrm{flow}}P/A_{\mathrm{d}} \ll 1$) only the next two of the remaining three terms in each need to be retained, giving

$$Q = \frac{A_{\mathrm{d}}^2}{8\pi\eta L}(P(0) - P(L))(1 + \frac{c_{\mathrm{flow}}}{A_{\mathrm{d}}}(P(0) + P(L)). \tag{8.19}$$

This is Poiseuille's Law (7.24) with a correction for compliance. So, for a rigid wall vessel ($c_{\mathrm{flow}} = 0$) the flow rate Q is linear with the pressure drop, but when compliance is included, the variation with pressure drop is between linear and quadratic. This relation says that for a given pressure drop, the flow rate is increased due to the compliant nature of the vessel and that a smaller pressure drop is needed to achieve a given flow rate. (The smaller pressure drop in veins than in arteries is explained both by their larger radii and their larger compliances.) The next term in

the expansion within the brackets is $(c_{\text{flow}}^2/3A_{\text{d}}^2)((P(0))^2 + P(0)P(L) + (P(L))^2)$ (see Problem 8.26).

Equation (8.19) can also be expressed in terms of the vessel radius

$$Q = \frac{\pi r_{\text{d}}^4}{8\eta L}\,(P(0) - P(L))\left(1 + \frac{C'_{\text{flow}}}{r_{\text{d}}}\,(P(0) + P(L))\right), \tag{8.20}$$

which can also be obtained directly by integrating Poiseuille's Law expressed in terms of the radius (8.14) and using (8.7) (see Problem 8.27). The next term in this expression obtained by integrating Poiseuille's Law expressed in terms of the area, as given above, and that by integrating the form using the radius (Problem 8.28) are different. At first this may be surprising, but there is a subtle difference. In the former approach c_{flow} is assumed to be constant (as is C_{flow}) and in the latter C'_{flow} is assumed to be constant. Both compliances cannot be constant because of how they are defined (8.4)–(8.7), but they are approximately equal for small vessel expansions. This demonstrates how the details of a model can be very important.

The resistive pressure drop in a vessel also affects the volume of a compliant vessel [40]. With $\mathrm{d}V = A(x)\mathrm{d}x = A(P)(\mathrm{d}x/\mathrm{d}P)\mathrm{d}P = -(1/8\pi\eta Q)A(P)^3\mathrm{d}P$ (using (8.14)) and with $\mathrm{d}P = (\mathrm{d}P/\mathrm{d}A)\mathrm{d}A = (1/c_{\text{flow}})\mathrm{d}A$ (using (8.6)), leads to $\mathrm{d}V = -(1/8\pi c_{\text{flow}}\eta Q)A^3\mathrm{d}A$. Integrating from 0 to L gives

$$V = (1/32\pi c_{\text{flow}}\eta Q)(A(P(0))^4 - A(P(L))^4) \tag{8.21}$$

or

$$V = (1/32\pi c_{\text{flow}}\eta Q)((A_{\text{d}} + c_{\text{flow}}P(0))^4 - (A_{\text{d}} + c_{\text{flow}}P(L))^4). \tag{8.22}$$

Using (8.18) and $V_{\text{d}} = A_{\text{d}}L$ gives

$$\frac{V}{V_{\text{d}}} = \frac{3}{4}\left(\frac{(1 + (c_{\text{flow}}/A_{\text{d}})P(0))^4 - (1 + (c_{\text{flow}}/A_{\text{d}})P(L))^4}{(1 + (c_{\text{flow}}/A_{\text{d}})P(0))^3 - (1 + (c_{\text{flow}}/A_{\text{d}})P(L))^3}\right), \tag{8.23}$$

and for small pressure drops this becomes

$$\frac{V}{V_{\text{d}}} \simeq 1 + \frac{c_{\text{flow}}}{2A_{\text{d}}}(P(0) + P(L)) + \frac{c_{\text{flow}}^2}{6A_{\text{d}}^2}(P(0) - P(L))^2. \tag{8.24}$$

The second term is the effect of the average pressure in the vessel and the third term shows the effect of the pressure drop due to resistance. With no pressure drop, this relation reduces to (8.5) with P equal to the average pressure in the vessel, $(P(0) + P(L))/2$. (Why?)

The electrical analog of blood flow is described in Appendix D.

Table 8.5 Calculated tension in blood vessel walls

Vessel	Diameter (mm)	Wall thickness w (mm)	Internal pressure, ΔP (mmHg)	Wall tension, T (dyne/cm)	T/w (kPa)
Aorta	24.0	3.0	100	160,000	53
Large artery	8.0	1.0	97	52,000	52
Medium artery	4.0	0.8	90	24,000	30
Small artery	2.0	0.5	75	10,000	20
Arteriole	0.3	0.02	60	1,200	60
Capillary	0.008	0.001	30	16	16
Venule	0.02	0.002	20	27	13
Small vein	3.0	0.2	18	3,600	18
Medium vein	5.0	0.5	15	5,000	10
Large vein	15.0	0.8	10	10,000	12
Vena cava	30.0	1.5	10	20,000	13

Using data from [21, 31]
The wall thickness w is $R/5$ for arteries and $R/10$ for veins (where R is the vessel radius) and 1 μm for capillaries. Also see Table 8.2

The Strength of Blood Vessel Walls

The pressure inside blood vessel walls P exceeds that outside P_{ext}, by $\Delta P = P - P_{\text{ext}}$. *How large of a tension should the vessel walls be able to withstand to support this positive pressure differential?* Chapter 7 showed the answer is provided by the Law of Laplace for hollow cylinders (7.4). For a cylinder of radius of curvature R, this tension T is

$$\Delta P = \frac{T}{R}. \tag{8.25}$$

Table 8.5 shows that the tension capillaries need to withstand is very small because of their small radius. This circumferential stress, the tension (force per tube length) divided by the vessel wall thickness w, is surprisingly similar for these very different vessels. T/w can be compared to the UTS of such vessels.

Flow in Curving Arteries

In Chap. 7 we showed that the arterial walls feel a pressure due to the difference in hydrostatic pressure inside and outside the vessel. This pressure is felt equally around the wall. When blood flows in an artery that curves, a force equal to the centripetal force is felt on the arterial wall on the outer surface of the curve to change the direction (but not the magnitude) of the momentum vector of blood flow. *How large is this force? Is it comparable to the uniform hydrostatic pressure? Does it constitute a significant extra load on the arterial wall?*

For an artery of internal radius R that is turning with a radius $\mathcal{R}$ (Fig. 7.17), with blood of density ρ and average flow speed u, (7.57), this peak pressure is

$$P_{\text{cent}} = 2\rho u^2 \frac{R}{\mathcal{R}}. \tag{8.26}$$

This is largest for the fastest blood flow, which is in the aorta. Using $\rho = 1\,\text{g/cm}^3$, $R = 1.25\,\text{cm}$, $\mathcal{R} = 2\,\text{cm}$, and $u = 100\,\text{cm/s}$, we find $P_{\text{cent}} = 4.7\,\text{mmHg}$. This $\sim$5 mmHg is the extra pressure that must be supplied by the outer aorta wall to turn the blood around the aortic arch. This is small compared to the typical average aorta pressure of 100 mmHg, and does not likely promote pathological conditions such as aneurysms.

8.2.3 Capillaries and Osmotic Pressure

The purpose of systemic circulation is to supply blood to the capillary bed. We have seen that pressure is needed to bring the blood to the capillaries. There is diffusion and bulk flow between the blood in the capillaries and the interstitial fluid. Diffusion across the capillary wall transports oxygen, which is carried in red blood cells, and carbon dioxide, which is dissolved in the blood. (Not enough oxygen can be directly dissolved in the blood for our metabolic needs.) In the systemic capillaries there is net diffusion of oxygen out of the capillaries and carbon dioxide into them. In the pulmonary capillaries there is net diffusion of oxygen into the capillaries and carbon dioxide out of them. There is also bulk flow of fluid across the capillary walls due to the net pressure across the walls.

There are two forces driving this bulk transport: the force/area mechanical pressure P we have been discussing, which we will call *hydrostatic pressure* in this section, and a chemical driving force, called *osmotic pressure*, Π. The osmotic pressure characterizes the flow across a semipermeable membrane that occurs to equalize the concentrations of solutes on either side of the wall. This drives species that can permeate across the membrane, such as water, across it from the side of low concentration of solute to the side of high concentration. For low concentrations, the osmotic pressure is given by the van't Hoff equation

$$\Pi = n_s RT, \tag{8.27}$$

where n_s is the density (or concentration) of the solute in solution in moles per unit volume. This looks deceptively similar to the ideal gas law (7.2). It is the difference in the sum of these on either side of the capillary walls, $P + \Pi$, that drives the net transport across these walls.

As seen in Fig. 8.23, the hydrostatic force in the capillary is always greater inside the vessel than outside, and it decreases from about 36 mmHg at the arteriole side to 15 mmHg at the venule end because of viscosity. If this were the only driving force, there would be a large net flow of fluid from inside the capillary to the interstitial fluid! Small molecules, such as oxygen and carbon dioxide, are able to diffuse across the capillary wall, but larger molecules, such as proteins, cannot. Because there is

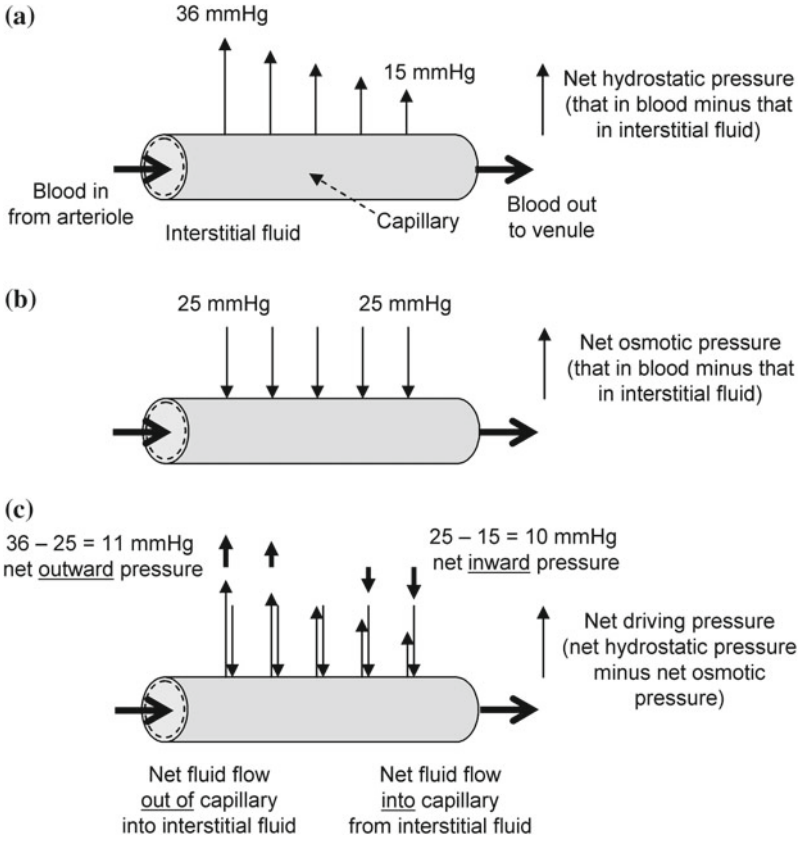


Fig. 8.23 Osmotic pressure in a capillary (Based on [66])

a much higher density of proteins in the blood than in the interstitial fluid, there is a chemical driving force of fluid into the capillary to try to equalize these densities. This leads to a net osmotic pressure of about 25 mmHg into the capillary. In Fig. 8.23, the net pressure is 36 mmHg − 25 mmHg = 11 mmHg outward at the arteriole end and 15 mmHg − 25 mmHg = −10 mmHg inward at the venule end. Therefore, there is net flow out of the capillary in the arteriole end and net flow into the capillary in the venule end. There is a small imbalance in this and a small net bulk flow out of the capillary.

We have assumed that the flow in all blood vessels, including the capillaries, is laminar. This cannot be really true for capillaries because many capillaries have an inner diameter of 5 or 6 μm and the red blood cells have a diameter of about 7.5 μm. The red blood cells deform to pass through the capillary and the resulting flow is called *bolus flow*. The red blood cells form plugs and the blood plasma is trapped in the regions between these plugs and moves in streamlines. Nowhere else in the body is the multicomponent nature of blood more apparent.

One major function of this capillary blood flow is the transfer of oxygen to the cells, leaving oxygen-depleted blood in the veins. As in (6.18), the rate of body consumption of O_2, dV_{O_2}/dt equals the product of the cardiac output Q_t (see below) and the difference in the oxygen partial pressure in the arteries and veins, $p_a - p_v$

$$\frac{dV_{O_2}}{dt} = Q_t(p_a - p_v). \tag{8.28}$$

If the lungs are bringing in air fast enough, then $p_a - p_v$ is fixed, and during aerobic exercise dV_{O_2}/dt increases linearly with Q_t. For a person with average fitness, the maximum blood flow rate is ≈19 L/min, for a highly fit person it is ≈25 L/min, and for an elite athlete it can be 35 L/min.

Oxygen combines with hemoglobin in the red blood cells in the lungs where the partial pressure of oxygen is high, about 100 mmHg. It is transported in the arteries to the tissues where it is released because the partial pressure of oxygen is low—and it is then used. The blood in the veins is then depleted in oxygen. Figure 8.24a shows the hemoglobin–oxygen dissociation curve. Clearly, hemoglobin is over 90% saturated with O_2 for partial pressures above 60 mmHg O_2. Increased CO_2 levels, increased temperature, and decreased pH all shift this curve to the right (Fig. 8.24b),

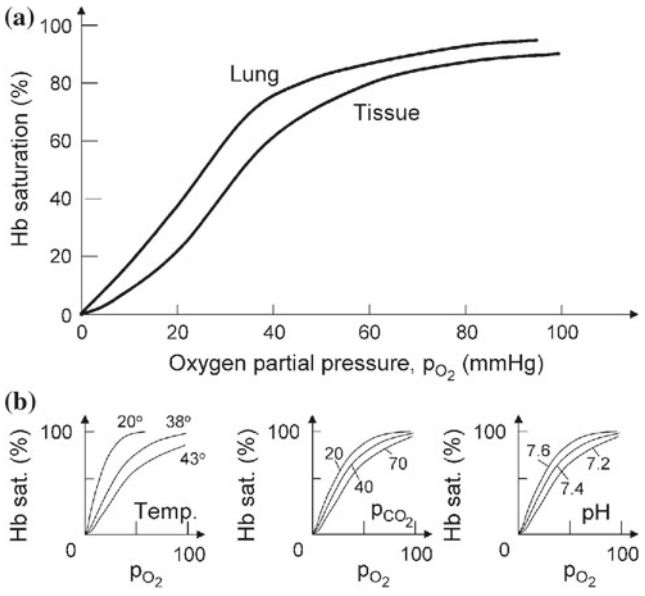


Fig. 8.24 **a** Hemoglobin–oxygen equilibrium in the lungs and tissue. During exercise oxygen intake is improved by the lung curve moving to the *left* and the tissue curve (exercising muscle) moving to the right due to increasing temperature, CO_2 partial pressure, and 2,3-diphosphoglycerate (DPG) (an end-product of red blood cell metabolism), and decreasing pH. **b** The hemoglobin–oxygen equilibrium shifts to the *right* with increasing temperature, increasing CO_2 partial pressure, and increasing DPG (not shown), and decreasing pH (Based on [60, 69])

which improves body performance. In Fig. 8.24a the dissociation curve for the lung is seen to be to the left of that in the tissues because the pH is higher and the CO_2 level is lower in the lung, increasing oxygen binding in the lungs relative to that in the tissues. During exercise, the muscle tissue pH falls and the local partial pressure of CO_2 and the local temperature increase. All of these changes move the curve to the right and this leads to more oxygen release (Fig. 8.24b). During exercise $p_a - p_v$ increases from the resting value of about 50 mL of oxygen per L of blood to 150 mL/L in normal people at $(dV_{O_2}/dt)_{max}$ (and to 160–170 mL/L in very fit people), in part because blood flow is being diverted from the organs to the muscles, where oxygen extraction is higher because of the exercise (see Fig. 8.24).

8.2.4 Blood Flow Rates and Speeds

The heart pumps about 80 mL (= 80 cm^3) of blood per contraction; this quantity is called the *stroke volume* V_{stroke}. The pump rate is the *heart beat rate* F of about 60/min or 1/s = 1 Hz. The *cardiac output* or total volumetric flow rate Q_t is the product of these two

$$Q_t = F V_{stroke} \tag{8.29}$$

or about 80 $cm^3/s = 4.8$ L/min. The total volume of blood is about 4.5–5.0 L, so all the blood is pumped throughout the body every minute. The flow rate in the arteries, arterioles, capillaries, venules, and veins are all the same because of the continuity of flow. (Q_t is actually a little less in the venules and veins because of the net fluid loss in the capillaries.) During the full term of pregnancy, cardiac output increases by ~30%, from ~5.0 to 6.7 L/min due to increased stroke volume and heart rate, and the blood volume increases by ~40–50% [47, 55].

The total flow in each of these vessel systems, Q_t, equals the total cross-sectional area A times the blood speed u, $Q_t = Au$. The parameters A and u are plotted in Fig. 8.16, which shows this inverse relationship for a flow rate of 90 cm^3/s. The cross-sectional area of the aorta is 3 cm^2, so in the aorta $u = (90\,cm^3/s)/3\,cm^2 = 30$ cm/s. In the capillaries the flow speed is much slower, $(90\,cm^3/s)/4{,}000\,cm^2 = 0.022$ cm/s $= 0.22$ mm/s. The net cross-sectional area in the capillaries is larger (~4,000 cm^2) even though they are very small (~3.5 μm in radius) because there are so many of them (~10^{10}). In the vena cava the flow speed is relatively fast $(90\,cm^3/s)/18\,cm^2 =$ 5 cm/s. In the arterial and venous systems, the smaller the vessel radius, the larger the total cross-section of all vessels in that order and the slower the blood speed. These are actually average blood flows during each cycle.

The maximum Reynolds number ($Re = \rho u d/\eta$, (7.11)) over a cardiac cycle ranges from ~6,000 in the heart and aorta to $<10^{-3}$ in the capillaries. The nominal lower threshold for turbulent flow is $Re \sim 2,000$, so it is possible that flow in the aorta is turbulent.

The overall flow in the systemic arterial system can be described by relating the total cardiac output Q_t to the systemic arterial pressure P_{sa}, by

$$P_{sa} = (\text{TPVR})\, Q_t, \tag{8.30}$$

where TPVR is the total peripheral vascular resistance—which is due to the combined effect of all the organ beds of systemic circulation (mostly arterioles and capillaries). (P_{sa} should really be replaced by the pressure drop in the system. See Problem 8.21.) Normal values for the systemic system range from 700 to 1,600 dyne-s/cm^5, and analogous normal values for the pulmonary system range from 20 to 130 dyne-s/cm^5. Equation (8.2) applies to an individual vessel, while this relation describes the entire systemic system. The body regulates P_{sa} by controlling the cardiac output and this peripheral resistance. When we lie down, a large volume of blood is transiently stored in the lower extremities and abdomen, and so when we stand there is initially less flow of blood to the heart and a drop in blood pressure, which can make you faint. Even though Q_t decreases, P_{sa} drops only mildly because there is a prompt reflex that increases the TPVR (vasoconstriction). In contrast, when blood pressure rises suddenly, feedback tends to decrease the overall vascular resistance, to restore a lower blood pressure. These are two examples of body feedback and control, as described in Chap. 13.

We can also evaluate the overall compliances of the vascular systems, such as those of the systemic arterial and venous systems. The compliance is the reciprocal of the slope of a pressure–volume curve in Fig. 8.25. The smooth muscles surrounding a large vessel can change the volume of the vessel at a given pressure, either decreasing it (by stimulating the muscles) or increasing it (by inhibiting the muscles).

During even moderate exercise the blood flow rate increases substantially, as seen in Fig. 8.26, and the absolute and relative distribution of blood to different parts of the body also changes radically. Figure 8.27 shows an example in which the flow rate increases from 5 to 12.5 L/min during exercise. There are extremely large increases

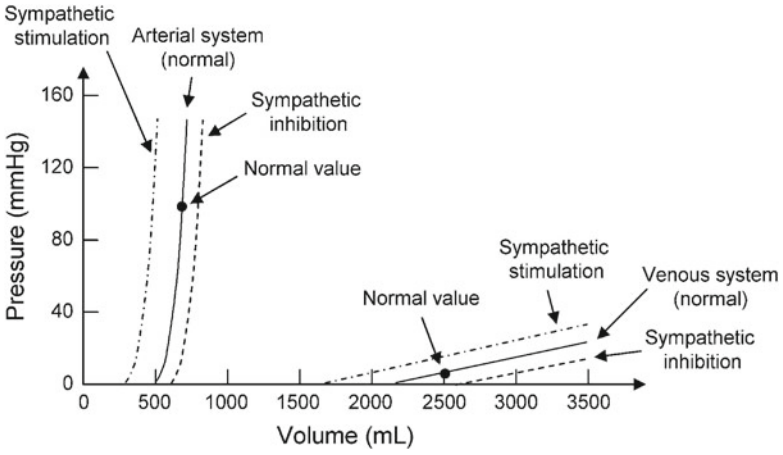


Fig. 8.25 Volume–pressure curves for the systemic arterial and venous systems, for normal conditions and for sympathetic stimulation and inhibition (Based on [29])

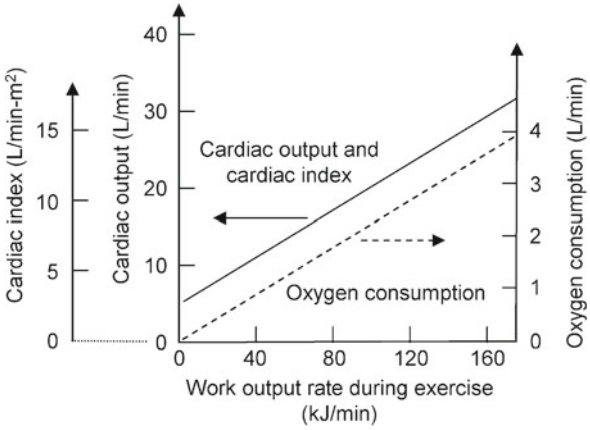


Fig. 8.26 Variation in cardiac output (and cardiac index) and oxygen consumption needed during varying levels of exercise with work output (The cardiac index is the cardiac output divided by the person's surface area.) (Based on [28, 29])

of blood flowing to the skeletal muscle to supply oxygen for aerobic metabolism (up 1,066%), to the heart so it can pump faster (up 367%), and to the skin (up 370%) to assist cooling (which increases the skin temperature, thus accelerating radiative and convection conduction from the body). Blood flow to the brain is unchanged. In contrast, blood flow to the digestive track, liver, and kidneys decrease by a factor of ~2. These changes are also seen in Tables 8.6 and 8.7. Figure 8.28 shows that the blood flow to the calf during rhythmic exercise is higher than normal and it varies with time.

This increase in cardiac output during exercise occurs because of increases in both the heart rate and the stroke volume; the blood speed also increases because $Q_t = Au$. The faster the heart rate, the shorter is diastole, while the duration of systole does not change. For short term (5–10 min) submaximal exercise, the cardiac output increases from 5 L/min to a new steady-state value in about 2 min. For a steady-state cardiac output of 18 L/min, the stroke volume increases from about 70–120 mL/beat and the heart rate F from about 70–150 beats/min. The cardiac output returns to the resting value in 5–10 min after exercise.

For a longer submaximal workout (30–60 min), the new steady-state cardiac output is maintained, but the stroke volume slowly decreases and the heart rate gradually increases with time, particularly in warmer environments. This *cardiovascular drift* is caused by a decrease in the venous return of blood to the heart, which decreases the stroke volume and so the heart rate must increase to maintain the same cardiac output. This decrease in venous return is caused by two factors. (1) During such exercise more blood flows under the skin to help lower the increase in body core temperature caused by the increased metabolic activity (Chap. 6), and this lowers the steady-state flow of blood back. (2) During exercise water flows from the blood to the surrounding cells and tissues because of increased arterial pressure and the compression of venules due to muscle action. This produces a steady-state decrease in blood plasma and blood volume, and a steady-state decrease in blood returning to the heart and the stroke volume.

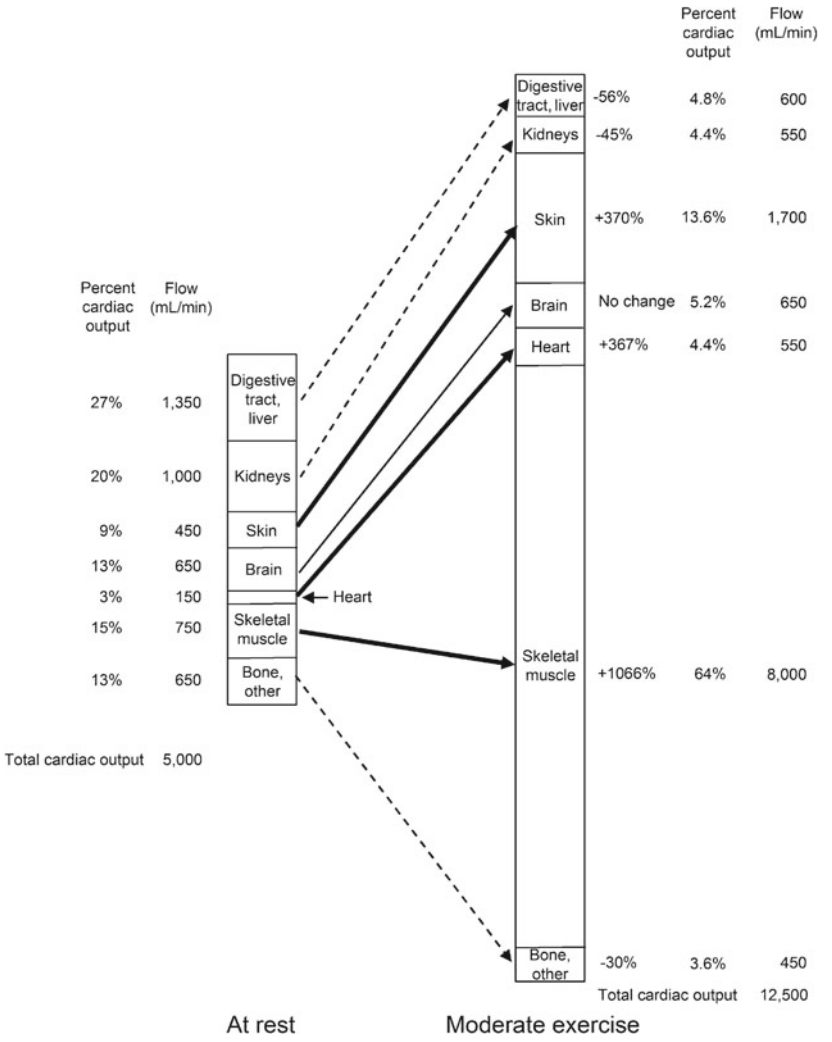


Fig. 8.27 Blood flow to different organs at rest and during moderate exercise, showing no change to the brain, increases to the skin, heart, and skeletal muscles (*thick arrows*), and less blood flow elsewhere (*dashed arrows*) (Based on [66])

Stroke volume and cardiac output are determined by the preload and afterload conditions (see below), contractility (ability to contract), and heart rate. The cardiac output is not directly regulated, but there is a feedback and control system that regulates arterial pressure that affects the heart rate and contractility, as well as afterload and other factors that control the preload [64].

The Frank–Starling mechanism (or Starling's Law of the heart) states that the larger the end-of-diastole volume or pressure (the *preload*), the larger the stroke

Table 8.6 Total cardiac flow (mL/min) for organs during exercise, including percentage of total flow

Organ	Rest	Light exercise	Heavy exercise	Maximal exercise
Brain	750 (13%)	750 (8%)	750 (4%)	750 (3%)
Heart	250 (4%)	350 (3.5%)	750 (4%)	1,000 (4%)
Muscle	1,200 (21%)	4,500 (47%)	12,500 (72%)	22,000 (88%)
Skin	500 (8.5%)	1,500 (16%)	1,900 (11%)	600 (2.5%)
Kidney	1,100 (19%)	900 (9.5%)	600 (3.5%)	250 (1%)
Abdomen	1,400 (24%)	1,100 (11.5%)	600 (3.5%)	300 (1.2%)
Other	600 (10.5%)	400 (4%)	400 (2%)	100 (0.4%)
Total	5,800 (100%)	9,500 (100%)	17,500 (100%)	25,000 (100%)

Using data from [20]

Table 8.7 Approximate blood flow (perfusion) for tissues and organs, per gram

Tissue type	Location or organ	Specific blood flow rate (mm^3/s-g)
Adipose tissue	Abdomen, ~20 mm thick	0.51
	Abdomen, >40 mm thick	0.31
	Thigh, ~20 mm thick	0.33
Bone	Humerus, marrow flow only	0.055
Connective tissue	Typical basal (max)	0.50 (2.5)
Muscle	Typical basal (max)	0.50 (10)
Organ	Brain, basal (max)	9.1 (18.3)
	Gastrointestinal track, basal (max)	6.7 (26.7)
	Heart, basal (max)	13.7 (64.0)
	Kidney, basal (max)	68 (100)
	Liver, basal (max)	12 (54)
	Lung, basal (max)	90 (490)
Skin	Typical resting flow (max)	1.7 (25.0)

Using data from [21]
In some cases the basal rate is given, along with the maximum rate in parentheses

volume, as is seen in Fig. 8.29. Furthermore, the larger the aortic pressure, the less blood can be ejected by the left ventricle (the *afterload*), as is seen in Fig. 8.30. An increase in the heart rate also increases cardiac output, however, the increase is sublinear because the stroke volume decreases (Fig. 8.31) due to the above preload and postload factors. With greater cardiac output there is less blood in the veins to return to the heart for diastole (lower preload) and the arterial pressure is higher so the heart can eject less blood (higher afterload), assuming the peripheral resistance is constant.

For the most part, these are changes in the systemic system. Similar increases in cardiac output have to occur in the pulmonary system. (Why?) This occurs by an increase in the number of open capillaries in the lung, by up to a factor of three, and by a distending of all the pulmonary capillaries, which increases the flow in each

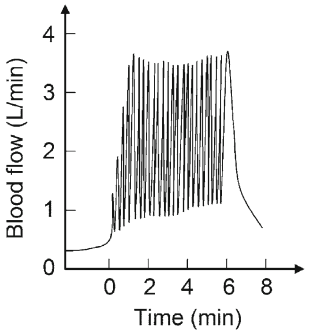


Fig. 8.28 Blood flow to the calf during rhythmic contraction exercises, showing less blood flow during contractions than between them (Based on [3, 29])

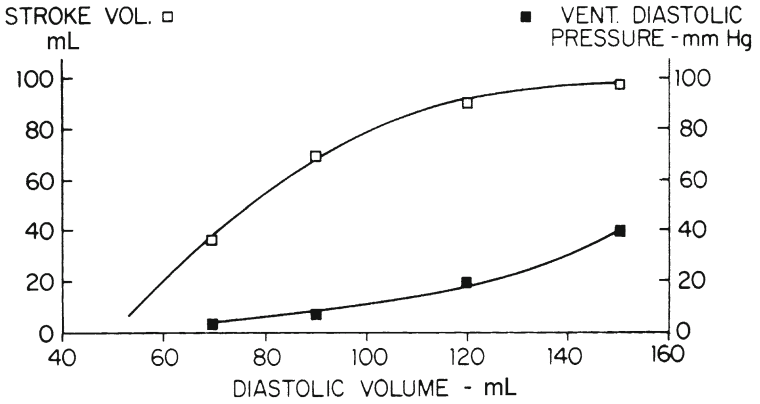


Fig. 8.29 The stroke volume increases with diastolic volume and pressure, as seen with data for four heart beats. This dependence, along with the explanation of it, is known as the Frank–Starling mechanism (Reprinted from [64]. Used with permission of Elsevier)

capillary by up to a factor of two, with very little change in the pulmonary arterial pressure (Fig. 8.32).

The maximum heart rate $F_{\max}$ (in beats/min) depends on age Y (in years). It is usually given as

$$F_{\max} = 220 - Y, \tag{8.31}$$

but recent work [56] suggests that

$$F_{\max} = 211 - 0.64Y \tag{8.32}$$

is more accurate. The standard error in these relations is ± 10 beats/min, which means 67% of people have a maximum rate ± 10 beats/min within the value predicted by

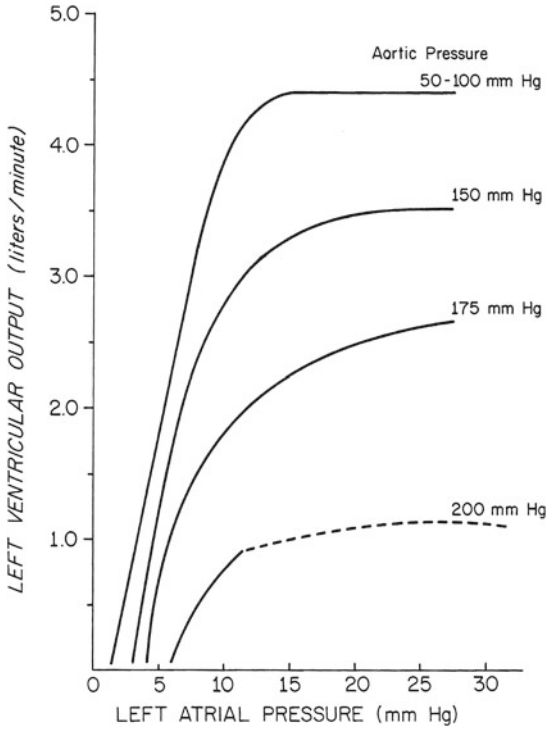


Fig. 8.30 The left ventricular output (the cardiac output) versus left atrial pressure for different aortic pressures. It increases with this atrial pressure and then levels off, and decreases with increasing aortic pressure (Reprinted from [64]. Used with permission of Elsevier; adapted from [62])

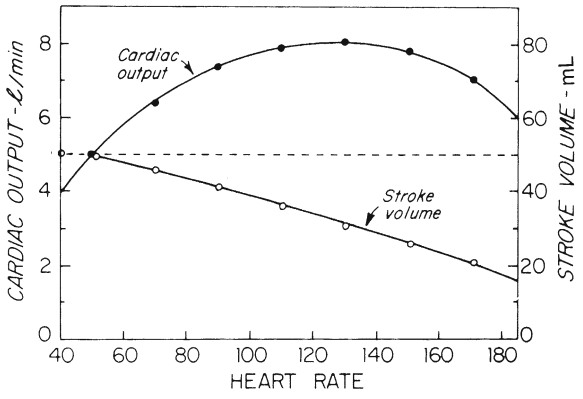


Fig. 8.31 Typical dependence of cardiac output and stroke volume on heart rate (Reprinted from [64]. Used with permission of Elsevier)

these relations and 95% of all people have a rate that is ±20 beats/min within the predicted values.

Because F does not depend on the level of fitness, athletes increase their maximum cardiac output by increasing their stroke volumes. Stroke volumes in untrained athletes are 50–70 mL at rest and reach 80–110 mL during heavy activity. For trained and highly trained athletes these stroke volumes increase to 70–90 mL and 90–110 mL at rest and 110–150 mL and 150–220 mL during heavy activity.

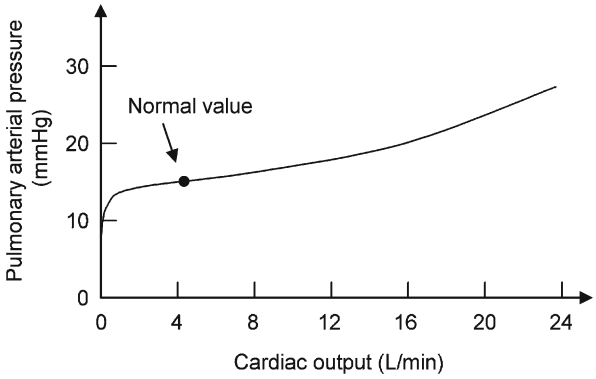


Fig. 8.32 The pulmonary arterial pressure versus cardiac pressure, showing it does not change much during exercise (Based on [29])

In steady state, the cardiac output Q_t must equal FV_{stroke} (8.29), as well as P_{sa}/TPVR (8.30). During heavy exercise, Q_t increases from 5 to 20 L/min, so not only must F and V_{stroke} increase, as we have described, but P_{sa}/TPVR must increase accordingly. Diastolic pressure changes little during exercise, remaining within ±10 mmHg of the resting value. The systolic pressure increases to about 200 mmHg for men and 180 mmHg for women. Using (8.1), the mean arterial pressure then increases only to 140–150 mmHg, which cannot account for most of the increase in blood flow rate. During heavy exercise the systemic vascular resistance TPVR decreases to about 40% of its resting value because of the widening of muscular vascular beds that are normally constricted at low levels of activity.

With advancing age, the cardiac output and ejection fraction do not change for a healthy person, but the maximum cardiac output decreases with age. At maximal exercise in endurance-trained men, it decreases by ~20% from ~27.0 L/min at 28 years of age to ~21.7 L/min at 60 years of age due to the ~10% decreases in stroke volume, from ~147 to ~132 mL, and heart rate, from ~184 beats/min to ~165 beats/min [1, 19, 67]. (As seen below, the ejection fraction is the stroke volume divided by the ventricular volume at the end of diastole, which is normally 55–75%.)

Figure 8.33 shows that blood pressure typically increases with age. Hypertension begins with systolic pressure ≥140 mmHg or diastolic pressure ≥90 mmHg. In essential hypertension this blood pressure is heightened for no obvious reason. The average blood pressure is the product of the total peripheral vascular resistance and the cardiac output, $P_{sa} = (\text{TPVR})\,Q_t$ (8.30). In people under 40 years of age, hypertension is driven by increased cardiac output, with normal TPVR. In older people, the cardiac output is normal or reduced, but the TPVR is high.

In the fetus, the combined cardiac output is the sum of the cardiac output from the right and left ventricles; it increases from ~210 mL/min at 20 weeks to ~1900 mL/min at 38 weeks, and is very roughly equally distributed between the ventricles [41]. (After birth, the cardiac output is clearly defined as the total amount from either ventricle.) Near the end of gestation ~20 and ~40% of this total is shunted through the foramen ovale and the ductus arteriosis. The blood flow through the umbilical cord is ~35 mL/min at 20 weeks and ~240 mL/min at 40 weeks of

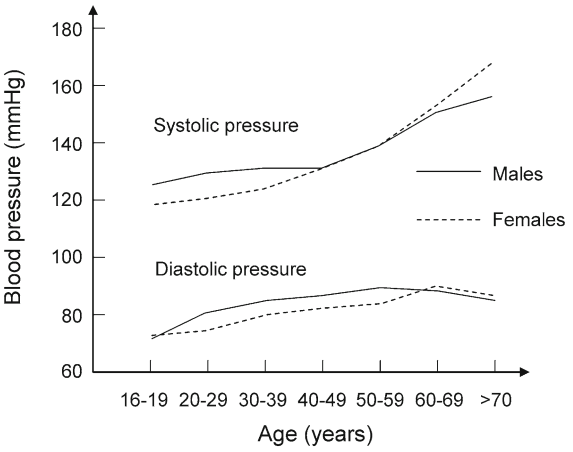


Fig. 8.33 Systolic and diastolic blood pressure is shown for males and females, averaged over age groups. The trend is to increased blood pressure with age (Based on [43, 48])

gestation, a small but significant fraction of the combined cardiac output. The fraction of blood returning from the umbilical cord shunted to the ductus venosus decreases from ~30% at mid-gestation to ~20% near the end of it, with the rest going to the liver. The weight of blood in the fetus is ~10–12% of body weight (including the placenta, with ~33% in the placenta at birth), which is a bit higher than the ~7–8% in adults; excluding the placenta, this corresponds to ~80 mL blood per kg fetal mass. The systemic systolic pressure in the fetus increases to ~30–40 mmHg at 28 weeks gestation.

Circulatory shock in people occurs when there is too little blood flowing generally in the body, and this results in tissue damaged from the inadequate delivery of oxygen and nutrients to the cells and the inadequate removal of waste products. Such shock can occur from inadequate pumping of blood by the heart or by inadequate venous return of blood to the heart, such as due to diminished blood volume (*hypovolemia*), decreased capillary vasomotion, or obstructed circulation. *Hemorrhage* is often the cause of the diminished blood volume. Figure 8.34 shows that cardiac output and arterial pressure can withstand a ~10% blood loss—if this were not so you would not be able to donate blood—but decrease for larger losses and approach zero with 35–45% blood loss. The localized loss of blood flow to the brain, *strokes*, is discussed below. This occurs due to clogged arteries (as we will see now and again below) and hemorrhaging (as we will see below). The localized loss of blood flow to the heart results in a *myocardial infarction* (heart attack).

8.2.5 Consequences of Clogged Arteries

Atherosclerosis (a-thear-oh'-scler-oh-sis) occurs when a deposit or atheroma (a-thear-oh'-ma) (or plaque) forms on an arterial wall (Fig. 8.35). The smaller

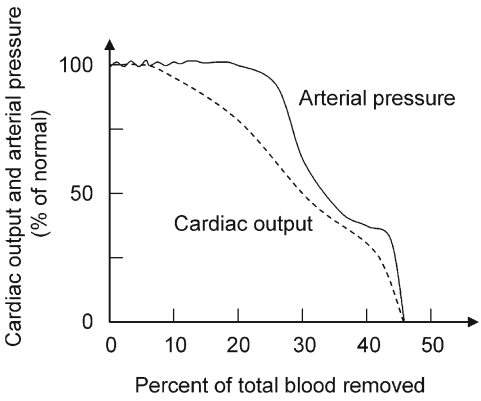


Fig. 8.34 Cardiac output and arterial pressure with decreased blood volume from hemorrhaging (Based on [29])

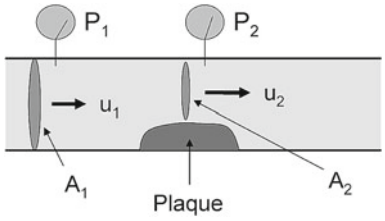


Fig. 8.35 Sketch of flow in a clogged artery

cross-sectional area at this site, because $A_2/A_1 < 1$, leads to a faster flow speed due to continuity of flow, with

$$u_2 = \frac{A_1}{A_2} u_1 \tag{8.33}$$

from (7.16). For $A_2/A_1 = 1/3$, we find that $u_2 = 3u_1$. Also from Bernoulli's equation (7.18) we find that

$$P_2 - P_1 = \frac{1}{2}\rho u_1^2 \left(1 - \left(\frac{A_1}{A_2}\right)^2\right) \tag{8.34}$$

and so $P_2 < P_1$ and for $A_2/A_1 = 1/3$ we see that $P_2 - P_1 = -4\rho u_1^2$. This pressure drop increases with blood speed and so it is expected to increase with increased physical activity. This flow is not necessary laminar in the occluded region, as is seen in Fig. 8.36.

In 1954 Arturo Toscanini was conducting the NBC Symphony Orchestra. (This orchestra was pretty prestigious then, performing on radio and the then-new television, but it does not exist now.) He was vigorously waving his arms, as conductors often do, and he fainted. Why? Equation (8.34) contains the answer and Fig. 8.37 illustrates why. He suffered a *transient ischemic attack*, or TIA [26, 42]. *Ischemia* (iss-kee'-mee-uh) is the local decrease in blood flow. TIA is the temporary loss of blood to the brain by the "subclavian steal syndrome." It results in temporary

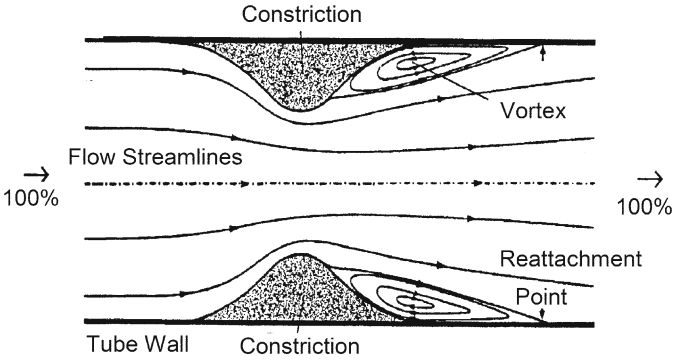


Fig. 8.36 The flow is partially turbulent in clogged arteries (From [21]. Adapted from [39]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com)

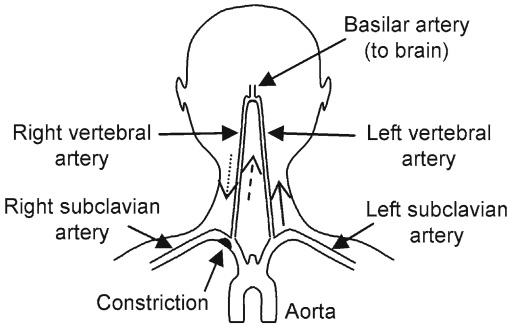


Fig. 8.37 Anterior view of the blood flow to the brain with the subclavian steal syndrome, resulting in a transient ischemic attack. Blood flow in the left vertebral artery is shown by the *unbroken line arrow*. Without the constriction, the blood flow in the right vertebral artery is normal, as shown by the *dashed arrow*, so there is normal blood flow to the basilar artery. With the constriction, there can be blood flow from the left vertebral artery to the right vertebral artery (*dotted arrow*), and there is no blood flow into the basilar artery (Based on [26])

dizziness, double vision, headache, and weakness in the limbs. By its nature it is only temporary, but it indicates a more severe problem.

The left and right carotid arteries are two major arteries supplying blood to the anterior brain. The left and right vertebral arteries supply blood to the posterior part of the brain. They branch off from the subclavian arteries that also supply blood to the arms. The internal carotid and vertebral systems join with each other at the base of the brain, forming the circle of Willis (Fig. 8.38, also see Fig. 8.42 below). Posteriorly, the flow in the left and right vertebral arteries merge to form a single basilar artery to the brain.

Say there is a constriction in the right subclavian artery near where the vertebral artery branches off (Fig. 8.37). The pressure before the constriction and in the same region in the normal right side is P_1. Because of the constriction, the pressure on the

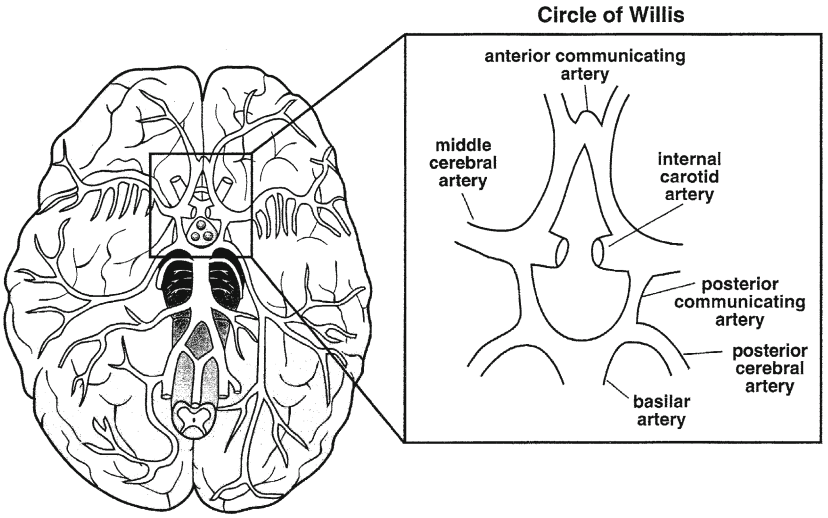


Fig. 8.38 Circle of Willis in the brain (From [31])

left side is $P_2 < P_1$. This difference increases with faster blood flow. With vigorous motion of the arms, there is a need for greater blood flow in the subclavian arteries to supply more oxygen to the skeletal muscle in the shoulder and arms, and the blood speed in these arteries u_1 increases. If u_1 increases enough, P_2 becomes so much smaller than P_1 that flow in the left vertebral artery is diverted to the right vertebral artery—and does not flow to the basilar artery to the brain. When this happened to Toscanini, he fainted. He stopped waving his hands, of course. Blood flow to his arms then slowed down. The difference in P_1 and P_2 decreased to its usual smaller value (even with this constriction). Blood from both vertebral arteries then flowed to his basilar artery. His brain started receiving a normal flow of blood again, and he regained consciousness. All was fine—but this constriction had to be removed. (Toscanini never conducted again. The underlying reason for his fainting spell was not known in 1954; TIA was first explained in 1961.)

How are such obstructions formed? Low-density lipoproteins (LDLs, bad cholesterol) transport cholesterol and triglycerides from the liver to other organs. They are the building materials in forming plaque on artery walls that lead to artherosclerosis (heart disease). High-density lipoproteins (HDL, good cholesterol) seem to help dissolve the plaque and bring cholesterol back to the liver. The full story is complex and currently not clear, but it is clear that good diet and exercise (as well as good genes) seem to help lessen the net formation of plaque and heart disease [6]. (Good diet and exercise, in addition to good genes, seem to avert the severe decree for many potential diseases.) Unsaturated fatty acids, such as from fish and nuts, have fewer hydrogen atoms because all carbon-carbon bonds are not single bonds and are liquids at room temperature. They seem to cause the liver to form HDLs. Saturated fatty acids, such as from red meat, have more hydrogen atoms because all carbon-carbon bonds are

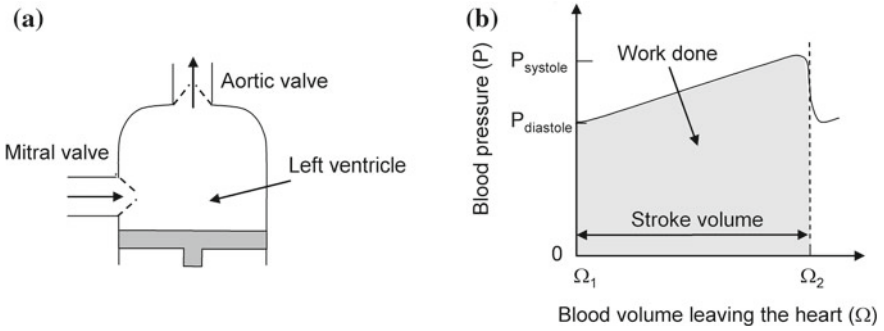


Fig. 8.39 **a** Schematic of the heart left ventricle as a pump, **b** and the pressure and volume of the left ventricle during systolic contraction during systole (from t_1 to t_2), showing the work done by the heart (*shaded area*) (Based on [4])

single bonds and are solids at room temperature. They seem to cause the liver to form LDLs. Unsaturated fats that have been converted into saturated fats industrially, trans fats, raise LDL and lower HDL levels [44].

More examples of the effects of obstructions in arteries will be examined later in this chapter.

8.2.6 *Work Done by the Heart and the Metabolic Needs of the Heart*

How much work is done by the heart? Consider the left ventricle, which is a pump during systole, as diagrammed in Fig. 8.39.

Work is done by this pump, with a force pushing the piston from the initial to final positions L_i to L_f to give

$$W = \int_{L_\mathrm{i}}^{L_\mathrm{f}} F \,\mathrm{d}L = \int_{L_\mathrm{i}}^{L_\mathrm{f}} (F/A)(A \,\mathrm{d}L) = \int_{V_\mathrm{i}}^{V_\mathrm{f}} P \,\mathrm{d}V, \tag{8.35}$$

where the force/area is the pressure, P, and the distance times the area is the ventricular volume, V. During systole the ventricular pressure increases from a very low value (that in the left atrium), to $P_\mathrm{diastole} \sim 80\,\mathrm{mmHg}$ (at time t_1), and then up to $P_\mathrm{systole} \sim 120\,\mathrm{mmHg}$, and it stays at this value until the end of systole (at time t_2). The aortic valve first opens when this pressure rises above P_diastole (at t_1), and blood is pumped out until systole is over (at t_2). At t_1, the volume of pumped blood is $\Omega_\mathrm{i} = 0$ and at t_2 it is $\Omega_\mathrm{f} = V_\mathrm{i} - V_\mathrm{f}$. Figure 8.39 shows this evolution of the ventricular pressure and pumped volume from t_1 to t_2.

The area under this curve is the work done and so $W = P_\mathrm{av} V_\mathrm{stroke}$, where P_av is average pressure during this cycle (averaged over the volume displaced and which

does not necessarily scale linearly with time, and assuming no back pressure), and $V_{\text{stroke}} = \Omega_f$, the stroke volume. Clearly, $P_{\text{diastole}} < P_{\text{av}} < P_{\text{systole}}$. For a linear variation, we see that $P_{\text{av}} = (P_{\text{diastole}} + P_{\text{systole}})/2 \sim 100\,\text{mmHg} = 1.3 \times 10^4\,\text{N/m}^2$. (This averaging is different from that in (8.1) because of the simplicity of this model. This leads to an ~20% uncertainty.) We take $V_{\text{stroke}} = 80\,\text{cm}^3 = 8 \times 10^{-5}\,\text{m}^3$, so $W = (1.3 \times 10^4\,\text{N/m}^2)(8 \times 10^{-5}\,\text{m}^3) = 1.04\,\text{J}$ per cycle. With a heart rate of 60/min = 1/s, the rate the left ventricle does work is $P_{\text{power,mech,av}} = (1.04\,\text{J per cycle})(1\,\text{cycle/s}) = 1.04\,\text{W}$.

The efficiency, ϵ, of converting metabolic energy into this mechanical work is approximately 20% (and sometimes this range is given as 12–30%), and so the metabolic power needed to run the left ventricle is $P_{\text{power,metab,av}} = P_{\text{power,mech,av}}/\epsilon = 5\,\text{W}$.

The heart pumps for about 1/3 of the cardiac cycle and rests for the other 2/3 of the time. Therefore the peak powers are higher than these average values by a factor of 3, with $P_{\text{power,mech,peak}} = 1.5\,\text{W}$ and $P_{\text{power,metab,peak}} = 15\,\text{W}$.

The energy consumed to run the left ventricle is $(86{,}400\,\text{s/day})(5\,\text{W}) = 4.32 \times 10^5\,\text{J/day} = 104\,\text{kcal/day}$. So far we examined the work done by only the left ventricle. The right ventricle pumps the same volume per cardiac cycle (to maintain the steady-state flow throughout), but at a pressure 1/5 times that of left ventricle, so the work and all of these powers are smaller by a factor of five. This increases the required metabolic power by 20%. Similarly, the pressures for the two atria are also relatively very small. Overall, with 20% muscle efficiency we expect to need ~125 kcal/day to run the heart; with 10% muscle efficiency it would be ~250 kcal/day.

The experimental value for the BMR contribution for the entire heart is ~117 kcal/day (Table 6.17), which is close to our estimate. The biggest uncertainty here is the efficiency of the cardiac muscle.

What happens with strenuous exercise? The blood pressure can increase by 50% and the blood flow rate can increase by a factor of 5. Therefore, the mechanical power exerted by the heart and the associated metabolic requirements can increase by a factor of 7.5.

8.3 Strokes and Aneurysms

Cerebral blood flows from the internal carotid and vertebral arteries and through the circle of Willis at the base of the brain (Fig. 8.38), and then permeates the brain through a complex series of capillaries.

Any severe restriction of blood to the brain is called a *stroke* [30, 31]. *Ischemia* means there is a lack of blood flow. The nearby tissue becomes deficient in oxygen and metabolites, and has excessive metabolic waste products. (We saw an example of ischemia earlier this chapter with the Toscanini TIA. Because TIA is transient, there is an oxygen deficiency due to the stroke but the patient can still recover with little or no brain damage.) *Hypoxia* means a lack of oxygen, and it can result from ischemia or other causes, such as high altitude (see Chap. 9) or CO poisoning. *Infarction* means

that the stroke causes permanent brain damage. (The terms *ischemia* and *infarction* are actually more general and also apply to tissues outside the brain, with a myocardial infarction in the heart being one example.) The transition from the reversible event to the irreversible infarct with the formation of necrotic tissue occurs when the stroke is particularly long or of particularly large magnitude. (Necrotic tissue is dead tissue that did not die in a manner programmed by the body, and so is in contrast to apoptotic tissue which the body kills as part of the life cycle.) The occurrence of a transient ischemic attack, or ministroke, sometimes means a more damaging stroke is imminent. We now address hemorrhagic and ischemic strokes, which are due to ruptured and blocked blood vessels, respectively.

8.3.1 Hemorrhagic Strokes

Hemorrhagic strokes are due to a ruptured vasculature (blood vessels) within the brain, attributed to an aneurysm or weakened blood vessel (Fig. 8.40). An *aneurysm* is an enlarged blood vessel. In addition to the loss of blood to the targeted regions, during hemorrhages blood fills the surrounding spaces and compresses the surrounding tissues. This accumulating blood compresses other blood vessels, decreasing their diameters and reducing the flow of blood to other parts of the brain, and increases the intracranial pressure, which leads to neurological complications. About 20% of all strokes are hemorrhagic; they occur mostly in the young and middle-aged, due to vascular *lesions* such as arteriovenous malformations and aneurysms. (Lesions are entities of diseased or abnormal tissue.) In the elderly, blood vessels are brittle and less distensible due to atherosclerotic deposits, and this can lead to possible spontaneous rupture of these vessels in the brain and hemorrhagic stroke. In *atherosclerosis* ("hardening of the arteries") lipid or fatty deposits in the blood accumulate on the inner vessel wall and eventually form hard arterial plaque.

The majority of cerebral aneurysms are *saccular (or berry) aneurysms* that most often occur where large cerebral arteries bifurcate (Figs. 8.41 and 8.42). In *fusiform*

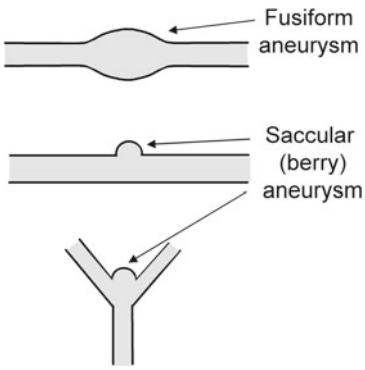


Fig. 8.40 Fusiform and saccular/berry aneurysms, the latter in a vessel and for bifurcated flow

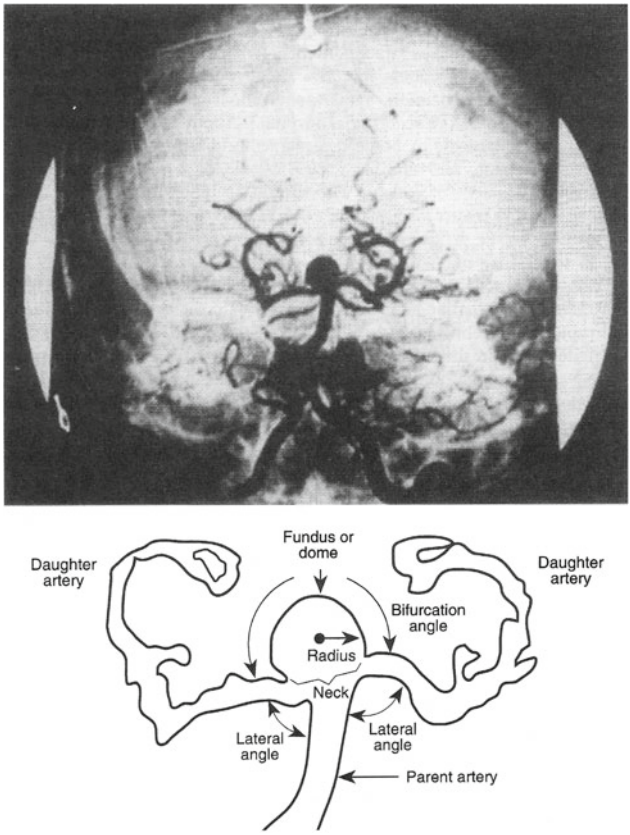


Fig. 8.41 Saccular (berry) aneurysm at an apex of a branching vessel, showing an angiographic projectional image (*top*) and a model (*bottom*) (From [31])

aneurysms there is uniform ballooning of the circumference of the vessel walls, instead of in localized regions of the vessels as in saccular aneurysms; this leads to ellipsoidal or football shaped aneurysms. (Fusiform means tapering at each end. Here it indicates a cylindrically symmetric aneurysm that tapers to the normal vessel at either end, while for the fusiform muscles in Chap. 5 it indicates a cylindrically symmetric muscle that tapers to tendons on each end.) Fusiform aneurysms are less common in the brain than are saccular aneurysms, but are common elsewhere in the body. The abdominal aortic aneurysm, which develops along the aorta in the abdominal or gut region, is the most common aneurysm found in the body and is a fusiform aneurysm (Fig. 8.43).

Healthy arteries contain the structural proteins elastin and collagen. Collagen has the larger Young's modulus (Table 4.2) and is expected to dominate the elastic properties of arteries. Still, the elastin contributes to the distensibility of the artery. The resistance to stretching at low pressures seems to be due to the elastin fibers, at

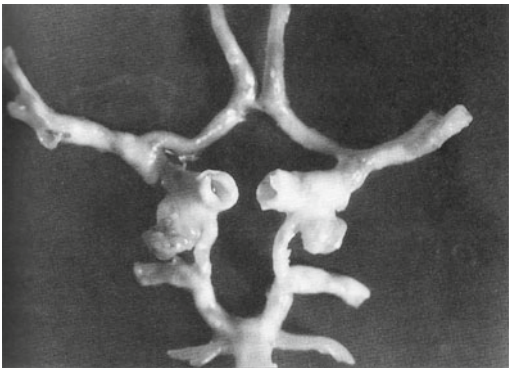


Fig. 8.42 Photograph of an inferior view of an excised human circle of Willis. Bilateral (i.e., on both sides) saccular aneurysms are seen near the junction between the internal carotid artery and the *circle*; the larger lesion (the one on the *left side* of the photograph) had ruptured (From [36])

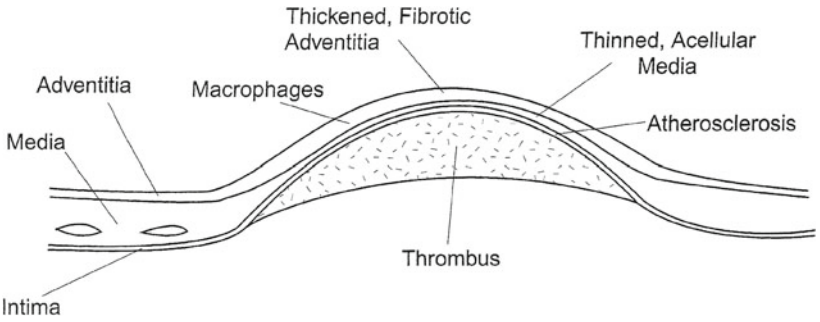


Fig. 8.43 Schematic of an abdominal aortic aneurysm, showing an attenuated media and an intraluminal thrombus (From [36])

normal physiological pressures it is due to elastin and collagen fibers, and at even higher pressures it is due to collagen. When the artery wall balloons or sacculates (i.e., it balloons in one circumferential part of the wall) as the aneurysm develops, the elastin becomes less effective in maintaining structural integrity of the artery and the collagen takes on most of the load. The artery becomes less distensible and this translates to greater stress for the same strain, thereby accelerating structural fatigue. This process accelerates the load on the arterial wall, and leads to rupture. A possible scenario for the formation and rupture of a saccular aneurysm is shown in Fig. 8.44. Figure 8.45 shows the equilibrium circumferential tension for a vessel assuming the Law of Laplace, for a normal artery, and one with an aneurysm. Figure 7.11 shows that the maximum blood flow velocity is in the center, where the wall shear stress is minimum. The minimum blood flow velocity is near the wall, where the wall shear stress is maximum.

There are (at least) four physical reasons why the larger radius of a fusiform aneurysm can lead to rupture. All are related to the stress relation $T/w = r(\Delta P)/w$,

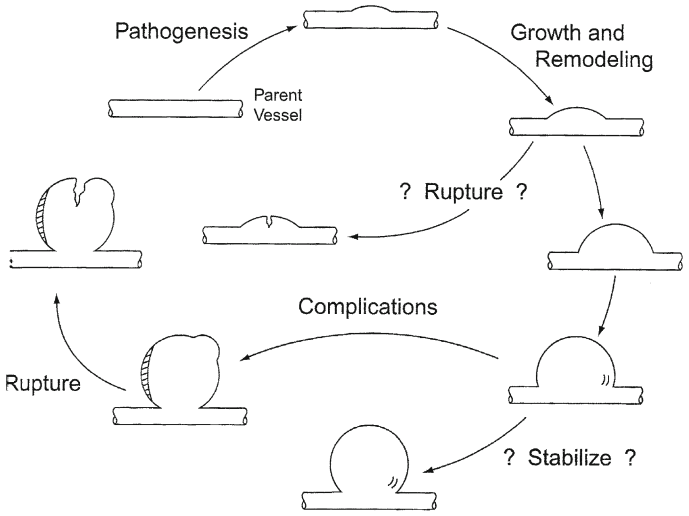


Fig. 8.44 A possible natural history for the development of a saccular aneurysm. A local weakening of the vessel wall, leading to a mild dilatation, can be caused by an initial "insult" from one of several causes (From [36])

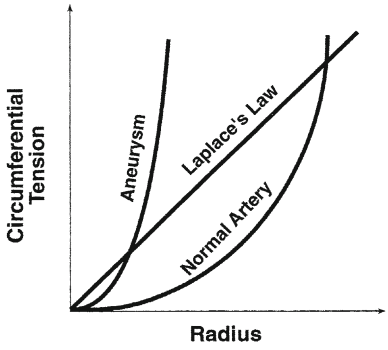


Fig. 8.45 Circumferential tension needed for equilibrium as a function of vessel radius, assuming the Law of Laplace, a normal artery, and an artery with an aneurysm (From [31, 32])

using the Law of Laplace for a cylinder (7.4), where w is the wall thickness and r is local vessel radius. (1) r is larger so more tension is needed to withstand even an unchanged pressure difference (because $T = r(\Delta P)$). (2) This increase in vessel radius can be accompanied by thinning walls (if the volume of vessel wall per unit length, $\sim 2\pi r w$, is relatively unchanged by the aneurysm), so the stress T/w increases even more. (3) With this wall thinning there may be damage that lowers the UTS locally, and the UTS needs to be $\gg T/w$ to avoid rupture. (4) A larger r, and the concomitant larger cross-sectional area A, leads to a slower blood speed u, through volumetric continuity (7.16). This in turn leads to a larger pressure P, through Bernoulli's equation (7.18). This increases the tension that the vessel must withstand. The magnitudes of several of these effects are evaluated in Problem 8.38.

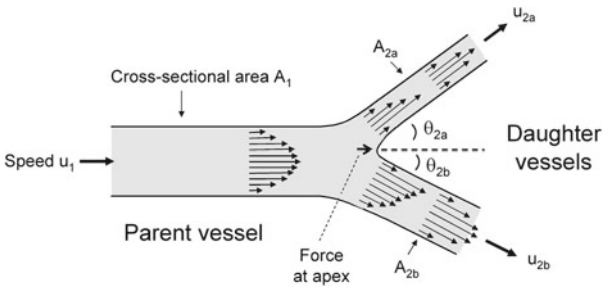


Fig. 8.46 Diagram of asymmetric bifurcation in vessel, with bifurcation angle $\theta_{2a} + \theta_{2b}$. For the symmetric bifurcation described in the text the "a" and "b" labels can be ignored, and the half-angle is called θ (See Problems 8.34–8.37)

The first three reasons also apply to saccular aneurysms, after replacing the Law of Laplace for a cylinder with that for a sphere (7.9).

Arterial Bifurcations and the Formation of Saccular Aneurysms

Two of the reasons for the formation of saccular aneurysms are the forces on the arterial walls caused by the change of momentum (like the fire hose effect in curving arteries) and shear stress. Figure 8.46 depicts a "parent" artery with cross-sectional area A_1 in which blood flows at an average speed u_1, which divides into two "daughter" vessels at an angle θ to the parent (and 2θ to each other). (The half-angle is shown as θ_2 in the figure.) Each daughter vessel has cross-sectional area A_2 and blood speed u_2. The angle 2θ usually ranges from 30 to 120° (also see Problems 8.34–8.37). The apex of the bifurcation (Fig. 8.46) is the site of maximum stress due to the impact, deflection, and separation of the flow, and possible turbulence and vortex formation (Fig. 8.47). Conservation of flow rate Q (volume flow/time) means

$$Q = u_1 A_1 = 2u_2 A_2. \tag{8.36}$$

In these vessels, usually $2A_2 > A_1$, so $u_2 < u_1$.

How large is the force on the arterial wall at the apex? The linear momentum per unit volume of blood in the parent artery (which we will say is in the x direction) is ρu_1, where ρ is the blood mass density. The momentum per unit volume carried in each daughter artery is ρu_2, of which $\rho u_2 \cos\theta$ is along the x direction. This change in momentum causes a force on the arterial wall. The force this flow exerts on an imaginary screen across the vessel in the parent artery is the change of this momentum per unit time, which equals this linear momentum per unit volume $\times$ the flow rate

$$F_{z,\text{parent}} = \rho u_1 Q. \tag{8.37}$$

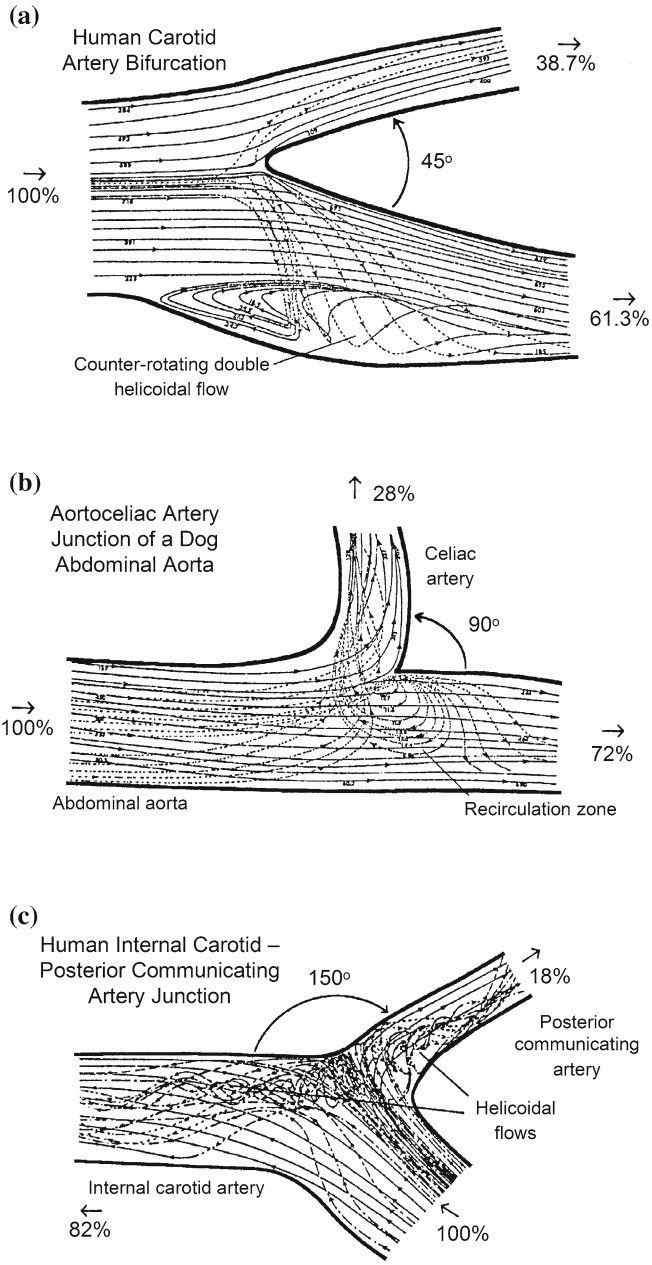


Fig. 8.47 Disturbed flow streamlines for progressively large angle bifurcations at bifurcations, and approach to steady flow afterward (From [21], based on **a** [54], **b** [38], and **c** [39]. Courtesy of Robert A. Freitas Jr., Nanomedicine, Vol. 1 (1999), http://www.nanomedicine.com)

Because the flow rate in each vessel is $Q/2$ and there are two of them, the force of the flow in the daughter arteries is

$$F_{z,\text{daughters}} = 2\,\frac{\rho u_2 \cos\theta\; Q}{2} = \rho u_2 \cos\theta\; Q. \tag{8.38}$$

The difference of these forces is

$$\begin{aligned} F_{\text{arterial wall}} &= F_{z,\text{daughters}} - F_{z,\text{parent}} \\ &= \rho u_2 \cos\theta\; Q - \rho u_1 Q = \rho Q(u_2 \cos\theta - u_1) \end{aligned} \tag{8.39}$$

or with (8.36)

$$F_{\text{arterial wall}} = \rho Q u_1 \left(\frac{A_1}{2A_2}\cos\theta - 1\right) = \rho A_1 u_1^2 \left(\frac{A_1}{2A_2}\cos\theta - 1\right). \tag{8.40}$$

This has a maximum negative value of $-\rho Q u_1 = -\rho A_1 u_2^2$ when $\theta = 90°$ (which is a bit larger than the typical maximum angles). Because pressure is force/area, if this force is exerted on the vessel wall of cross-sectional area A_1, as in Fig. 8.41, there is a pressure on the arterial wall of

$$P_{\text{arterial wall}} = \frac{\rho Q u_1}{A_1} = \frac{\rho A_1 u_1^2}{A_1} = \rho u_1^2 \tag{8.41}$$

(which is also the kinetic energy per unit volume). This can lead to the formation of an aneurysm there. This mechanism of momentum change is the same as that causing a force on an arterial wall during flow in a curving artery. The saccular aneurysm gets larger and larger until it ruptures.

The shear stress near the bifurcation apex due to viscosity is another factor in the formation of a saccular aneurysm. Figure 8.46 shows that in steady flow the velocity is small near the walls and the velocity gradient and consequently the shear stress is large near the walls. After bifurcation and momentum transfer to the apex, the blood velocity, velocity gradient, and shear are larger near the vessel walls after the apex. The shear stress is large, until the flow pattern rearranges to give (7.40) (Fig. 7.11). This is also seen in Fig. 8.47 for bifurcations at increasingly large angles. Equation (7.54) can be used to estimate the distance from the apex for steady-state flow. Note that the Law of Laplace for spheres applies to saccular aneurysms.

Equation of Motion of Arteries and Aneurysms During Pulsatile Flow (Advanced Topic)

For steady flow, the tension in the arterial wall balances the pressure difference inside and outside the artery, as described by the Law of Laplace. Because the pressure inside the artery really varies within every cycle (during the arterial pulse), this balance needs to be evaluated to account for these cyclic pressure variations. Let us

model a saccular aneurysm as a sphere of radius R, with a thin wall of thickness w. (We could similarly model a fusiform aneurysm as a cylinder of radius R.)

If the change in radius due to this time-varying pressure is r, then the circumference of the sphere changes from $2\pi R$ to $2\pi(R + r)$. The circumferential strain ϵ is $(2\pi(R + r) - 2\pi R)/2\pi R = r/R$, and the circumferential stress σ is related to this strain by

$$\sigma = Y\epsilon = Y\frac{r(t)}{R}. \tag{8.42}$$

The response of the arterial wall has a component that balances the average pressure P_{av}—which is a time-weighted average of the systolic and diastolic pressures for larger arteries—and a time varying component—due to the pulse $P_{\text{pulse}}(t) = P_{\text{p}} \cos \omega_{\text{p}} t$, where P_{p} is a time-weighted difference of the systolic and diastolic pressures for larger arteries and ω_{p} is the pulse (radial) frequency. This time-varying pressure is $P(t) = P_{\text{ss}} + P_{\text{p}} \cos \omega_{\text{p}} t$.

The inertial response force ma, per unit area on the sphere is the mass per unit area of the spherical shell, ρw, where ρ is the mass density, times the radial acceleration, $\mathrm{d}^2 r/\mathrm{d}t^2$, or $\rho w\, \mathrm{d}^2 r/\mathrm{d}t^2$. The hydrostatic pressure term $P(t)$ tends to increase the radius, while the stress due to the wall resists this change, so the inertial response is

$$\rho w \frac{\mathrm{d}^2 r_{\text{p}}(t)}{\mathrm{d}t^2} = P_{\text{av}} + P_{\text{p}} \cos \omega_{\text{p}} t - Y\frac{r(t)}{R} \tag{8.43}$$

because $\mathrm{d}^2 r_{\text{av}}/\mathrm{d}t^2 = 0$.

The perturbation $r(t)$ has a component in response to the average pressure P_{av} and a time-varying part in response to the pulse pressure $P_{\text{p}} \cos \omega_{\text{p}} t$. Substituting $r(t) = r_{\text{av}} + r_{\text{p}}(t)$ into (8.43), we get

$$\rho w \frac{\mathrm{d}^2 r_{\text{p}}(t)}{\mathrm{d}t^2} = P_{\text{av}} + P_{\text{p}} \cos \omega_{\text{p}} t - Y\frac{r_{\text{av}} + r_{\text{p}}(t)}{R}. \tag{8.44}$$

Equating the time-varying terms to give one equation and the average terms to give another leads to

$$\rho w \frac{\mathrm{d}^2 r_{\text{p}}(t)}{\mathrm{d}t^2} = P_{\text{p}} \cos \omega_{\text{p}} t - Y\frac{r_{\text{p}}(t)}{R} \tag{8.45}$$

and

$$0 = P_{\text{av}} - Y\frac{r_{\text{av}}}{R}. \tag{8.46}$$

Equation (8.46) is related to the Law of Laplace for a sphere (7.9) ($\Delta P = 2T/R$). Equation (8.45) can be rewritten as

$$\frac{\mathrm{d}^2 r_\mathrm{p}(t)}{\mathrm{d}t^2} + \omega_0^2 r_\mathrm{p}(t) = \frac{P_\mathrm{p}}{\rho w}\cos\omega_\mathrm{p} t, \tag{8.47}$$

where $\omega_0^2 = Y/R\rho w$. This looks like the equation of motion for a simple harmonic oscillator of frequency ω_0 plus an extra term (the last one), which is due to the pulse driving force at frequency ω_p. The steady-state solution to this equation is

$$r_\mathrm{p}(t) = \frac{P_\mathrm{p}/\rho w}{\omega_0^2 - \omega_\mathrm{p}^2}\cos(\omega_\mathrm{p} t), \tag{8.48}$$

which can be verified by substitution (see Appendix C). If r_p is large, it could lead to rupture. In principle, r_p can become larger as the aneurysm develops because Y, R, ρ, or w change (in $\omega_0^2 = Y/R\rho w$). Without the driving term ($P_\mathrm{p} = 0$), the solution is the usual harmonic solution: $r_\mathrm{p}(t) = A\cos(\omega_0 t + \phi)$.

If ω_0 were to approach ω_p, (8.48) indicates that the change in radius would become very large because of this resonance. If this were to occur (and it does in some examples of driven oscillators but it really does not for aneurysms), the viscous (or damping) properties of the vessel wall would have to be included through a term $\gamma\,\mathrm{d}r_\mathrm{p}/\mathrm{d}t$, leading to the new equation of motion

$$\frac{\mathrm{d}^2 r_\mathrm{p}(t)}{\mathrm{d}t^2} + \gamma\frac{\mathrm{d}r_\mathrm{p}(t)}{\mathrm{d}t} + \omega_0^2 r_\mathrm{p}(t) = \frac{P_\mathrm{p}}{\rho w}\cos\omega_\mathrm{p} t \tag{8.49}$$

with steady-state solution

$$r_\mathrm{p}(t) = \frac{\left(\omega_0^2 - \omega_\mathrm{p}^2\right) P_\mathrm{p}/\rho w}{\left(\omega_0^2 - \omega_\mathrm{p}^2\right)^2 + (\gamma\omega_\mathrm{p})^2}\cos(\omega_\mathrm{p} t). \tag{8.50}$$

(See Appendix C.) This now includes the viscoelastic properties of the arterial wall, which dampens the resonance a bit.

Far above the $\omega_0 \simeq \omega_\mathrm{p}$ resonance (with $\omega_0 \gg \omega_\mathrm{p}$), (8.48) and (8.50) give

$$r_\mathrm{p}(t) \simeq \frac{P_\mathrm{p}}{\rho w \omega_0^2}\cos(\omega_\mathrm{p} t). \tag{8.51}$$

Without the driving term ($P_\mathrm{p} = 0$), the solution is that of a damped harmonic oscillator: $r_\mathrm{p}(t) = A\exp(-\gamma t/2)\cos(\omega_0 t + \phi)$, for $\omega_0 \gg \gamma$. This harmonic oscillation damps in a time $\sim 1/\gamma$, which corresponds to about $\omega_0/(2\pi\gamma)$ cycles; this last number is often called the quality factor Q of the system, as is discussed in the Chap. 10 discussion of acoustic resonances and in Appendices C and D.

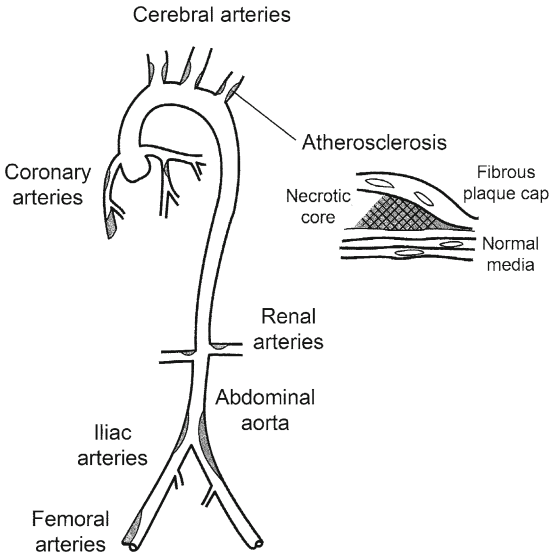


Fig. 8.48 Clogging of arteries by plaque. A schematic of the atherosclerosis is shown, along with some of the preferred sites of atherosclerosis in the vasculature (blood vessels) (From [36])

8.3.2 Ischemic Strokes

During an *ischemic stroke* there is cessation of blood flow in arteries transporting blood to the brain due to a luminal obstruction or clogging. (As alluded to earlier, the *lumen* is the opening of a blood vessel.) About 80% of all strokes are ischemic. An *embolus* is a gaseous (air bubble), particulate matter, or blood clot that travels within a blood vessel and causes the obstruction of blood flow. For example, artherosclerotic lesions (in the brain and elsewhere in the body) cause an irregular inner vessel surface and blood platelet aggregation due to turbulence, that can produce emboli that are platelet aggregates. Such emboli can be formed outside the brain, in the heart, lungs, and systemic circulation, and travel to the brain until they reach vessels too small for further travel; this prevents blood flow to more distal (downstream) regions in the brain. A *thrombus* is blood coagulation that can produce a local fibrin clot; this can also cause an ischemic stroke. (If the thrombus forms and moves elsewhere, such as to a smaller diameter vessel or a partially occluded vessel, it is an embolus.)

Strokes can occur by *stenosis* (narrowing or closure of lumens) or *occlusions* (closures or obstructions) (Fig. 8.48). They are most commonly due to artherosclerotic lesions. Such lesions are irregularly distributed masses of calcified fatty deposits that narrow the arterial lumen. If the normal inner diameter of the vessel is d_{norm} and the minimum diameter due to the stenotic lesion is d_{sten}, then the % stenosis is defined as: $((d_{\text{norm}} - d_{\text{sten}})/d_{\text{norm}}) \times 100\%$. They are characterized as being mild (1–39%), moderate (40–59%), severe (60–79%), critical (80–99%), and occluded (100%). Poiseuille's Law (7.24) shows that the flow decreases with decreasing lumen diameter for a given initial pressure. Flow decreases dramatically above the onset of critical stenosis, and the pressure drop across the stenosis increases, resulting in a need for a

greater blood pressure to maintain the same flow rate. The physics of clogged arteries was analyzed earlier in this chapter (Fig. 8.35).

This calcified lesion stiffens the vessel and abruptly changes the flow pattern as the blood flows from an elastic, distensible region of a vessel to this rigid and narrower region and back to a distensible vessel. Overall blood flow is slower because of the stenosis (unless the pressure increases), which can lead to clotting. The resulting clot (or thrombus) does not adhere well to the vessel wall and can move to elsewhere in the flow stream and this embolus can lead to a stroke. Within the stenosis itself, the blood flow is faster than just before it, from Bernoulli's Principle. This increased blood flow has several negative consequences. As this increased kinetic energy (per unit volume) in the stenotic region decreases in the poststenotic normal vessel (due to Bernoulli's Principle), there can be structural fatigue in this latter region leading to distention and this possible dilatation can lead to a fusiform aneurysm. The increased stenotic flow speed and irregular geometry can increase the Reynolds number, resulting in turbulent flow and the eventual disengaging of arterial plaque, which then becomes a particulate embolus. The increased speed also leads to higher shear stresses on the lesion.

8.4 Modeling the Circulatory System and the Heart

The branching in the circulatory system is very complex. Still, there is an orderly transition from larger to smaller arteries and then from smaller to larger veins, and so we can imagine an overall model of the circulation with all arteries or veins of a given diameter combining to form a subsystem. We have seen that larger vessels can be modeled quite well as ideal compliance vessels, while smaller vessels can be modeled as resistance vessels. We will now use these models to develop a comprehensive model of circulation, which can handle steady-state flow and changes that depend on time, like the arterial pulse. It can be used to understand the control of circulation. For a complete circulation model, we need to include the action of the heart, which is really two separate pumps: the right heart for the pulmonary system and the left heart for the systemic system. We first develop static and dynamic models of the left and right hearts.

8.4.1 Model of the Heart

Let us consider the left ventricle, which is the major pump in the left heart; the treatment of the right heart is analogous. During systole, the mitral valve is closed and the aortic valve is open. The pressure that develops is essentially that in the systemic arteries (sa) P_{sa} because of the very small pressure drop. During diastole the aortic valve is closed and the mitral valve is open. The left ventricle receives

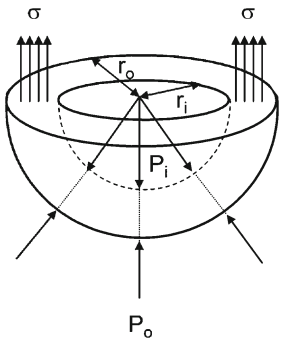


Fig. 8.49 Model of the left ventricle (Based on [50])

blood from the left atrium at a pressure that is pretty low, and is essentially equal to that in the pulmonary veins (pv), $P_{\text{pv}} \sim 5\,\text{mmHg}$, that feeds the left atrium.

Static Model of the Ventricles

We will model the left (or right) ventricle as a hemispherical shell, with an inner radius r_{i} and outer radius r_{o}, with a very thick wall of thickness $r_{\text{o}} - r_{\text{i}}$. Assume the open side is facing upward, as in Fig. 8.49. The (gauge) pressure inside P_{i} pushes the ventricle down and reaches a maximum during systole of 120 mmHg. The pressure outside (acting on the round bottom surface) P_{o} is from the pericardium and pushes the ventricle up and is approximately the pleural pressure, which can be negative (i.e., less than an atmosphere). The circumferential wall stress σ acts vertically and pushes the ventricle up. These arguments are the same as those for the Law of Laplace for a sphere (7.9), except we are now assuming the wall has finite thickness.

Force balance in the vertical direction gives

$$(\pi r_{\text{i}}^2) P_{\text{i}} = (\pi r_{\text{o}}^2) P_{\text{o}} + \left(\pi (r_{\text{o}}^2 - r_{\text{i}}^2)\right) \sigma \tag{8.52}$$

and so the circumferential wall stress is

$$\sigma = \frac{P_{\text{i}} r_{\text{i}}^2 - P_{\text{o}} r_{\text{o}}^2}{r_{\text{o}}^2 - r_{\text{i}}^2}. \tag{8.53}$$

The first two terms in (8.52) are not just the surface areas of the respective areas, $2\pi r_{\text{i,o}}^2$, times the hydrostatic pressure, $P_{\text{i,o}}$, but half of that, because only part of the force due to the pressure is in the vertical direction (as in the Law or Laplace for spheres and Problem 7.12). Because $P_{\text{i}} \gg P_{\text{o}}$

$$\sigma \simeq \frac{P_{\text{i}}}{(r_{\text{o}}/r_{\text{i}})^2 - 1}. \tag{8.54}$$

Because the material in the heart wall is incompressible, the volume of the heart wall V_{wall} does not change with pressure. From the difference of the volumes of

the outer and inner hemispherical shells, we know that $V_{\text{wall}} = 2\pi(r_{\text{o}}^3 - r_{\text{i}}^3)/3$ and so $(r_{\text{o}}/r_{\text{i}})^2 = (1 + V_{\text{wall}}/V_{\text{i}})^{2/3}$, where the inner volume of the left ventricle is $V_{\text{i}} = 2\pi r_{\text{i}}^3/3$. Therefore we see

$$\sigma \simeq \frac{P_{\text{i}}}{(1 + V_{\text{wall}}/V_{\text{i}})^{2/3} - 1}. \tag{8.55}$$

Using the expansion $(1+x)^n \simeq 1 + nx$ for $| x | \ll 1$ and the fact that the internal volume of the left ventricle is much larger than the volume of the heart wall, we find

$$\sigma \simeq \frac{3V_{\text{i}}}{2V_{\text{wall}}} P_{\text{i}}. \tag{8.56}$$

This shows how excessive systolic pressure or the enlargement of the left ventricle (and of the heart) will lead to excessive cardiac wall stress (for a constant wall thickness).

Dynamic Model of the Ventricles

We will now model a ventricle as a compliance vessel with a compliance that changes with time [34], so we use (8.5)

$$V(t) = V_{\text{d}} + C(t)P(t). \tag{8.57}$$

(We will simply call the flow resistance R_{flow} and compliance C_{flow}, respectively R and C in this section.) Fig. 8.50 shows how the compliance of the ventricle changes during a cardiac cycle. Note that in this and related plots a larger compliance corresponds to a smaller slope because P is plotted as a function of V. During systole the compliance becomes low, which causes a high pressure to develop because the volume remains pretty constant. In diastole the ventricle relaxes, which is associated with a large compliance that induces a low pressure.

The changes of volume and pressure in the ventricle during a cardiac cycle are modeled in Fig. 8.51. Stage A is when the inflow valve (which is the mitral valve for

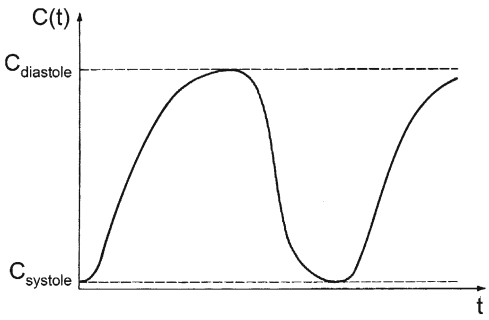


Fig. 8.50 Changes in ventricle compliance during the cardiac cycle, from a small value while it is contracting during systole to a large value when it is relaxing during diastole (From [34])

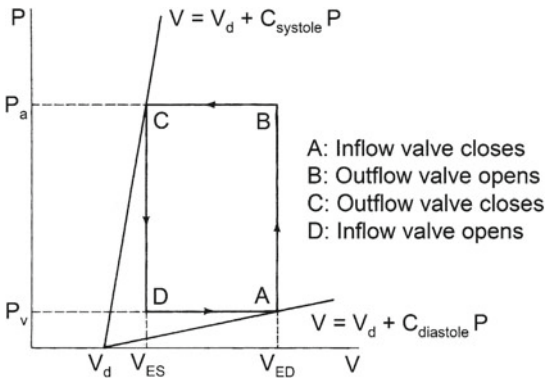


Fig. 8.51 Cycle of pressure and volume in either ventricle is given by the *ABCD rectangle* in this simplified model. The *slanting lines* radiating from the dead volume are the ventricle pressure–volume relationships at the end of systole and diastole. (Similar *slanting lines* with different slopes (not shown) characterize the ventricle at other times.) The venous pressure is the same as the inflow atrial and venous pressures during diastole. The arterial pressure is the outflow ventricle pressure during systole. For the *left ventricle* P_a is P_{sa} and P_v is P_{pv}; for the *right ventricle* P_a is P_{pa} and P_v is P_{sv} (From [34])

the LV) closes, which marks the end of diastole (ED) and the beginning of systole. The ventricle volume is a maximum, with

$$V_{ED} = V_d + C_{diastole} P_{pv} \tag{8.58}$$

for the left ventricle. During systole, the compliance decreases isovolumetrically, so the pressure increases. At Stage B, the outflow valve (which is the aortic valve for the LV) opens; the pressure remains constant as blood leaves the ventricle and concomitantly the volume of the ventricle decreases. Stage C is when the outflow valve closes, which marks the end of systole (ES) and the beginning of diastole. The ventricle volume is a minimum, with

$$V_{ES} = V_d + C_{systole} P_{sa} \tag{8.59}$$

for the left ventricle. During diastole, the compliance increases isovolumetrically, so the pressure decreases. At Stage D, the inflow valve opens; the pressure then remains constant as blood enters the ventricle and concomitantly the volume of the ventricle increases—until the end of diastole is reached, Stage A again.

The stroke volume is

$$V_{stroke} = V_{ED} - V_{ES} = C_{diastole} P_{pv} - C_{systole} P_{sa} \tag{8.60}$$

for the left ventricle. Because $C_{systole} \sim 0$, we can take

$$V_{\text{stroke}} = C_{\text{diastole}} P_{\text{pv}}. \tag{8.61}$$

With a heart rate F, the volumetric flow is

$$Q = F V_{\text{stroke}} = F C_{\text{diastole}} P_{\text{pv}}. \tag{8.62}$$

Calling $K = F C_{\text{diastole}}$ the pump coefficient, we can model the left ventricle by

$$Q_{\text{L}} = K_{\text{L}} P_{\text{pv}}. \tag{8.63}$$

Analogously for the right heart, the right ventricle, which is fed by the systemic veins (sv), is modeled by

$$Q_{\text{R}} = K_{\text{R}} P_{\text{sv}}. \tag{8.64}$$

8.4.2 Model of the Overall Flow in the Circulatory System

We will model the eight subsystems shown in Fig. 8.52, two hearts (L and R), large arteries in the systemic and pulmonary systems (sa and pa), large veins in the systemic and pulmonary systems (sv and pv), and the small vessels (arterioles/capillaries/venules) in the systemic and pulmonary systems [34, 40, 59]. Blood flows from the left ventricle successively through the systemic large arteries, small vessels, and large veins, and to the right ventricle. Then blood goes from the right ventricle successively through the pulmonary large arteries, small vessels, and large veins, and to the left ventricle.

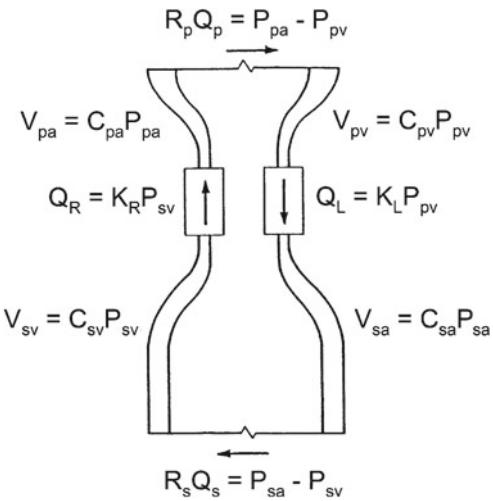


Fig. 8.52 The eight subsystems in systemic and pulmonary flow, with their model equations (From [34])

The flows through the ventricles are determined by the pressures in the veins feeding them (through the atria)

$$Q_{\mathrm{L}} = K_{\mathrm{L}} P_{\mathrm{pv}} \qquad \text{and} \qquad Q_{\mathrm{R}} = K_{\mathrm{R}} P_{\mathrm{sv}}. \tag{8.65}$$

The volumes in the large arteries are determined by their compliances and the pressures within them

$$V_{\mathrm{sa}} = C_{\mathrm{sa}} P_{\mathrm{sa}} \qquad \text{and} \qquad V_{\mathrm{pa}} = C_{\mathrm{pa}} P_{\mathrm{pa}}, \tag{8.66}$$

as are the volumes of the veins

$$V_{\mathrm{sv}} = C_{\mathrm{sv}} P_{\mathrm{sv}} \qquad \text{and} \qquad V_{\mathrm{pv}} = C_{\mathrm{pv}} P_{\mathrm{pv}}. \tag{8.67}$$

(A more refined model would include the dead volume in these large vessels.) The pressure drop across the small vessels is the difference in pressure between the large arteries and large veins. For the respective systemic and pulmonary systems, the pressure drop is determined by their resistances (R_{s} and R_{p}) and the flows through them (Q_{s} and Q_{p})

$$P_{\mathrm{sa}} - P_{\mathrm{sv}} = R_{\mathrm{s}} Q_{\mathrm{s}} \qquad \text{and} \qquad P_{\mathrm{pa}} - P_{\mathrm{pv}} = R_{\mathrm{p}} Q_{\mathrm{p}}. \tag{8.68}$$

We see a symmetry here. There are three types of variables, the flow Q, volume V, and pressure P. Each subsystem depends on relations between two of them: Q and P for the ventricles, V and P for the large vessels, and P and Q for the small vessels.

There are 12 unknowns: four flows: Q_{R}, Q_{L}, Q_{s}, Q_{p}; four pressures: P_{sa}, P_{sv}, P_{pa}, P_{pv}; and four volumes: V_{sa}, V_{sv}, V_{pa}, V_{pv}. We need 12 equations to solve for these 12 unknowns, but have only eight here and need four more. However, for steady-state flow, the flow in each region is the same, so $Q_{\mathrm{R}} = Q_{\mathrm{L}} = Q_{\mathrm{s}} = Q_{\mathrm{p}}$. These are really the three independent equations

$$Q_{\mathrm{R}} = Q_{\mathrm{L}}, \qquad Q_{\mathrm{L}} = Q_{\mathrm{s}}, \qquad \text{and} \qquad Q_{\mathrm{R}} = Q_{\mathrm{p}}, \tag{8.69}$$

with each flow rate equal to Q. The fourth equation describes the constant total volume and is

$$V_0 = V_{\mathrm{sa}} + V_{\mathrm{sv}} + V_{\mathrm{pa}} + V_{\mathrm{pv}}. \tag{8.70}$$

We can now solve these equations. From the pump (8.63) and (8.64), the venous pressures are

$$P_{\mathrm{sv}} = \frac{Q}{K_{\mathrm{R}}} \qquad \text{and} \qquad P_{\mathrm{pv}} = \frac{Q}{K_{\mathrm{L}}}. \tag{8.71}$$

Inserting these venous pressures into the resistance (8.68) gives

$$P_{\mathrm{sa}} = \frac{Q}{K_{\mathrm{R}}} + R_{\mathrm{s}} Q \qquad \text{and} \qquad P_{\mathrm{pa}} = \frac{Q}{K_{\mathrm{L}}} + R_{\mathrm{p}} Q. \tag{8.72}$$

Inserting these venous and arterial pressures into the compliance (8.66) and (8.67) gives

$$V_{\mathrm{sa}} = Q C_{\mathrm{sa}} \left(\frac{1}{K_{\mathrm{R}}} + R_{\mathrm{s}} \right) \qquad \text{and} \qquad V_{\mathrm{pa}} = Q C_{\mathrm{pa}} \left(\frac{1}{K_{\mathrm{L}}} + R_{\mathrm{p}} \right) \tag{8.73}$$

$$V_{\mathrm{sv}} = Q C_{\mathrm{sv}} \left(\frac{1}{K_{\mathrm{R}}} \right) \qquad \text{and} \qquad V_{\mathrm{pv}} = Q C_{\mathrm{pv}} \left(\frac{1}{K_{\mathrm{L}}} \right). \tag{8.74}$$

These can be expressed as

$$V_{\mathrm{i}} = T_{\mathrm{i}} Q \qquad \text{for i} = \text{sv, pv, sa, pa,} \tag{8.75}$$

with

$$T_{\mathrm{sa}} = C_{\mathrm{sa}} \left(\frac{1}{K_{\mathrm{R}}} + R_{\mathrm{s}} \right) \qquad \text{and} \qquad T_{\mathrm{pa}} = C_{\mathrm{pa}} \left(\frac{1}{K_{\mathrm{L}}} + R_{\mathrm{p}} \right) \tag{8.76}$$

$$T_{\mathrm{sv}} = C_{\mathrm{sv}} \left(\frac{1}{K_{\mathrm{R}}} \right) \qquad \text{and} \qquad T_{\mathrm{pv}} = C_{\mathrm{pv}} \left(\frac{1}{K_{\mathrm{L}}} \right). \tag{8.77}$$

Inserting these equations in (8.70) for the total blood volume gives

$$V_0 = V_{\mathrm{sa}} + V_{\mathrm{sv}} + V_{\mathrm{pa}} + V_{\mathrm{pv}} \tag{8.78}$$

$$= Q(T_{\mathrm{sa}} + T_{\mathrm{sv}} + T_{\mathrm{pa}} + T_{\mathrm{pv}}) \tag{8.79}$$

or

$$Q = \frac{V_0}{T_{\mathrm{sa}} + T_{\mathrm{sv}} + T_{\mathrm{pa}} + T_{\mathrm{pv}}}. \tag{8.80}$$

All the volumes are obtained from this and (8.75)

$$V_{\mathrm{i}} = \frac{T_{\mathrm{i}} V_0}{T_{\mathrm{sa}} + T_{\mathrm{sv}} + T_{\mathrm{pa}} + T_{\mathrm{pv}}} \tag{8.81}$$

and all of the pressures from $P_{\mathrm{i}} = V_{\mathrm{i}}/C_{\mathrm{i}}$ (from (8.66) and (8.67))

$$P_{\mathrm{i}} = \frac{1}{C_{\mathrm{i}}} \frac{T_{\mathrm{i}} V_0}{T_{\mathrm{sa}} + T_{\mathrm{sv}} + T_{\mathrm{pa}} + T_{\mathrm{pv}}}. \tag{8.82}$$

Table 8.8 Normal resting parameters of the circulation model

	Systemic system	Pulmonary system
Resistance, mmHg/(L/min)	$R_s = 17.5$	$R_p = 1.79$
Compliance, L/mmHg		
Arterial	$C_{sa} = 0.01$	$C_{pa} = 0.00667$
Venous	$C_{sv} = 1.75$	$C_{pv} = 0.08$
Heart	$K_L = 1.12$	$K_R = 2.8$
Total volume: $V_0 = 5.0$ L		

Using data from [34]

With known values of the flow rate and the total volume, the model parameters can be determined—see Table 8.8.

This model can be adapted to blood circulation in the developing fetus by including flows through the ductus arteriosus (the shunt from the pulmonary artery to the aorta, represented by the systemic arteries here), Q_d, and foramen ovale (the shunt to allow flow from the right atrium to the left), represented by flow from the right to left heart here), Q_f [34, 40]. (The fetal systemic system includes vessels that go through the umbilical cord to contact the placenta in this model.) Earlier, continuity of flow gave us $Q_R = Q_L = Q_s = Q_p$, which led to three independent equations for flow (8.69). There are still four flow equations (Problem 8.55):

$$Q_R = Q_p + Q_d \tag{8.83}$$

$$Q_d + Q_L = Q_s \tag{8.84}$$

$$Q_s = Q_R + Q_f \tag{8.85}$$

$$Q_p + Q_f = Q_L, \tag{8.86}$$

which are respectively for the pulmonary and systemic arteries and the systemic and pulmonary veins. Adding them gives an identity (Problem 8.56), so we still have only three independent flow equations. Because we added two new variables, we need two new equations. These are provided by modeling the ductus arteriosus as a resistance vessel with resistance R_d

$$P_{pa} - P_{sa} = R_d Q_d \tag{8.87}$$

and the foramen ovale as an ideal valve that is either open

$$Q_f \geq 0 \quad \text{and} \quad P_{sv} = P_{pv} \tag{8.88}$$

(the normal condition for the developing fetus) or closed

$$Q_f = 0 \quad \text{and} \quad P_{sv} \leq P_{pv} \tag{8.89}$$

(the normal condition for the baby right after birth).

We will outline key points of the overall the solution here and discuss the model in more detail in Problems 8.54–8.59. The flow equations can be rearranged to give

$$Q_s = Q_L + Q_d \tag{8.90}$$

$$Q_p = Q_R - Q_d \tag{8.91}$$

$$Q_f = Q_L - Q_R + Q_d. \tag{8.92}$$

The second equation shows that there must be flow through the ductus arteriosus to decrease flow in the pulmonary system (and the lungs). The third equation shows that there must be comparable flows through both shunts (and moreover that there must be two shunts and not one), because the left and right hearts in the developing fetus are about the same size and are designed to have approximately the same flows.

Before birth we can (1) assume the pulmonary resistance is very high to prevent blood flow through the fetal lungs (which are not involved in oxygenation of blood), and so $R_p > R_s$ (whereas after birth $R_p < R_s$), (2) set $K_R = K_L = K$ because the left and right hearts in the fetus are about the same size, (3) set $R_d = 0$ because the ductus arteriosus is wide open and therefore $P_{sa} = P_{pa} = P_a$, and (4) assume the foramen ovale is open, so $P_{sv} = P_{pv} = P_v$. The solution of the fetal circulation model with these assumptions confirms that $Q_f \geq 0$, so this assumption and the solution are self-consistent.

Right after birth and the first breath, the lungs of the baby expand and then $R_p < R_s$, so our assumption (1) is then no longer valid. Problem (8.57) shows that then our $Q_f \geq 0$ assumption (4) is also no longer valid, which means the foramen ovale is now closed. The model must be solved again with it assumed to be closed (which can be verified in the solution), keeping only assumptions (2) and (3) (Problem 8.59). This solution shows that after birth, flow in the ductus arteriosus reverses direction, with blood now flowing from the aorta (systemic arteries) to the pulmonary arteries, and that the foramen ovale is indeed closed. As the baby grows, both shunts remain closed for good in healthy babies and the left heart grows faster than does the right heart.

8.4.3 The Arterial Pulse

This model is capable of analyzing much more than just this idealized average flow. We can use it to understand the arterial pulse, which is the periodic deviation of the systemic arterial pressure from the diastolic value due to systole (Fig. 8.53). This pulse pressure is

$$P_{pulse} = P_{systole} - P_{diastole}, \tag{8.93}$$

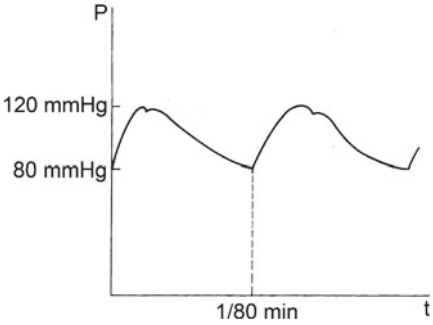


Fig. 8.53 The systemic arterial pulse. Also see the aortic pressure in Fig. 8.5 (From [34])

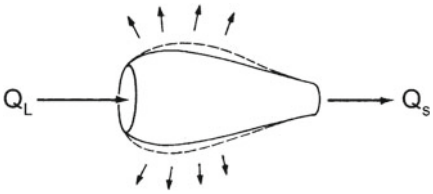


Fig. 8.54 During pulsatile flow, the change of arterial volume is equal to the volumetric inflow minus the outflow, as required by the conservation of volume (mass) (From [34])

which is ∼40 mmHg (= 120 mmHg −80 mmHg). These steady-state assumptions we have been making are no longer appropriate for this *pulsatile* flow.

Consider the systemic artery system, which has volume V_{sa} and pressure P_{sa}. As a compliance vessel its volume can change, and it will change when the flow into it does not equal to the flow leaving it (Fig. 8.54). The flow into it is that from the left heart Q_L and the flow out of it goes into the (noncompliant) small vessel system, which has flow Q_s. Conservation of volume gives

$$\frac{dV_{sa}(t)}{dt} = Q_L(t) - Q_s(t). \tag{8.94}$$

For this compliance vessel we know that

$$V_{sa}(t) = V_{sa,d} + C_{sa}P_{sa}(t), \tag{8.95}$$

where we have now included the dead volume. Differentiating this equation with respect to time and combining it with the previous equation gives

$$\frac{dV_{sa}(t)}{dt} = C_{sa}\frac{dP_{sa}(t)}{dt} = Q_L(t) - Q_s(t). \tag{8.96}$$

Using (8.68) gives

$$R_s Q_s = P_{sa} - P_{sv} \sim P_{sa} \tag{8.97}$$

since $P_{sv} \ll P_{sa}$. Therefore we arrive at

$$C_{sa}\frac{dP_{sa}(t)}{dt} = Q_L(t) - \frac{P_{sa}}{R_s}. \tag{8.98}$$

This determines the time dependence of the systemic artery pressure, and consequently the arterial pulse, if the flow rate out of the left ventricle is known.

We will assume that systole occurs very fast (for a very small fraction of the cardiac cycle of duration T, with $T = 1/F$) with a very large $Q_L(t)$ for $t \sim 0$, and that it is zero for the rest of the cardiac cycle. This idealized model is amenable to simple analysis. Although it is not very accurate, some of the features it predicts are significant and accurate.

When $Q_L(t) = 0$, (8.98) is

$$C_{sa}\frac{dP_{sa}(t)}{dt} = -\frac{P_{sa}}{R_s} \tag{8.99}$$

or

$$\frac{dP_{sa}(t)}{dt} = -\frac{P_{sa}}{R_s C_{sa}} = -\frac{P_{sa}}{\tau}, \tag{8.100}$$

where $\tau = R_s C_{sa}$. Therefore

$$P_{sa}(t) = P_{sa}(0)\exp(-t/\tau), \tag{8.101}$$

as is depicted in Fig. 8.55 (see Appendix C). In this model, at the start of the cardiac cycle $P_{sa}(0) = P_{systole}$ and at the end of the cardiac cycle $P_{sa}(T) = P_{diastole} = P_{systole}\exp(-T/\tau)$. The pressure of the arterial pulse is $P_{pulse} = P_{systole} - P_{diastole}$.

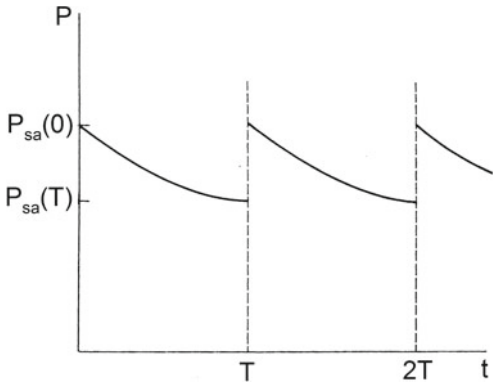


Fig. 8.55 Systemic arterial pulse from idealized model (From [34])

We can determine these pressures in terms of the stroke volume using (8.58)–(8.60) and (8.66)

$$V_{\mathrm{sa}}(0) = V_{\mathrm{sa,d}} + C_{\mathrm{sa}} P_{\mathrm{sa}}(0) \tag{8.102}$$

$$V_{\mathrm{sa}}(T) = V_{\mathrm{sa,d}} + C_{\mathrm{sa}} P_{\mathrm{sa}}(T). \tag{8.103}$$

The difference is the stroke volume

$$\begin{aligned} V_{\mathrm{stroke}} &= V_{\mathrm{sa}}(0) - V_{\mathrm{sa}}(T) = C_{\mathrm{sa}}(P_{\mathrm{sa}}(0) - P_{\mathrm{sa}}(T)) \\ &= C_{\mathrm{sa}} P_{\mathrm{sa}}(0)[1 - \exp(-T/\tau)]. \end{aligned} \tag{8.104}$$

Consequently, we find

$$P_{\mathrm{systole}} = P_{\mathrm{sa}}(0) = \frac{V_{\mathrm{stroke}}}{C_{\mathrm{sa}}[1 - \exp(-T/\tau)]} \tag{8.105}$$

$$P_{\mathrm{diastole}} = P_{\mathrm{sa}}(T) = \frac{V_{\mathrm{stroke}} \exp(-T/\tau)}{C_{\mathrm{sa}}[1 - \exp(-T/\tau)]}. \tag{8.106}$$

With (8.93) we see

$$P_{\mathrm{pulse}} = P_{\mathrm{systole}} - P_{\mathrm{diastole}} = \frac{V_{\mathrm{stroke}}}{C_{\mathrm{sa}}} \tag{8.107}$$

and so P_{systole}, P_{diastole}, V_{stroke}, the heart rate $1/T$, C_{sa}, and R_{s} $(= \tau/C_{\mathrm{sa}})$ are all interrelated.

Let us consider a numerical example. For $V_{\mathrm{stroke}} = 70\,\mathrm{cm}^3 = 0.070\,\mathrm{L}$ and $P_{\mathrm{pulse}} = 40\,\mathrm{mmHg}$, (8.104) gives $C_{\mathrm{sa}} = V_{\mathrm{stroke}}/P_{\mathrm{pulse}} = 0.00175\,\mathrm{L/mmHg}$. Using R_{s} from Table 8.8 of 17.5 mmHg/(L/min), we know that $\tau = R_{\mathrm{s}} C_{\mathrm{sa}} = 0.0306\,\mathrm{min}$. With $T = 1/80\,\mathrm{min} = 0.0125\,\mathrm{min}$, (8.105) and (8.106) give the last remaining parameter, which could be either P_{systole} or P_{diastole}, because they are related by $P_{\mathrm{pulse}} = P_{\mathrm{systole}} - P_{\mathrm{diastole}}$. From (8.107), we get $P_{\mathrm{systole}} = 120\,\mathrm{mmHg}$. As expected from self-consistency, $P_{\mathrm{diastole}} = P_{\mathrm{systole}} \exp(-T/\tau) = 120\,\mathrm{mmHg} \times \exp(-0.0125\,\mathrm{min}/(-0.0306\,\mathrm{min})) = 80\,\mathrm{mmHg}$.

A better model for the arterial pulse is described in Problem 8.62, in which the flow of blood from the left ventricle to the systemic arteries occurs with a finite (nonzero) duration in the cardiac cycle. It gives predictions that agree with Fig. 8.53 better than do those in Fig. 8.55. Even this improved model does not explain the small increase in pressure just after systole that is seen in Figs. 8.5 and 8.53, which is known as the "incisura" or "dicrotic notch." When the semilunar valve closes, some backward flowing blood bounces off the elastic aorta walls, setting up a pressure wave in the aorta; this pressure blip can be included with even more refinement in the model.

8.4.4 Windkessel Model

This simplified model of the arterial pulse is also known as the Windkessel Model [8], which was the first real model of blood flow. In this model blood flows from the left ventricle at a rate $Q_{\text{in}}(t)$ into an elastic chamber of compliance C (i.e., the larger arteries) of volume $V(t)$ and leaves it at a rate $Q_{\text{out}}(t)$ to enter a resistive element of resistance R_{p} (peripheral resistance of the arterioles and capillaries). It assumes that all pressure changes in the arteries occur at the same time. Because conservation of volume for an incompressible fluid gives

$$Q_{\text{in}}(t) = \frac{\mathrm{d}V(t)}{\mathrm{d}t} + Q_{\text{out}}(t), \tag{8.108}$$

with $C = \mathrm{d}V(t)/\mathrm{d}P(t)$ (8.57) and $Q_{\text{out}}(t) = P(t)/R_{\text{p}}$ (8.68), (8.108) becomes

$$Q_{\text{in}}(t) = C\frac{\mathrm{d}P(t)}{\mathrm{d}t} + \frac{P(t)}{R_{\text{p}}}. \tag{8.109}$$

The formal solution to this is

$$P(t) = \exp(-(t/\tau))\left(P(0) + \frac{1}{C}\int_0^t \exp(t'/\tau)Q_{\text{in}}(t')\mathrm{d}t'\right), \tag{8.110}$$

where $\tau = R_{\text{p}}C$. This can be shown to be the solution by substitution and by the method shown in Appendix C. Still it is simple and instructive to examine what happens after ventricular ejection, starting at $t = 0$ when $Q_{\text{in}}(t) = 0$ and $P = P(0)$, until the next cycle starts at $t = T$. Then

$$\frac{\mathrm{d}P(t)}{\mathrm{d}t} + \frac{P(t)}{R_{\text{p}}C} = 0 \qquad \text{for } 0 < t < T. \tag{8.111}$$

The solution is

$$P(t) = P(0)\exp(-(t/\tau)) \qquad \text{for } 0 < t < T, \tag{8.112}$$

which is the same as (8.101) (see Appendix C).

This two-element classic Windkessel Model is formally equivalent to the electrical circuit, where the electrical resistance R_{elect} maps into the flow resistance in the capillaries and the electrical capacitance C_{elect} maps into the flow compliance (Table D.1, Appendix D). (This is how the model is expressed in Fig. 8.56a.) The voltage V_{elect} in the electrical model corresponds to the pressure P in the flow model and the current I corresponds to the volumetric flow rate Q. Ohm's Law relates V_{elect} and I across the resistor by $V_{\text{elect}} = I R_{\text{elect}}$. The voltage across the capacitor is $V_{\text{elect}} = q/C_{\text{elect}}$, where the charge on the capacitor is q. The voltages across both

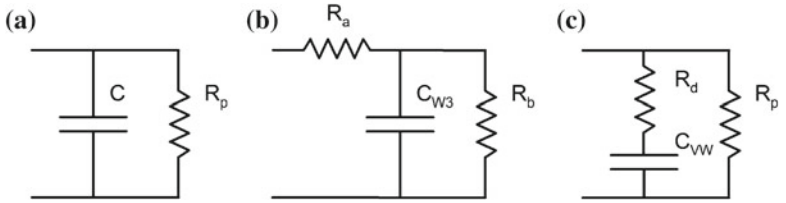


Fig. 8.56 Electrical analogs of the **a** classic or elastic Windkessel Model (W2), **b** three-element (or improved) Windkessel Model (W3), and **c** viscoelastic Windkessel Model (VW). The electrical resistances represent flow resistances and the electrical capacitances represent flow compliances (See Appendix D)

elements sum to zero (Kirchhoff's 2nd Law, Chap. 12), so $IR_{\text{elect}} + q/C_{\text{elect}} = 0$. Because $I = \mathrm{d}q/\mathrm{d}t$, we have $(\mathrm{d}q/\mathrm{d}t)R_{\text{elect}} + q/C_{\text{elect}} = 0$ or with $V_{\text{elect}} = q/C_{\text{elect}}$

$$\frac{\mathrm{d}V_{\text{elect}}}{\mathrm{d}t} + \frac{V_{\text{elect}}}{R_{\text{elect}}C_{\text{elect}}} = 0. \tag{8.113}$$

The two-element Windkessel Model can be refined by adding more elements to the electrical analog. The three-element Windkessel Model shown in Fig. 8.56b predicts a more realistic arterial pulse. The vascular resistance R_a of the aorta is added in series, and the value of the resistance in parallel is now R_b, which equals $R_p - R_a$. The viscoelastic Windkessel Model (Fig. 8.56c) represents a different type of improvement of the two-element Windkessel Model. The compliant arterial systems are represented by a capacitance (in the electrical analog of flow compliance) that is in series with a resistor R_d, which represents the viscous wall motion to more fully represent the viscoelasticity of the aorta.

8.4.5 Modeling the Malfunctioning Heart

There are many reasons and ways a heart can malfunction [46]. For example, single mutations that truncate the sarcomere protein titin (Chap. 5) can lead to changes in heart size, shape and function (which is also known as cardiac remodeling), and can cause early heart failure and death, including by dilated cardiomyopathy [33]. These conditions can be characterized quantitatively and, in principle, can be included in our models of the heart and circulation. Cardiac performance can be characterized by several parameters, such as the stroke volume V_{stroke} and cardiac output $Q_t = FV_{\text{stroke}}$, described earlier.

Figure 8.57 shows cardiac performance as a function of the conditions at the end of diastole in the left ventricle. The middle curve describes the operation of that person's heart, and point *a* represents that normal person at rest. In systolic heart failure, such as after a myocardial infarction, the curve shifts down due to lessened

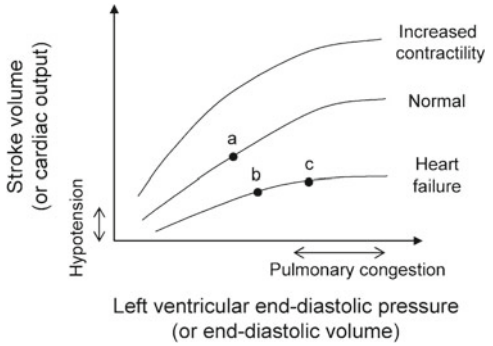


Fig. 8.57 Left ventricular performance (Frank–Starling) curves relate cardiac performance to preconditions (preloading), such as the pressure or volume of the left ventricle after diastole (Based on [18])

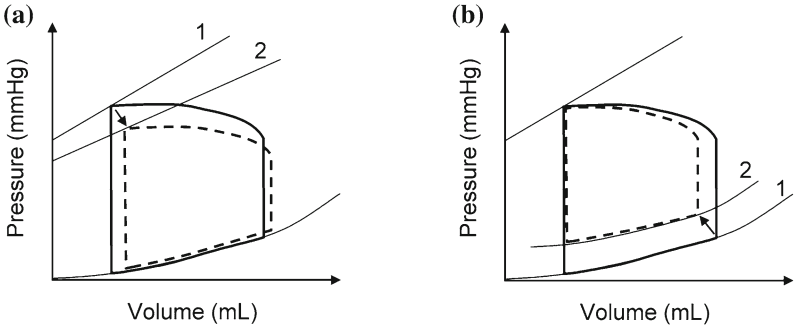


Fig. 8.58 Pressure–volume curves for the left ventricle for normal performance (*solid loop*), systolic dysfunction (*dashed loop* in (**a**)), and diastolic dysfunction (*dashed loop* in (**b**)) (Based on [18])

left ventricle contractility. Point *b* represents the person described by point *a* after heart failure. Increased circulatory volume in this person is represented by point *c*. The stroke volume is increased due to increased contractility in the uppermost curve, perhaps caused by the infusion of the drug norepinephrine.

The solid line pressure–volume loops in Fig. 8.58 represent normal heart function, and are more accurate representations than the simplified rectangular one depicted in our model in Fig. 8.51. In systolic dysfunction, the pressure-volume curve at the end of systole (also known as the "end systolic pressure-volume relationship", ESPVR) shifts from 1 to 2 in (a) due to decreased cardiac contractility. As seen in the dashed loop, the volume at the end of systole increases. In diastolic dysfunction, the pressure-volume curve at the end of diastole ("end diastolic pressure-volume relationship", EDPVR) shifts from 1 to 2 in (b) due to increased stiffness (decreased compliance). As seen by the dashed loop, the ventricular pressure is higher than normal at any diastolic volume. With advancing age, there is little change in systolic function, but there are major changes in diastolic function. There is reduced early diastolic filling,

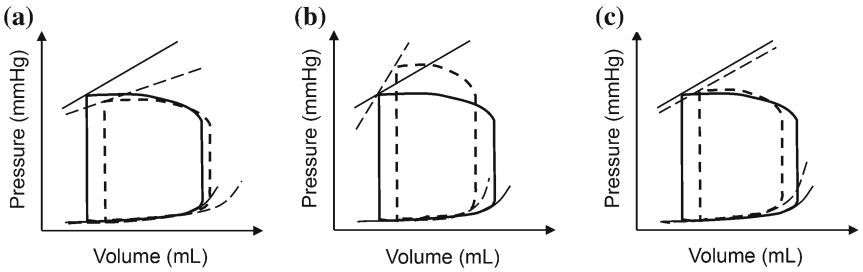


Fig. 8.59 Pressure-volume curves for the left ventricle for normal performance (*solid loop*) and for several heart abnormalities (*dashed loop*). The pressure-volume curves at the end of systole (*upper*) and diastole (*lower*) are shown in (**a**–**c**) for the normal (*solid curve*) and diseased (*dashed curve*) heart. In dilated cardiomyopathy (DCM) (**a**), the ESPVR shifts to the *right* and has decreased slope, while the EDPVR shifts to the *right*. In hypertrophic cardiomyopathy (HCM) (**b**), the ESPVR shifts to the *left* and has increased slope, while the EDPVR shifts to the *left*. In restrictive cardiomyopathy (RCM) (**c**), the ESPVR shifts to the right and has an unchanged slope, while the EDPVR has increased slope (Based on [35])

which is compensated for by increased filling at the end of diastole. This can make people more likely to develop congestive heart failure [1, 19].

The dashed pressure-volume loops for several cardiac disorders are shown in Fig. 8.59. The three cardiomypathies described in (a–c) are characterized by changes in the pressure-volume curves at the end of systole and diastole, as in the plots of Fig. 8.58. In dilated cardiomyopathy (DCM), the left ventricle becomes dilated, without necessarily becoming thicker. As seen in (a), the end systolic and diastolic ventricular volumes are larger than those of a normal heart, with the end systolic volume increasing more, leading to a net decrease in stroke volume and the pressures changing little. The Law of Laplace shows that a larger tension is needed in the ventricle wall to maintain the same pressure for blood ejection because the ventricle radius is larger in DCM, so someone with DCM uses more power to pump the same volume of blood [40]. In hypertrophic cardiomyopathy (HCM) (a cause of left ventricular hypertrophy), the mass of the left ventricle increases, sometimes along with its size. As seen in (b), this leads to a shift of the end-of systole curve and decreased systolic compliance, which cause the pressure to increase and decreased diastolic compliance at larger volumes. The stroke volume decreases because the volume at the end of diastole decreases and that after systole increases. In restrictive cardiomyopathy (RCM), the left ventricle stiffens, but does not necessarily become thicker, and this is sometimes due to scar tissue. As seen in (c), the ventricular systolic pressure remains normal, the diastolic pressure increases, and the stroke volume decreases. (Such cardiomyopathies have been responsible for the early deaths of several high-profile athletes.)

The effects of narrowed aortas (aortic stenosis) and mitral valve orifices (mitral stenosis) are shown in Fig. 8.60. In aortic stenosis, the left ventricular pressure is much greater that the aortic pressure during left ventricular ejection. The volume after systole and the volume and pressure after diastole increase and the stroke volume

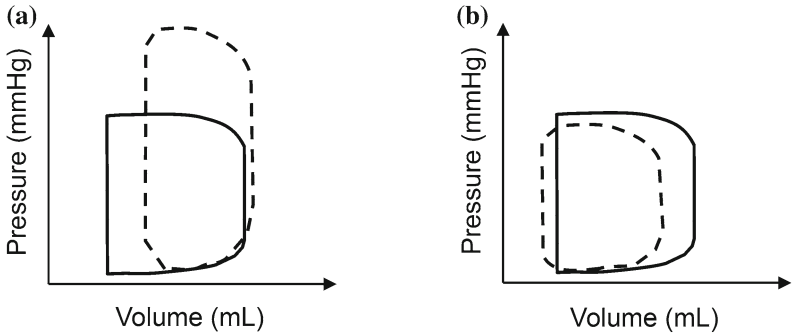


Fig. 8.60 Pressure-volume curves for the left ventricle for normal performance (*solid loop*) and for the valve diseases **a** aortic and **b** mitral stenosis (*dashed loop*) (Based on [35])

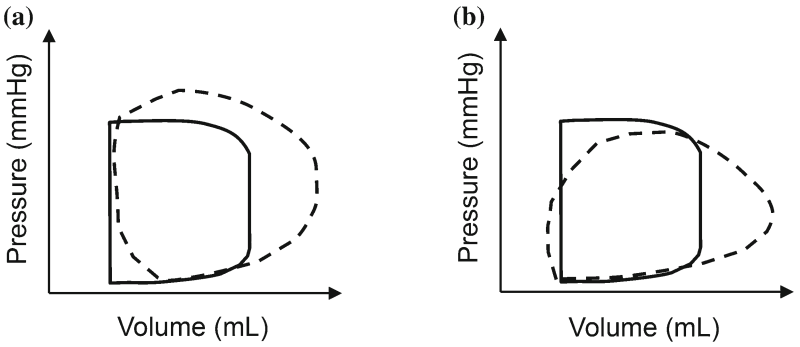


Fig. 8.61 Pressure-volume curves for the left ventricle for normal performance (*solid loop*) and for **a** aortic and **b** mitral regurgitation (*dashed loop*) (Based on [35])

decreases. In mitral stenosis, there is reduced diastolic filling, which leads to decreased stroke volume and aortic pressure.

Several series of problems occur when there is leakage between heart chambers. In valvular regurgitation there is backflow through one of the heart valves, such as aortic regurgitation and mitral regurgitation in the left heart valves (Fig. 8.61). The regurgitant fraction for mitral regurgitation is the volume of blood flowing back through the mitral valve divided by the left ventricle stroke volume. There can also be openings in the septum between the atria or the ventricles, called a septal defect [12]. For example, a ventricular septal defect is a congenital condition in which there is an opening between the left and right ventricles (Fig. 8.62). During systole blood flows from the left to right ventricle because of the higher pressure in the former. This leads to increased blood return to the left side of the heart which causes the left atrium and ventricle to enlarge. The ejection fraction (EF) is the stroke volume divided by the ventricular volume at the end of diastole (when the ventricle is full). It is normally 55–75%, but can be less with a septal defect.

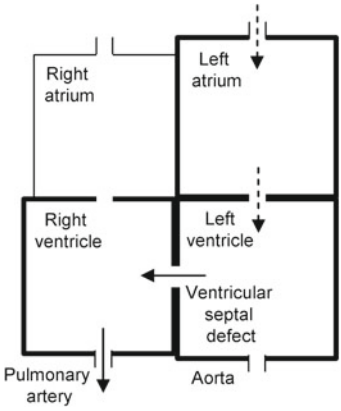


Fig. 8.62 With a ventricular septal valve defect blood flows from the left to right ventricle (Based on [12])

8.5 Summary

The flow of blood in the vessels of the circulation system can be modeled using the resistive and compliant nature of the flow. This can be combined with a model of the heart to understand the flow in the entire circulation system. These models can also be used to understand the physical consequences of a malfunctioning heart and problems in the circulation system, such as clogged and weakened arteries, that can lead to strokes and aneurysms. The energy needed to operate the heart can also be modeled.

Problems

Blood Pressure

8.1 (a) The brain in a human is 55 cm above the heart. If the average blood pressure in the major arteries near the heart is 100 mmHg, what is the blood pressure in major arteries in the brain (in both mmHg and cmH_2O) when a person is either lying down or standing up?
(b) Repeat part (a) for an erect human on the moon ($g = g_{\text{Earth}}/6$) and on Jupiter ($g = 2.34 g_{\text{Earth}}$).
(c) A pilot coming out of a dive experiences an upward centripetal acceleration a of magnitude v^2/r, where v is the speed of the jet and r is the radius of curvature of the trajectory, that adds to gravity (effectively increasing g to $g + a$). What is the arterial pressure in the pilot's brain for $v = 200$ m/s and $r = 2$ km? What could happen to the pilot during this recovery from the dive? Would you expect dizziness because of a lack of blood to the head? (See Fig. 8.63.)
(d) What must the pressure in the aorta in a giraffe be (on Earth) for its brain to receive blood? (How can you estimate the elevation of its brain above its aorta?)

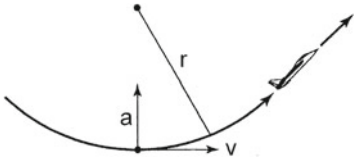


Fig. 8.63 Trajectory of a pilot coming out of a dive, with the acceleration and velocity vectors shown for the low point of the dive. For Problem 8.1 (From [4])

8.2 In Problem 8.1d we saw that the pressure of the blood leaving the heart of a giraffe and entering its systemic system must be much larger than that for humans because of its long neck.
(a) Would a blood pressure of 280 mmHg/180 mmHg account for pumping the blood up this long neck in large arteries and then for the pressure drop that occurs in the very small arteries in the brain?
(b) The giraffe has this relatively high blood pressure because of this long neck, which is usually nearly vertical. However, we could expect that there would be a rush of blood to the brain because of this high pressure when the giraffe lowers its head by almost 7 m when it bends down to drink water, and that this could lead to rupture of the arteries in the brain. Why?
(c) Such artery rupturing does not occur because of compensating effects. Explain this by considering the following: (i) The elastic walls of the long giraffe carotid artery help force blood upward (which is a peristaltic action), and this also means that this artery can swell to absorb excess blood when the head is lowered because it is very compliant. (ii) The giraffe jugular vein contains a series of one-way valves that prevent back-flow of the blood when the giraffe's head is down.

8.3 When you stand on your head, why does your head become red and why do your legs become pale?

8.4 Why is blood pressure measured using major arteries in the upper arm, rather than those in the lower arm or leg?

8.5 You are told that your blood pressure is 880 mmHg/840 mmHg. You are quite understandably concerned because these values are astronomically high, but you are told not to worry because your blood pressure is normal. Should you be concerned?

8.6 (a) An intravenous infusion is made under gravity. If the fluid to be delivered has a density of 1.0 g/cm^3, at what height above the vein, h, should the top surface of the fluid in the bottle be positioned so the fluid just barely enters the vein? The gauge pressure in the vein is 18 mmHg. (Assume the needle entering the vein has a "large" inside diameter.) (See Fig. 8.64.)
(b) If this needle has a "very small" inside diameter, should the bottle be placed higher, lower, or at the same height? Why?
(c) Why are such infusions performed intravenously and not intra-arterially?

8.7 You are lying down and are injured in such a way that blood from a major artery squirts upward. How high can it spurt?

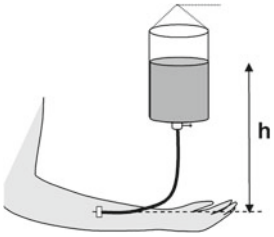


Fig. 8.64 Intravenous infusion under gravity. (Based on [26].) For Problem 8.6

8.8 Twirl one arm around as fast as you can many times until you see your fingers on that arm turn red. (Continue even if you do not see them getting redder.) [2]
(a) Estimate the centrifugal acceleration at the end of your finger tips, $v_{\text{radial}}^2/r = v_{\text{radial}}^2/l_{\text{arm}}$, where l_{arm} is your arm length (to your fingertips) and v_{radial} is the radial speed of your finger tips. (Why is $v_{\text{radial}} = 2\pi l_{\text{arm}}/T$, where T is the period for a complete cycle of this motion?)
(b) Express this acceleration in units of g.
(c) Calculate the effective pressure pushing your blood to your fingers by this motion. This is the apparent outward force divided by the cross-sectional area of your fingers, A_{fingers}, or $(m_{\text{fingers}} v_{\text{radial}}^2/l_{\text{arm}})/A_{\text{fingers}}$. Because $m_{\text{fingers}} = \rho_{\text{fingers}} l_{\text{fingers}} A_{\text{fingers}}$, this pressure is $\rho_{\text{fingers}} v_{\text{radial}}^2 (l_{\text{fingers}}/l_{\text{arm}})$.
(d) Compare this to the systolic pressure 120 mmHg and explain why your fingers (could have or should have) turned red.

Blood Pressure Drop During Flow

8.9 For smaller arteries, how much larger is the total cross-sectional area of the two daughter vessels than the parent?

8.10 Calculate the pressure drop (in mmHg) across the following arterial systems using Poiseuille's Law with $\eta_{\text{blood}} = 4 \times 10^{-3}$ Pa-s, for a total flow of 80 cm^3/s across each system:
(a) aorta (internal radius $r = 1.25$ cm, length $L = 10$ cm, all of the flow in this one aorta)
(b) large arteries ($r = 0.2$ cm, $L = 75$ cm, $n = 200$ of them, each with equal flow and the same dimensions)
(c) arterioles ($r = 30\,\mu$m, $L = 0.6$ cm, $n = 5 \times 10^5$)
(d) capillaries ($r = 3.5\,\mu$m, $L = 2$ mm, $n = 10^{10}$).

8.11 In estimating pressure drops across the different arterial branches we assumed specific numbers of arteries of given diameters and lengths. There is really a wide range of arterial diameters and lengths. How does this affect the pressure drops in the systemic arterial system?

8.12 In estimating pressure drops across the arterioles we assumed a specific number of arterioles with the same diameter and length.

(a) Let us say that all arterioles have the same radius, but their lengths (instead of all being L) range between $0.8L$ and $1.2L$ (with equal probability throughout). How does this change the overall resistance of the arteriole system?
(b) Let us say that all arterioles have the same length, but their radii (instead of all being r) range between $0.8r$ and $1.2r$ (with equal probability throughout). How does this change the overall resistance of the arteriole system?
(c) The pressure drop across each arteriole in the system must be the same because each is fed by the large arteries, whose pressure is set by the left ventricle, and by the arterial side of the capillaries, whose pressure is also set. If the overall pressure drop across the arterioles is unchanged (by the changes in (a) or (b)), how is the overall flow rate in the arteriole system changed and what is the flow in each arteriole, for the situations alternately described in (a) and (b)?
(d) If you wanted the flow rate to stay the same in each arteriole in (a) and (b) for the given distributions of lengths and radii, how would you have to change the distributions of radii and lengths in each, respectively?

8.13 Find the pressure drop across the arterioles in Problem 8.10c, if—with the same total flow in both cases—and either
(a) all the arterioles become clogged in such a way that their radii decrease to 28 μm or
(b) the number of the arterioles decreases to 4×10^5.
(c) By how much would the pressure in the main arteries need to change if the body responded to either change by maintaining the same flow rate?

8.14 Assume that the diameter of each blood vessel in a person is doubled and the total volumetric flow rate is not changed.
(a) What is the new total volume of blood? (Assume the base line parameters in the chapter.)
(b) What is the new circulation time for blood (which is the total blood volume/total volumetric flow rate)?
(c) How do the resistances of the arterioles and capillaries change?
(d) How do the pressure drops across the arterioles and capillaries change?
(e) How does the work done by the heart change?

8.15 Repeat Problem 8.14 if instead the length of each blood vessel is doubled.

8.16 The length of a blood vessel is doubled and its diameter is doubled.
(a) How does the flow resistance change?
(b) If the flow through it is unchanged, how does the pressure drop change?
(c) How does the flow through it change if instead the pressure drop is unchanged?

8.17 Your internal body temperature increases from 37 to 40 °C. Assuming that the only thing that changes is the viscosity of blood, how must the blood pressure change to ensure the flow rate remains unchanged?

8.18 Use (8.20) to relate $P(L)$ to $P(0)$ and other flow terms.

8.19 (a) Estimate how much the flow is changed in small arteries by including the influence of compliance by using (8.20) and assuming the same pressure drop.
(b) Repeat this estimate for how much the pressure drop changes with this analysis assuming the same flow rate.

8.20 Express the flow resistance units of N-s/m^5 in terms of (N/m^2)/(cm^3/s), dyne-s/cm^5, and PRU (with 1 PRU = 1 mmHg-s/mL).

8.21 (a) Calculate the total peripheral vascular resistance in the systemic and pulmonary systems for someone with a steady-state blood flow rate of 5 L/min, and with 120 mmHg/80 mmHg blood pressure in the systemic system and 25 mmHg/8 mmHg pressure in the pulmonary system. Express your answer in the units of dyne-s/cm^5, which are CGS units and those that are often used by cardiologists.
(b) The expression given in the text for this vascular flow resistance should be corrected because it uses the average pressure at the beginning of the system instead of the pressure drop across the system. For the systemic system the final pressure is that at the right atrium (2 mmHg) and for the pulmonary system it is that at the left atrium (5 mmHg). How does using the actual pressure drop affect your calculations in part (a) (both qualitatively and quantitatively)?

8.22 (a) The pulmonary vascular resistance changes with lung volume. Figure 8.65 shows that it increases much with larger lung volumes, in part because the larger alveoli stretch the pulmonary capillaries. It also increases at very small lung volumes because these capillaries surrounding the alveoli become narrow. Calculate the range of pulmonary blood flow rates (in L/min), assuming this range of resistances and assuming that the pulmonary pressures are the same as in Problem 8.21.
(b) The pulmonary pressures actually change with lung volume. If they changed in a manner to keep the average flow the same as it is for a 110 mL lung volume, determine this change. (For the purpose of this calculation, assume that the ratio of systolic and diastolic pressures remains a constant and that the left atrium pressure remains the same.)

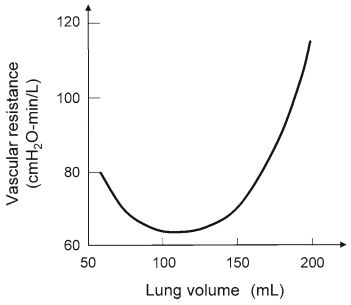


Fig. 8.65 Effect of lung volume on pulmonary vascular (blood flow) resistance (From an animal lobe preparation). (Based on [69].) For Problem 8.22

8.23 (a) Determine the overall compliances of the systemic arterial and venous systems by using Fig. 8.25.
(b) Is the ratio of these two compliances reasonable, given our model for compliance and the data for the vessels in both groups? (Consider only the large vessels in both groups.)

8.24 Does Fig. 8.25 suggest that sympathetic stimulation and inhibition change the vessels' compliances, dead volumes, or both?

8.25 Relate the distensibility (introduced in Chap. 7) to C_{flow}, c_{flow} and to C'_{flow}.

8.26 Show that the correction term to (8.19) is $(c_{\text{flow}}^2/3A_{\text{d}}^2)((P(0))^2 + P(0)P(L) + (P(L))^2)$.

8.27 (a) Show that (8.20) can be obtained from (8.19).
(b) Show that it can also be derived by integration of (8.14), expressed in r, using (8.7).

8.28 (a) Show that the correction term to (8.20), obtained by integrating (8.14) expressed in r and using (8.7), is $(1/2)(C'\text{flow}^2/r_{\text{d}}^2)((P(0))^2 + P(0)P(L) + (P(L))^2)$.
(b) Show that when the correction determined from the area approach (Problem 8.26) is expressed in terms of r, it becomes $(1/3)(C'\text{flow}^2/r_{\text{d}}^2)((P(0))^2 + P(0)P(L) + (P(L))^2)$.
(c) Why does this differ from the result in (a)?

Flow and Pressure

8.29 What is the average time blood spends in a capillary?

8.30 The *cardiac index* is the cardiac output divided by the person's surface area. It normally ranges from 2.6 to 4.2 (L/min)/m^2. Use this to determine the cardiac output of a standard human. How does this value compare to the normal cardiac output we have assumed?

8.31 An artery with radius r_1 and blood speed u_1 divides into n arteries of equal radius. Find the radius r_2 and blood speed u_2 in these daughter arteries assuming that the pressure drop per unit distance $\text{d}P/\text{d}x$ is the same in the initial artery and each daughter artery.

8.32 Four veins with radius r_1 and flow speed u_1 combine to form one vein with radius $r_2 = 4r_1$. Find the flow speed in the larger vein.

8.33 The radius of each section of the aorta and the external iliac, femoral, subclavian, and brachial arteries each tapers as $\exp(-kx)$, where k is the tapering factor and x the distance along that vessel. Use Table 8.4 to find the tapering factor k for each vessel [57].

8.34 The design of blood vessels is sometimes optimized by minimizing a "cost function," F, which is the sum of the rate work is done on the blood in the vessel and the rate that energy is used by metabolism through the blood in the vessel [24]. The first term is $Q(\Delta P)$, for flow rate Q and pressure drop ΔP, and the second term is assumed to be proportional to the volume of the vessel of radius r and length L, $K\pi r^2 L$, where K is a constant. Consequently, the cost function can be written as

$$F = Q(\Delta P) + K\pi r^2 L. \tag{8.114}$$

(a) For a resistive vessel, show that this becomes

$$F = \frac{8\eta L}{\pi r^4}\, Q^2 + K\pi r^2 L. \tag{8.115}$$

(b) For a fixed vessel length and flow rate, show the optimal radius is

$$r_{\text{opt}} = \left(\frac{16\eta}{\pi^2 K}\right)^{1/6} Q^{1/3} \tag{8.116}$$

and the minimum cost function is

$$F_{\text{min}} = \frac{3\pi}{2} K L r_{\text{opt}}^2. \tag{8.117}$$

8.35 (advanced problem) We will use the cost function defined in Problem 8.34 to optimize the flow in a vessel of radius r_0 and length L_0 with flow rate Q_0, that bifurcates into a vessel of radius r_1 and length L_1 with flow rate Q_1 at an angle θ to the first vessel and one with radius r_2 and length L_2 with flow rate Q_2 at an angle ϕ to the first vessel, as seen in Fig. 8.66a [24]. Having straight, coplanar vessels minimizes the vessel lengths.
(a) Show that the total cost function is

$$F_{\text{min}} = \frac{3\pi K}{2}(r_0^2 L_0 + r_1^2 L_1 + r_2^2 L_2). \tag{8.118}$$

(b) We can optimize the lengths and angles of the vessels by considering how the displacements of point B in Fig. 8.66b–d change the cost function. Show that any such movement of point B causes length changes δL_0, δL_1, and δL_2 that lead to a change of the cost function of

$$\delta F_{\text{min}} = \frac{3\pi K}{2}(r_0^2(\delta L_0) + r_1^2(\delta L_1) + r_2^2(\delta L_2)). \tag{8.119}$$

This is optimized by setting $\delta F_{\text{min,opt}} = 0$.

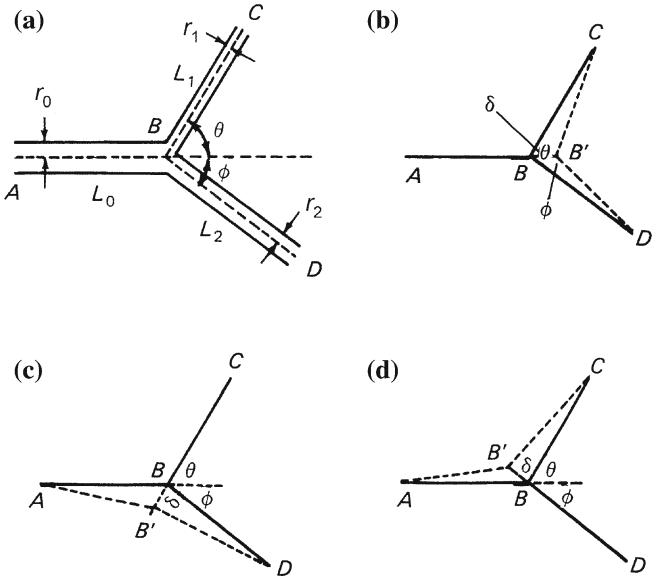


Fig. 8.66 **a** A planar, bifurcating vessel. Determining variations in the length of each vessel for small planar displacements of B to B' in the **b** AB, **c** AC, and **d** DB directions (Based on [24].) For Problem 8.35

(c) Show that moving point B along AB to B' by a distance δ as shown in Fig. 8.66b gives $\delta L_0 = \delta$, $\delta L_1 = -\delta \cos\theta$, and $\delta L_2 = -\delta \cos\phi$, and $\delta F_{\text{min,opt}} = (3\pi K\delta/2)(r_0^2 - r_1^2 \cos\theta - r_2^2 \cos\phi) = 0$, and so it is optimized by

$$r_0^2 = r_1^2 \cos\theta + r_2^2 \cos\phi. \tag{8.120}$$

(d) Show that moving point B along CB to B' by a distance δ as shown in Fig. 8.66c gives $\delta L_0 = -\delta \cos\theta$, $\delta L_1 = \delta$, and $\delta L_2 = \delta \cos(\theta + \phi)$, and $\delta F_{\text{min,opt}} = (3\pi K\delta/2)(-r_0^2 \cos\theta + r_1^2 + r_2^2 \cos(\theta + \phi)) = 0$, and so it is optimized by

$$-r_0^2 \cos\theta + r_1^2 + r_2^2 \cos(\theta + \phi) = 0. \tag{8.121}$$

(e) Show that moving point B along DB to B' by a distance δ as shown in Fig. 8.66d gives the optimization condition

$$-r_0^2 \cos\phi + r_1^2 \cos(\theta + \phi) + r_2^2 = 0. \tag{8.122}$$

(Note the symmetry in the last two equations.)

(f) Show that (8.120)–(8.122) can be solved to give

$$\cos\theta = \frac{r_0^4 + r_1^4 - r_2^4}{2r_0^2 r_1^2}, \tag{8.123}$$

$$\cos\phi = \frac{r_0^4 - r_1^4 + r_2^4}{2r_0^2 r_2^2}, \tag{8.124}$$

and

$$\cos(\theta + \phi) = \frac{r_0^4 - r_1^4 - r_2^4}{2r_1^2 r_2^2}. \tag{8.125}$$

(g) Use continuity of flow and (8.116) to show that

$$r_0^3 = r_1^3 + r_2^3. \tag{8.126}$$

(h) Show that (8.123) then becomes

$$\cos\theta = \frac{r_0^4 + r_1^4 - (r_0^3 - r_1^3)^{4/3}}{2r_0^2 r_1^2} \tag{8.127}$$

and find the analogous relations for (8.124) and (8.125).

8.36 Use Problem 8.35 to show that for optimized bifurcating vessels if [24]
(a) $r_1 = r_2$, then $\theta = \phi$,
(b) $r_2 > r_1$, then $\theta > \phi$,
(c) r_2 is much greater than r_1, then r_2 approaches r_0 and ϕ approaches $\pi/2$,
(d) $r_1 = r_2$, then $r_1/r_0 = 2^{-1/3} = 0.794 = \cos\theta$ and so $\theta = 37.5°$.

These results generally agree with observations.

8.37 Use Problem 8.36d to show that it would take ~30 generations of symmetric bifurcations starting with a vessel with the aorta radius of 1.5 cm to arrive at a vessel with the capillary radius of 5×10^{-4} cm [24]. (Note, however, that such arterial divisions are usually not simple symmetric bifurcations.)

8.38 There is a fusiform aneurysm in an aorta where the internal radius increases from r_1 (=1.25 cm) in the normal section to $r_2 = 1.3r_1$ in the diseased section, while staying at the same vertical height. The speed of blood flow is $v_1 = 0.4$ m/s in the normal section and the (gauge) pressure P_1 is 100 mmHg. The blood density is 1,060 kg/m^3.
(a) Find the speed of blood flow v_2 in the aneurysm.
(b) Find the pressure P_2 in the aneurysm.
(c) Use the Law of Laplace to find the tensions required in the normal part of the aorta and in the aneurysm to maintain the pressure difference (from inside to outside the vessel). Compare these values.

(d) Describe how this increase in the tension needed in the aneurysm wall and the decreased strength of the wall (due to the thinner aorta wall in the aneurysm) can lead to an unstable situation.

8.39 The normal inner radius of a large artery is 2 mm. It is 75 cm long, and the flow through it is 1/200 of the total blood flow. How would the pressure drop across it change if the flow through it were unchanged and there were severe stenosis in the artery
(a) across its entire length or
(b) across 5 cm of it?
(c) Alternately for (a) and (b), if the pressure at the beginning of the artery were 75 mmHg, would the pressure drop be severe enough to affect flow in the arterioles and capillaries?
(d) Alternately for (a) and (b), what added pressure would be needed at the beginning of the artery to maintain an unchanged flow in these arterioles and capillaries?

8.40 Arteriosclerotic plaque narrows down a section of an artery to 20% of its normal cross-sectional area. What is the pressure in that section if immediately before it the pressure is 100 mmHg and the flow speed is 0.12 m/s?

8.41 The osmotic pressure of blood is 25 mmHg higher than that of interstitial fluid because it has a higher density of proteins. What is the difference in their densities of proteins that accounts for this?

8.42 (advanced problem) The blood hematocrit is usually higher nearer the center of a blood vessel than at the blood vessel wall and has a distribution that we will take as $h(r) = H[1 - (r/R)^2]$ from $r = 0$ to R [51]. (The reason why flowing suspended particles, such as red blood cells, have higher concentrations near the center, called the Fahraeus–Lindquist effect, is not obvious.)
(a) The volume flow of a cylindrical shell in the vessel is $2\pi r v(r)\mathrm{d}r$, where $v(r) = 2u\left(1 - r^2/R^2\right)$ from (7.40), so this flow weighted for the hematocrit is $2\pi r h(r) v(r)\mathrm{d}r$. Therefore the average hematocrit in the transported blood is $h_{\mathrm{av}} = \int_0^R 2\pi r h(r) v(r)\mathrm{d}r / \int_0^R 2\pi r v(r)\mathrm{d}r$. Show that $H = 3h_{\mathrm{av}}/2$.
(b) Now find the average value of the hematocrit at any given time in the blood vessel by calculating $\int_0^R 2\pi r h(r)\mathrm{d}r / \int_0^R 2\pi r\,\mathrm{d}r$. Show that for the parabolic distribution of hematocrit this volume-averaged hematocrit is $3h_{\mathrm{av}}/4$.
(c) The result in (b) states that the average hematocrit of the blood in the vessel at any given time is less than that in the blood that is being transported. Is this a contradiction?

8.43 (advanced problem) Repeat parts (a) and (b) in Problem 8.42 assuming $h(r) = H\exp(-r/R)$ and show the volume-averaged hematocrit is $0.88h_{\mathrm{av}}$ [51].

8.44 (advanced problem) The analysis in Problems 8.42 and 8.43 assumed that the parabolic $v(r)$ we derived earlier assuming a constant viscosity is still valid when the hematocrit—and consequently the viscosity—decreases with radius. This should not be true. Qualitatively, how would you expect the spatially varying hematocrit and viscosity to affect the parabolic flow rate?

The Heart and Circulation

8.45 Would you expect cardiac muscle to be most similar to Type I, IIA, or IIB skeletal muscle? Why?

8.46 Someone wants to donate two pints of blood instead of the usual (and allowed maximum of) one. What consequences could this have?

8.47 The cardiac output of a woman remains at 5 L as she ages from 25 to 65 years of age, while her blood pressure increases in the average way.
(a) How does her total vascular resistance change?
(b) What fractional changes in vessel radius do this correspond to? (Assume typical conditions for arterioles.)

8.48 The volumetric flow rate out of a ventricle has been described in terms of the heart rate F and stroke volume V_{stroke} as $Q = FV_{\text{stroke}}$, while flow rates have also been expressed in terms of the vessel cross section A and flow speed u as $Q = Au$. Explain why these two characterizations are either consistent or inconsistent.

8.49 If the cardiac output is 5 L/min and heart rate is 1 Hz, determine the volume of the left ventricle at its peak if the ejection fraction is 65%.

8.50 (a) If the inner volume of the left ventricle is $100\,\text{cm}^3$ and the wall volume is $30\,\text{cm}^3$, find the inner radius, outer radius, and wall thickness for the ventricle modeled as a hemispherical shell.
(b) Find the wall stress during systole.

8.51 When blood is pumped out of the left ventricle, it travels "upward" a distance of about 10 cm in the aorta during the ∼0.2 s duration of the peak of systole, stretching the walls of this very compliant vessel. There are no external forces on the body during this time, so the center of mass of the body does not move. Consequently, when the stroke volume of blood (of mass $m_{\text{blood}} \simeq 70\,\text{g}$) is ejected upward, the rest of the body (of mass $m_{\text{rest}} \simeq 70\,\text{kg}$) moves "downward" (ignoring gravity and frictional forces). This is the basis of the diagnostic method called *ballistocardiography*, in which a person rests horizontally on a light, very low friction horizontal suspension [4]. (Such devices have been used to develop methods that assess heart function, but are not in clinical use.) Assume the person is lying along the x direction on this "couch"—with his head pointing in the positive direction—and the center of mass of ejected blood is at x_{blood}, that of the rest of the body is at x_{rest}, and that of the entire body is x_{body}.
(a) Show that $x_{\text{rest}} = (x_{\text{blood}}m_{\text{blood}} + m_{\text{rest}}x_{\text{rest}})/(m_{\text{blood}} + m_{\text{rest}})$ always.
(b) Now let us call the positions in (a) those before systole. At the end of the main part of systole, the blood and rest of the body have moved by Δx_{blood} and Δx_{rest}, respectively. Show that the body has moved by $\Delta x_{\text{rest}} = -(m_{\text{blood}}/m_{\text{rest}})\Delta x_{\text{blood}}$ and that this is $-0.1\,\text{mm}$.
(c) Because the blood moves with a constant velocity in this motion in the aorta, show that the velocity of the body during systole is $-0.5\,\text{mm/s}$ in the x direction.
(d) What is the average of Δx_{rest} during a full cardiac cycle? Why?

8.52 Compare the total mechanical and metabolic powers needed by the left heart and the right heart to pump blood.

8.53 Determine all the pressures, volumes, and flow rates in the overall body circulation model using the data provided in the Table 8.8. Do your answers agree with your expectations, such as the values in Table 8.1?

8.54 Draw a diagram for the fetal circulation model, along the lines of Fig. 8.52.

8.55 Derive and explain the equations of blood flow rates in a fetus (8.83)–(8.86).

8.56 (a) Sum the four equations of blood flow rates in a fetus (8.83)–(8.86) and show they are an identity [34].
(b) Explain why this means there are only three independent flow rate equations.

8.57 (a) Show that the fetal circulation model, with the four stated assumptions and equations (8.83)–(8.86), leads to the following solutions of the blood flow (with $KP_v = Q_R = Q_L = Q$) [34]:

$$Q_d/Q = (R_p - R_s)/(R_p + R_s) \tag{8.128}$$

$$Q_f/Q = (R_p - R_s)/(R_p + R_s) \tag{8.129}$$

$$Q_s/Q = 2R_p/(R_p + R_s) \tag{8.130}$$

$$Q_p/Q = 2R_s/(R_p + R_s); \tag{8.131}$$

(b) Show that this means the foramen ovale is open, as assumed, for $R_p > R_s$.

8.58 Show that closing the lungs totally in the fetal blood circulation models, by using $R_p = \infty$, gives the limiting solution with $Q_f = Q_d = Q$, $Q_s = 2Q$, and $Q_p = 0$ [34], with Q defined as in Problem 8.57.

8.59 Immediately after birth, the left and right hearts are still the same size (so $K_R = K_L = K$, the fetal circulation model assumption (2)) and the ductus arteriosus is still open ($R_d = 0$, assumption (3)), but now the lungs are inflated, so $R_p < R_s$, and the foramen ovale is now assumed to be closed. Show there is a self-consistent solution with a closed foramen ovale, and find Q_d and show that it is now negative [34].

8.60 There is a hole in the septum that separates the left ventricle and right ventricle (Fig. 8.62).
(a) Explain why you would expect the pressure in the left ventricle to decrease and that in the right ventricle to increase.
(b) Explain why you would expect the stroke volume from the left ventricle to decrease because of this.
(c) Explain why the oxygenation of the blood in the left ventricle would decrease and that in the right ventricle would increase.

(d) If during systole the (gauge) pressure, stroke volume, and oxygenation levels (relative to that in the vena cavae) in the left ventricle each decreases by 10% as a result of this, explain quantitatively how the body could try to compensate for this?

8.61 Sketch the pressure-volume relationship, as in Fig. 8.58, for a disorder of the left ventricle that raises the overall pressure-volume curve at the end of systole and decreases systolic compliance. Describe how this affects stroke volume, cardiac output, and blood pressure.

8.62 (advanced problem)
(a) Solve (8.98) assuming that the flow $Q_{\mathrm{L}}(t)$ is a constant a from $t = 0$ to αT, and 0 from $t = \alpha T$ to T, where $0 \leq \alpha \leq 1$. (This repeats for every heart beat.) Note that the pressure at the beginning and end of each cardiac cycle is P_{diastole} and it becomes P_{systole} at $t = \alpha T$. (Hint: The analysis is similar to that for exciting an isometric muscle in Chap. 5 (see (5.8)–(5.10)) and temperature regulation in Chap. 13 (see (13.18)); also see Appendix C.)
(b) Show that $a = V_{\mathrm{stroke}}/\alpha T C_{\mathrm{sa}}$.
(c) Show the solutions from (a) lead to the relations

$$P_{\mathrm{diastole}} = P_{\mathrm{systole}} \exp(-(1-\alpha)T/\tau) \qquad \text{and} \tag{8.132}$$

$$P_{\mathrm{systole}} = V_{\mathrm{stroke}}\tau/\alpha T C_{\mathrm{sa}} + (P_{\mathrm{diastole}} - V_{\mathrm{stroke}}\tau/\alpha T C_{\mathrm{sa}}) \exp(-\alpha T/\tau). \tag{8.133}$$

(d) Sketch $P_{\mathrm{sa}}(t)$ for several heart beats for $\alpha = 1/3$. Compare this sketch with those from the simple model in Fig. 8.55 and the real pulse in Fig. 8.53. Is this model better? Why?
(e) Show that when $\alpha = 0$ the solutions in (a) and (c) give the results presented in the text for the simpler model.

8.63 The solution to the classic Windkessel Model for steady-state flow that is suddenly turned off is exponential decay of the flow, as we saw in the simple model of the arterial pulse. In the electrical analog in Fig. 8.56 this corresponds to tracking the current when a constant voltage is initially applied and is suddenly turned off. Analyze this electrical problem analog and show that it has the same solution as the flow problem.

8.64 (advanced problem) Solve the electrical analog in Problem 8.63 for the three-element Windkessel Model.

8.65 (advanced problem) Solve the electrical analog in Problem 8.63 for the viscoelastic Windkessel Model.

Scaling

8.66 Calculate the heart mass and heart beat rate (in beats per minute) for a man (70 kg), woman (50 kg), and an infant (5 kg) using the allometric relation parameters in Table 1.13.

8.67 The heart rate of mammals F is known to decrease with body mass as $m_b^{-1/3}$. This seems to be true interspecies and also within a species. The human heart rate is known to decrease from infancy, through childhood and to maturity in a manner described better by body mass than age. Derive this relation using the dimensional analysis methods presented in Chap. 1. Assume that the stroke volume scales as body mass. Assume that a primary function of circulation is to bring warm blood from the core to the body surface to keep it warm. This means that the total blood flow rate scales as the rate of heat loss from the body.

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